

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	BS-PSYS-1 CIP Code 42.2799 - Research and Experimental Psychology, Other.
Program name	BS Psychological Science BS-PSYS-1: Bachelor of Science in Psychological Science See also: BS-BIOL-3/BS-PSYC-1 BS in Biology & Psych Science (CIP Code 42.2799 - Research and Experimental Psychology, Other.) BS-BCHM-3/BS-PSYC-1 BS in Biochemistry & Psych Science (CIP Code 42.2799 - Research and Experimental Psychology, Other.) BS-BPSY – BS in Business & Psychological Science (CIP Code 30.1701 - Behavioral Sciences) BS-CIS-2/BS-PSYC-1 BS in Computer Science & Psych Science (CIP Code 42.2799 - Research and Experimental Psychology, Other.)
year	Program assessment plan updated 2025
Program assessment coordinator	Arlen Moller
Assessment plan year	2024-2025
No assessment this year per plan	☐
LO(s) assessed	L1 Students will appropriately apply basic research methods in psychology, including research design, data analysis, and interpretation. L2 Students will be able to describe key concepts, principles, and applications of psychology L3 Students will demonstrate effective communication to scientific and professional audiences
Presented at program authority meeting date: <i>Please attach minutes</i>	Discussed at the Dept of Psych Undergraduate Programs Meeting on Friday Oct 10 th , 2025. Minutes from that meeting are appended below.

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations *M*, **P** Proficient}

LO1 -- Students will appropriately apply basic research methods in psychology, including research design, data analysis, and interpretation.

Artifact #1: Final Papers in PSY204 “Research Methods in Behavioral Science” SP24 (total enrollment = 18; Psych Major enrollment 16 out of 18; Instructor: Prof. Robert Keating)

(1) A description of how the artifact is concordant with LO #1.

LO#1: Students will appropriately apply basic research methods in psychology, including research design, data analysis, and interpretation. The final paper in PSY204 was explicitly designed to assess LO#1. The assignment requires students to describe in written form a psychology research design (involving the replication of a previously published effect), collection of original data, data analysis, and interpretation of findings.

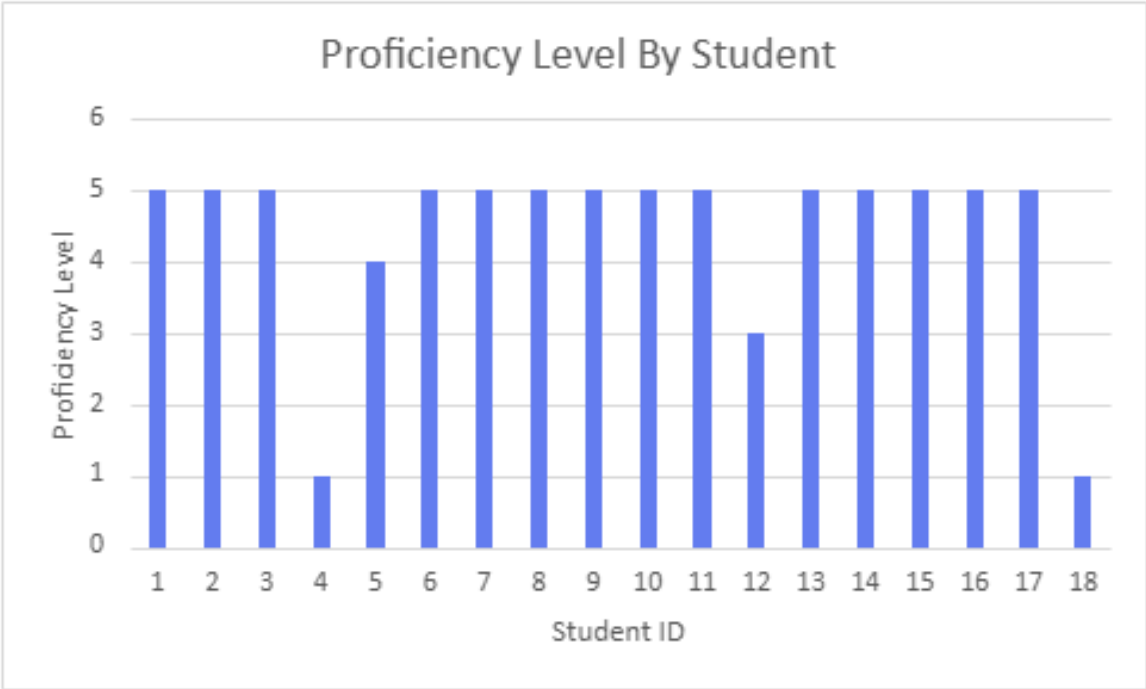
Below we provide the original grading rubric used by the instructor (Prof. Rob Keating), followed by the scoring rubric used by the program assessment coordinator (Prof. Arlen Moller) for the subset of PSY204 students who were also Majors in the BS in Psychological Science program.

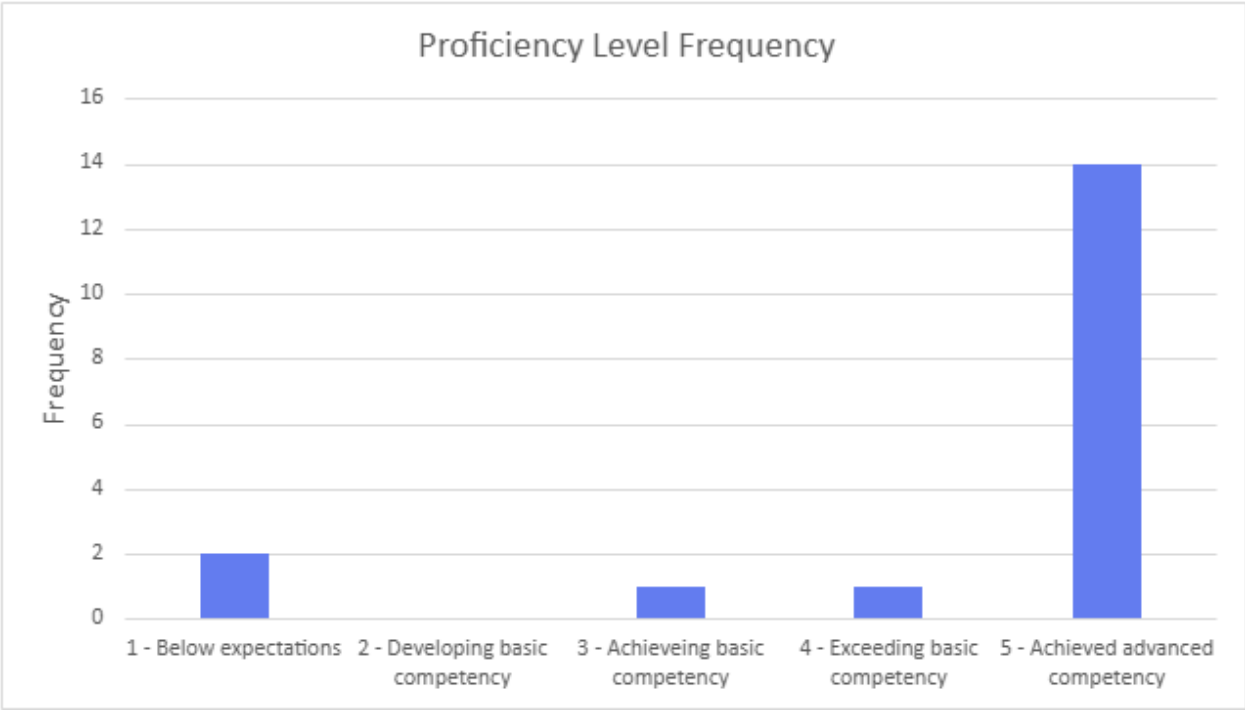
This PSY204 paper assignment included points allocated for each section of the paper (Instructors full grading rubric for this course assignment provided below). However, only some sections were relevant to LO1 (focused on research design, data analysis, and interpretation). Thus, our evaluation focused on only the following sections: Methods (20 points), Results (20 points), Discussion (20 points).

We converted scores for these sections into the following five levels of proficiency using our LO1 assessment rubric provided below. Below is a table with frequency statistics for achieving different levels of proficiency.

Levels of Proficiency Mapped onto PSY204 Research Methods Final Paper Artifact
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	Below expectations	Developing basic competency	Achieving basic competency	Exceeding basic competency	Achieved advanced competency
PSY204 Research Methods Final Paper	0-60%	61-70%	71-80%	81-90%	91-100%
Frequency of Majors in Psychology achieving different levels of proficiency in Spring 2024 (n = 21)	2 out of 18 (11.1%) *1 student did not turn in a final paper	0 out of 17 (0%)	1 out of 18 (5.6%)	1 out of 18 (5.6%)	14 out of 18 (77.8%)





FINAL PAPER GRADING RUBRIC | PSYC 204 RESEARCH METHODS – SPRING 2025

	EXCELLENT (100%-90%)	GOOD (89%-80%)	AVERAGE (79%-70%)	BELOW AVERAGE (< 70%)
Title Page <i>6 points</i>	Title includes variables and some expression of relationships (e.g., "difference between..."; "effects of x on y"). Running head (RH) shortened but complete within character limit. All relevant parts of the title page are included (i.e., header with running head and page numbering, title of paper, student's name, & college name). APA style is completely correct (double-spaced, Times New Roman, 12pt., 1" margins).	Title/RH is appropriate but may not be very concise. All relevant parts of the title page are included. APA style is completely correct.	Title/RH does not effectively convey all the variables in the study. Some relevant parts may be missing. Some elements of APA style may be incorrect.	Title/RH is not appropriate for a scientific paper. Most relevant parts are missing. Most elements of APA style are incorrect.
Abstract <i>8 points</i>	Abstract provides complete, accurate overview of paper, including research purpose, hypothesis, brief description of method (i.e., online experimental study, number and type of participants), major results, and implications/limitations of those results stated clearly and concisely within the word limit (100-250 words) APA style is completely correct (double-spaced, Times New Roman, 12pt., 1" margins, correct headings and paragraph alignment).	Abstract includes all essential information lacks conciseness. Word count falls slightly outside of requirements. APA style is completely correct.	Abstract contains incomplete information, contains inaccuracies, and/or lacks conciseness. Word count is below or above requirements. Some elements of APA style may be incorrect.	Abstract missing most essential information, contains extensive inaccuracies, and/or is highly unconcise. Word count is well below or above requirements. Most elements of APA style are incorrect.
Introduction Context & Topic <i>6 points</i>	Paper (i.e., first paragraph or two) begins in a broad manner and clearly explains the problem to be investigated.	Paper starts somewhat broadly and provides some theoretical or real-world context for the main concept in the study. An explanation of the key concept or	More clarity in the opening may be needed or the paper may begin with a definition of the topic but provide very little context for the idea (e.g., may begin immediately with	Paper provides very little or no context and provides no topic.

		question is provided, but it could be clearer.	review of previous research).	
Introduction Literature Review 7 points	Studies are described in enough detail so that their relation to other studies or to the relevant theoretical and methodological issues can be understood by the reader. The review is in the author's own words, and the focus is on the research, rather than the researchers. Citations are provided where appropriate. There is a specific, clear description of what is missing from this literature or what researchers do not yet know. A clear explanation of how the proposed study will answer this question or fill this research gap is included. APA style is completely correct (applies to entire Introduction; double-spaced, Times New Roman, 12pt., 1" margins, correct headings and paragraph alignment, in-text citations).	Studies are generally described in enough detail so that their relation to other studies or to the relevant theoretical and methodological issues can be understood by the reader (although some sections could be more specific). The review may include unnecessary quotations or poor paraphrases of the original articles. Some sentences may lack appropriate citations. The description of what is missing from this literature or what researchers do not yet know could be stated more clearly. An explanation of how the proposed study will answer this question or fill this research gap is included, but it could be more specific; or, the author makes a vague call for more research without specifying variables, populations, or methods. APA style is completely correct (applies to entire Introduction).	Some of the reviewed literature seems to be inappropriate or not well-linked to the topic. Literature may not be reviewed in enough detail for the reader to be sure of its relation to other studies or to the relevant theoretical or methodological issues or it may be one-sided, omitting contrasting viewpoints. The review may discuss key concepts from the literature without paraphrasing adequately (i.e., over-reliance on quotations). There is an abundance of uncited information. The description of what is missing from this literature or what researchers do not yet know could be stated more clearly. An explanation of how the proposed study will answer this question or fill this research gap is included, but it could be more specific; or, the author makes a vague call for more research without specifying variables, populations, or methods. Some elements of APA style may be incorrect (applies to entire Introduction).	Too few citations are included for the reader to be confident that that literature has been adequately reviewed. Much of the reviewed literature may be inappropriate or not reviewed in enough detail for the reader to be sure of its relation to other studies or to the relevant theoretical or methodological issues. Definition or discussion of key concepts may be improperly paraphrased. A brief summary of the literature is not provided. The description of what is missing from this literature or what researchers do not yet know is absent or very unclear. There is no discussion of why the proposed study will be important to this literature, or no study is proposed at this point. Most elements of APA style are incorrect (applies to entire Introduction).
Introduction Hypothesis 7 points	At least one hypothesis is clearly stated and describes a directional prediction between the study	A hypothesis is stated clearly and a directional prediction is made. The hypothesis	Variables in the hypothesis are stated, but no directional prediction is made. The hypothesis may	No hypothesis is presented, or the variables and/or directional prediction do not follow from

Results 20 points	Appropriate statistical analysis is performed (t-test) are appropriate statistics are reported (e.g., M_s and SDs, test statistic, p-value, effect size) and computed accurately. A figure is included on a separate page after the references and called out in text; contains relevant variables. APA style is completely correct (double-spaced, Times New Roman, 12pt., 1" margins, correct headings and paragraph alignment, correct formatting for reporting statistics).	The appropriate statistical test is computed and reported accurately. The figure may contain minor errors or confusing aspects, or is not called out in the text. Major APA style elements are correct, but here may be one or two minor errors in statistics formatting.	The appropriate statistical test is computed but may contain some reporting errors. The figure may contain minor errors or confusing aspects, is not called out in the text, or is presented in the body of the paper instead of on a separate page. Several APA style elements are incorrect.	The statistical test conducted was incorrect or computed and/or reported highly inaccurately. A figure is not included or contains extensive errors or the wrong information. Several APA style elements are incorrect.
Discussion 20 points	Discussion includes a restatement of the hypothesis and findings. Patterns in the data and relations among the variables are explained and conclusions do not go beyond the data. The explanation/interpretation is well connected to the hypotheses and to the broader psychological problem as represented in the introduction. Any discrepancies between the expected results and the actual data are explained. The take-home message is clearly summarized at the end. Author has considered to what extent the results are conclusive and can be generalized. Potential confounds or methodological limits (threats to main types of validity) are discussed as appropriate, and future research is suggested.	Discussion includes a restatement of the hypothesis and findings, but the analysis of their meaning may be weak or not well connected to the previous literature. There may be lack of consideration for the broader psychological problem, or the links to previous literature simply restate the introduction. Potential confounds or methodological limits are discussed but may be slightly incomplete or slightly under-detailed. Future research is suggested. Author has not considered to what extent the results are conclusive and can be generalized. APA style is completely correct.	The restatement of the results is not clear or is misleading. Links to previous literature simply restate the introduction. The author may inappropriately generalize beyond the data. Potential confounds or methodological limits are listed but not clearly discussed, and future research is not suggested or lacks detail. Author has not considered to what extent the results are conclusive and can be generalized. Some elements of APA style may be incorrect.	Discussion incorrectly states the results or is a rehash of the introduction without clearly presenting the current study. The overall message of the study is not clear. Potential confounds and methodological limits, and future research, may be listed but may be inaccurate, incomplete, or very unclear. Most elements of APA style are incorrect.

	variables. The hypothesis follows logically from the reviewed literature. It is clear what the experimental groups will be and what will be measured.	follows logically from the reviewed literature. It is somewhat unclear what the experimental groups will be or what will be measured.	not be well justified from the reviewed literature. It is unclear what the experimental groups will be or what will be measured.	the intended purpose of the study or the literature presented.
Method Design <i>5 points</i>	The design of the study is clear and complete and appropriate to test the hypothesis. Variables are appropriate and operationalized properly. APA style is completely correct (applies to entire Method; double-spaced, Times New Roman, 12pt., 1" margins, correct headings and paragraph alignment, in-text citations).	Design is complete and appropriate but not clearly described. Variables are appropriately operationalized but may be simplistic. APA style is completely correct (applies to entire Method).	Design is not complete or the operationalization of the variables is not clear. Some elements of APA style may be incorrect (applies to entire Method).	Design is not appropriate for the hypothesis; variables are not operationalized or not valid. Most elements of APA style are incorrect (applies to entire Method).
Method Participants <i>5 points</i>	Sample is adequately described, including initial and final sample size, reasons for exclusions, how participants were recruited, and summary of sample demographics.	Sample is described but may be missing one or two pieces of information (e.g., a demographic variable).	Sample is described but may be missing two or more pieces of information.	Sample is not described or is missing essential information (e.g., sample size, recruitment, exclusion, multiple demographic variables).
Method Materials <i>5 points</i>	Materials are described in full (manipulation and measures) with enough detail that a reader could replicate the study; materials should be appended if self-created, cited if not.	The description is adequate but could use more detail. The measures are appended or cited, as needed.	Materials are incomplete. The description is lacking in details but the measures are appended or cited, as needed.	Materials are not described, or the description is missing many details and the materials are not appended or cited, as needed.
Method Procedure <i>5 points</i>	Procedures is described in order with enough detail that a reader could replicate the study; instructions and protocol are included. Condition assignments are clear; randomization and any counterbalancing are explained as necessary.	Procedure is primarily complete and in order, but some minor details may be missing, or some procedural aspects could be explained more clearly.	The description is not in order, difficult to follow, or a few major details are absent.	Procedure is not described, or the description is very unclear, or many major details are absent.

	APA style is completely correct (double-spaced, Times New Roman, 12pt., 1" margins, correct headings and paragraph alignment, in-text citations).			
References <i>6 points</i>	Reference page includes all and only cited articles. The articles are appropriately scholarly and appropriate to the topic. Original articles/chapters were clearly read by the student. APA style is completely correct (double-spaced, Times New Roman, 12pt., 1" margins, hanging indent, correct reference components and formatting).	Reference includes all cited articles. The articles are appropriately scholarly but may be somewhat tangential and were likely read by the student. Major APA style elements are correct, but there may be a few errors with reference components and formatting.	Some cited sources may be missing from reference. Some references may not be appropriate for the assignment or scholarly. Key references are clearly cited from other sources and not likely read by the student. Several APA style errors with reference components and formatting.	Reference list is not included or is missing several cited articles. References may not be scholarly sources or otherwise not appropriate for the assignment (e.g., too many secondary sources). Most elements of APA style are incorrect.
General Formatting <i>Minus 2 points for each error</i>	Page requirement met (7-10 pages, not including Title, Abstract, References, Figure, and any Tables) – 11-15 pages total. All major paper sections included and in order (Title, Abstract, Introduction, Method, Results, Discussion, References, and Figure/Tables). Paper is prepared and submitted as a Microsoft Word Document	Page requirement met but one section is out of order. Paper is submitted as a word document.	Page requirement is over or under one page, and one or two sections are out of order. Paper may be not submitted as a word document.	Page requirement is over or under more than one page, and several sections are out of order or missing. Paper is not submitted as a word document.
Grammar, Punctuation, & Spelling <i>For each page, minus 1 point per error after first error</i>	Paper contains very few grammatical, punctuation and spelling errors. Less than one error per page.	Paper contains some grammatical, punctuation and spelling errors. No more than one error per page.	Paper contains multiple grammatical, punctuation and spelling errors per page. Sentences could be clearer and more precise.	Paper contains many grammatical, punctuation, and spelling errors throughout, and sentences have a pattern of being unclear.

Program Assessment Rubric:

PSYC Learning Outcome 1

Students will appropriately apply basic research methods in psychology, including research design, data analysis, and interpretation

Levels of Proficiency					
Components of Research Skills	Below expectations	Developing basic competency	Achieving basic competency	Exceeding basic competency	Achieved advanced competency
Methods skills: Recognition, evaluation, and implementation of appropriate methodology.	Does not rely on or demonstrate understanding of scientific method as recognized in the psychological and behavioral sciences	Recites steps in conducting research; articulates basic knowledge of common research designs (correlational and causal techniques) and validity; shows basic understanding of the value of controlled comparisons	Selects and applies appropriate method in simple projects; operationalizes and isolates variables; identifies influence of extraneous variables; able to identify relevant research from literature	Selects and applies appropriate method to maximize validity and reduce alternative explanations	Develops unique applications of research methods; establishes a research focus that identifies and builds on relevant primary research
Statistical reasoning: Recognition, evaluation, and application, of appropriate statistical techniques	Does not rely on statistical analysis to demonstrate relationships between variables in research and/or show understanding of the meaning of statistical values and concepts	Uses basic descriptive statistics; accepts inferential analysis without understanding statistical foundations	Selects and applies appropriate statistical processes to simple projects; evaluates success of projects based on statistics in a global sense	Selects and applies appropriate statistics with more independence; begins to use statistical reasoning as a basis for criticizing research results; distinguishes and correctly interprets measures of descriptive and inferential statistics and effect sizes	Uses statistical reasoning routinely for evaluating research; develops unique applications of statistics; shows strong command of distinctions among measures of descriptive and inferential statistics and effect sizes

Bias detection and management	Shows limited sensitivity to existence or effects of bias in designing and interpreting research	Can recognize some potential sources of bias when prompted	Begins to recognize influence of bias and confounds in framing research questions	Identifies the potential influences of bias and confounds; recognizes personal, professional, and cultural interests influence problem and method selection	Implements and demands high standards for adherence to scientific method to minimize complications of bias and confounds
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LO2 Students will be able to describe key concepts, principles, and applications of psychology

LO3 Students will demonstrate effective communication to scientific and professional audiences

Artifact 2: Final Paper Assignment in PSY312 “Human Motivation and Emotion” FA24

(enrollment = 29 total, 15 Majors in Psych Science; Instructor: Prof. Arlen Moller)

(1) A description of how the artifact is concordant with LO2 and LO3

LO2 Students will be able to describe key concepts, principles, and applications of psychology

This paper assignment directed students to describe key concepts and principles from motivation and emotion science, then apply those concepts and principles to a real-world context (options for application context included learning, video game design, and promoting environmental sustainability).

LO3 Students will demonstrate effective communication to scientific and professional audiences.

This paper assignment directed students to exercise their communication skills and directly assessed all four components of communication skills outlined in our rubric:

1. Quality of evidence
2. Organization and persuasiveness of argument
3. Clarity in communication
4. Use of APA format, grammar, appropriate supportive figures or visuals

15 Psych Science Majors completed this Assignment in Fall 2024.

Provide a rubric describing the method used to place individual student in at least 3 assessment categories; **D** Does not meet expectations, **M** Meets expectations *M*, **P** Proficient.

Below we provide the instructions and scoring rubric for the assignment with a numeric scale of 0 to 100. We translate those numeric scores to 5 (rather than the minimum of 3) assessment categories as follows.

Below expectations	Developing basic competency	Achieving basic competency	Exceeding basic competency	Achieved advanced competency
0-64	65-70	71-80	81-90	91-100

PSY312 Paper Assignment Instructions & Scoring Rubric

PSY312: Human Motivation & Emotion Fall 2024

Due by or before Friday Nov 1st 2024, at 5pm CST
Choose your own adventure

General Instructions (for all 3 topic options)

The Basics. This assignment will involve writing a structured 5-6 page paper using a topic of your choice (three options). The target length is 5-6 pages (not including cover page, references section, or optional figures), APA format, double-spaced, Times New Roman 12-point font, 1" margins. The reference section should include 4-10 references in addition to the course textbook, when citing research findings, the primary (original) source should be cited. Citations should appear in the body of the paper using APA format. There should be zero quotations in this paper, please paraphrase sources in your own words. First person pronouns ("I" and "me") are okay for this paper, and may be necessary. Otherwise, successful papers may take a wide variety of forms.

Option #1: Exploring your motivation to learn. Identify three motivational strategies that have been used by parents/guardians/teachers to help motivate you to learn. Analyze the relative strengths and weaknesses of each using concepts from this course. What have the consequences been for the quality and quantity of your motivation to learn in general, and in specific areas (college courses, this course, etc.)?

Option #2: Exploring motivation in video games. Use motivation and emotion science to recommend improvements to the makers of a favorite video game.

Option #3: Climate change risk mitigation strategies. Use motivation and emotion science to advance climate change risk mitigation strategies for the sake of your and future generations' quality of life on planet Earth.

All papers should be succinct and well written and should not contain spelling or grammar errors. Students who struggle with writing papers are highly encouraged to take their paper to the writing center for free 1-on-1 help before handing it in. For more information, go to <https://web.iit.edu/cac/student-resources/writing-guides/one-one-help-writing-assignments> (This free service is really valuable for writers of all skill levels)

Turning it in. Upload an electronic copy of a Word (.doc or .docx) file to Canvas using the assignment portal. If you use another word processor (e.g., .pages, etc.), you must convert the file to .doc/.docx before uploading it. For the file name, please use your first and last name followed by the option you selected (e.g., "Arlen_Moller_Option1.docx").

Formatting. APA style is required. If you haven't used APA style look for a guide, examples, and ask for help at the Writing Center. You should use Times New Roman 12-point font and your paper should have 1" margins on all four sides. In addition, your paper should have an APA style title page (does not count in the 5-to-6-page double-spaced length). You should also include an APA style references section (does not count in the 5-to-6-page double-spaced length). Remember to cite any information that is not common knowledge or did not originate from you. When you cite sources in the body of your paper,

please follow APA format. For help with using APA citation, please see the following guide:
http://www.iit.edu/arc/workshops/pdfs/APA_style.pdf

Here are some examples of the APA style of citation in the body of a paper:

Myers (2011) suggests that the formal definition of a group is two or more interacting individuals.

This scene did not include a group of individuals, which is often considered to be two or more interacting individuals (Myers, 2011).

Response deadline procedures are useful for establishing that a particular tendency might be non-strategic in nature in that participants have insufficient time for developing response strategies (Draine & Greenwald, 1998; as cited by Myers, 2011).

I have listed an example reference page below. Note that books and journal articles are listed in slightly different ways. Consult the APA Publication Manual (in the library) to learn how to cite information and list references according to APA style.

References

Draine, S. C., & Greenwald, A. G. (1998). Replicable unconscious semantic priming. *Journal of Experimental Psychology: General*, 127, 286-303.

Academic Integrity. The faculty and administration at IIT support an environment free from cheating and plagiarism. This writing assignment is meant to practice and demonstrate your abilities and understanding of the course material. Each student is responsible for being aware of what constitutes cheating and plagiarism and for avoiding both. The complete text of the IIT code of academic honesty can be found in the student handbook:
http://www.iit.edu/student_affairs/handbook/information_and_regulations/code_of_academic_honesty.shtml

Plagiarism and Quoting Sources. This paper must represent your own original work, and any uses of or references to others' work (articles, books, etc.) must be cited. In psychology, we typically avoid using direct quotes. If you review the assigned reading in this course (both the textbook and primary sources), you will find that quotations are used very rarely. Instead of quoting, in psychology we tend to paraphrase the words from our sources (i.e., we put them in our own words) and use a citation. If we do use a quote, we must use quotation marks and provide the page number from the source in the citation (see the APA manual for proper quotation formats). If you turn in a paper without following these rules, this is considered plagiarism. If you have questions at any time about how or when to cite or quote published material, please contact me. Evidence that you have plagiarized will automatically result in a zero, and further action may be taken as specified by the IIT Code of Academic Honesty.

SafeAssign. All written assignments will be submitted via Canvas and will be subject to review by SafeAssign software, which identifies content that appears in other sources, including papers by former students at Illinois Tech. When turning in assignments you will have access to the SafeAssign report and can see if your work has been flagged for possible plagiarism. If this software identifies that you have copied content from another source, without quoting and properly citing that content (i.e., plagiarism) you will receive **a zero for the assignment**.

Generative AI. There are many interesting and helpful applications for generative AI tools, in general, and for Large Language Models (LLMs) like ChatGPT, specifically. However, there are also many reasons for concern about using these tools, specifically, for the purpose of academic writing assignments, for both students and professional writers (van Rooij, 2022). Scholars including Bender et al. (2021) have alternatively referred to Large Language Models (LLMs) as “stochastic parrots” to help convey their limited “intelligence.” In particular, the tendency for stochastic parrots (e.g., ChatGPT, Bard, QuillBot, etc.) to generate academic writing that is systematically biased or false, often citing nonexistent empirical sources, is very problematic for this application. As such, compelling evidence that a stochastic parrot was used to generate content submitted for a writing assignment in this course, you will receive **a zero for the assignment**. This includes, but is not limited to, describing a study or citing a source in your reference section that does not exist. The high penalty is a reflection of this constituting a violation of [Illinois Tech’s Code of Academic Honesty](#). Faculty members who are aware of a violation, or who have good faith evidence of a violation, must report the infraction to the [Designated Dean of Academic Discipline](#). There is also a very important self-serving rationale for not using stochastic parrots for writing assignments in this course. The real purpose of these writing assignments is not to evaluate you but to help you learn – to improve your academic writing and ability to articulate your own perspectives on the material. To develop these skills, going through the process of writing and editing your own work is critical. In the long run, taking shortcuts is cheating yourself.

Finally, a discussion of ethics and generative AI should at least mention the fact that, at present, generative AI tools are having a very negative effect on the environment. Please consider this in the context of deciding whether to use these tools (See: https://pad.riseup.net/p/Against_AI_and_Its_Environmental_Harms-keep).

PSY312 Paper Grading Rubric Option 1

Option #1: Exploring your motivation to learn. Identify three motivational strategies that have been used by parents/guardians/teachers to help motivate you to learn. Three clearly labeled subsections of your paper should include: (i) the name of a motivation strategy using a term from class and its definition, (ii) what it looked like in your personal experience, (iii) how this motivation strategy has been studied (citing a primary source), (iv) strengths and/or weakness, (v) consequences for the quality and/or quantity of your motivation to learn in general, and in specific areas (college courses, this course, etc.). In the last section, a conclusion, identify a new (4th) motivation strategy you'd like to use on yourself and why you picked it.

The Basics (20 points)

_____ / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

_____ / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

_____ / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

_____ / 25pts. Motivational strategy #1. (i) the name of a motivation strategy using a term from class and its definition (5pts), (ii) what it looked like in your personal experience (5pts), (iii) how this motivation strategy has been studied by researchers with a primary source cited (10pts), (iv) strengths and weakness (5pts).

_____ / 25 pts. Motivational strategy #2. (i) the name of a motivation strategy using a term from class and its definition (5pts), (ii) what it looked like in your personal experience (5pts), (iii) how this motivation strategy has been studied by researchers with a primary source cited (10pts), (iv) strengths and weakness (5pts).

_____ / 25 pts. Motivational strategy #3. (i) the name of a motivation strategy using a term from class and its definition (5pts), (ii) what it looked like in your personal experience (5pts), (iii) how this motivation strategy has been studied by researchers with a primary source cited (10pts), (iv) strengths and weakness (5pts).

____ / 5 pts. Conclusions. Going forward in your journey as a lifetime learner, what new evidence-based strategy could you introduce to support your motivation to keep learning. Cite a primary source.

PSY312 Paper Grading Rubric Option 2

Option #2: Exploring motivation and emotion science in video games. Use motivation and emotion science to analyze strategies used in a video game of your choice, and make recommendations for improving it. Specifically, in this paper you will identify **three different strategies** used to motivate players in a video game of your choice. For each strategy / game feature, evaluate the relative strengths and weaknesses using concepts from this course and empirical evidence (findings from peer reviewed research studies on motivation and emotion, though not necessarily about video games, per se). Finally, offer some suggestions for how game developers could use evidence-based strategies to improve this video game in the future.

The Basics (20 points)

____ / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

____ / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

____ / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

____ / 5 pts. Briefly describe the video game and why you chose it.

____ / 20 pts. Motivational strategy #1 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

____ / 20 pts. Motivational strategy #2 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

____ / 20 pts. Motivational strategy #3 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

____ / 15 pts. Conclusions. Offer some suggestions for how game developers could use evidence-based strategies from motivation science to change/improve this video game in the future (15 points)

PSY312 Paper Grading Rubric Option 3

Use motivation and emotion science to advance climate change risk mitigation strategies for the sake of your and future generations' quality of life on planet Earth.

Option #3: Climate Change Risk Mitigation Strategies. Use motivation and emotion science to inform strategies for reducing climate change for the sake of your and future generations of people living on planet earth. As evidence that the planet is warming due to human behaviors increases, many scientists have put forward recommendations for governments, businesses, and individuals to help mitigate further downstream suffering. Yet, we are finding that many governments, businesses, and individuals lack sufficient motivation to follow those recommendations. A big challenge! To help address this challenge, identify **three different strategies** that you think could help motivate government officials, business leaders, or individual citizens to follow one or more of

these recommendations. Then, evaluate the relative strengths and weaknesses of each strategy using concepts from this course and empirical evidence (findings from peer reviewed research studies on motivation and emotion, though not necessarily about climate change mitigation, per se). Finally, **offer some suggestions** for how the U.S. federal government, Chicago Mayors' Office, Illinois Tech, or you could use evidence-based strategies to motivate climate risk reducing action better going forward.

The Basics (20 points)

____ / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

____ / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

____ / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

____ / 20 pts. Motivational strategy #1 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

____ / 20 pts. Motivational strategy #2 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss

consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

____ / 20 pts. Motivational strategy #3 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

____ / 20 pts. Conclusions. Offer some specific evidence-based suggestions for improving how the U.S. federal government, Chicago Mayors' Office, Illinois Tech, or you as an individual could use evidence-based strategies to motivate climate risk reducing action better going forward.

PSYC Learning Outcome 3: Students will demonstrate effective communication to scientific and professional audiences

Levels of Proficiency					
Components of Communication Skills	Below expectations	Developing basic competency	Achieving basic competency	Exceeding basic competency	Achieved advanced competency
Quality of evidence	Relies on popular press reports and anecdotal evidence; selects sources based on personal bias	Reproduces simplistic, textbook information; conducts limited literature review	Relates content from several sources, but tends to include resources nonselectively; limited literature review	Integrates content from multiple sources, acknowledging contradictory information; uses resources selectively and to the point	Evaluates relevant content from range of available sources; reflects how context influences judgment; shows refined and flexible use of published research
Organization and persuasiveness of argument	Argues based on common sense; accepts personal experience as conclusive	Can develop simple arguments; tends to argue from personal experience rather than from research evidence	Develops plausible arguments	Creates coherent and integrated arguments based on research evidence	Creates compelling arguments; anticipates and defends against criticism
Clarity in communication	Confusing and lacking in clarity	Shows limited awareness of engaging audience; does not define constructs or use examples	Demonstrates some awareness of audience by using some examples; assumes audience's knowledge is consistent with own	Engages audience with examples, definitions of constructs, and clear reasoning	Clearly conveys ideas for wide range of audiences (experts and laypersons alike); uses examples effectively to augment understanding
Use of APA format, grammar, appropriate supportive figures or visuals	Fails to use APA style, serious grammar mistakes, no visuals	Applies APA format with serious flaws; expresses ideas in informal language; ineffective or no visuals	Uses APA style inconsistently; shows increasing formality and professionalism in expression; uses supportive visuals	Uses APA format more consistently; practices professional, formal expression to improve communication; uses effective visuals	Uses APA format expertly with minimal errors; demonstrates sophisticated and clear expression; uses visuals that enhance understanding

Below expectations	Developing basic competency	Achieving basic competency	Exceeding basic competency	Achieved advanced competency
0-64	65-70	71-80	81-90	91-100
0	0	3	5	7
		20% (3/15)	33.3% (5/15)	46.7% (7/15)

Description of findings

100% of the 15 Majors in Psychological Science were Achieving Basic Competency or Better.

46.7% (7 of 15) achieved advanced competency

33.3% (5 of 15) were exceeding basic competency

And 20% (3 of 15) were achieving basic competency.

Continuous Improvement, CI, plans

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

CI Plan 2024-2025

CI Plans	Strategy to implement these plans in 2024-2025	Were the CI plans implemented?
CI Plan 1. Developing a Dept sponsored 1-hour workshop for UG Psych Majors on academic writing and academic poster making for psychology audiences	The Program Assessment Coordinator, Prof. Arlen Moller (who is also the Dept's Director of Director of UG Programs) will coordinate the development of a 1-hour Psych Communication Skills workshop focused on academic writing and poster making to be schedule in late March 2025. This will be done with input from the campus Writing Center, and materials (slides and handouts) will be shared with Writing Center Coaches for use with Psych Majors in the future.	Materials for a 1-hour communication skills workshop for Psych Science Majors was developed in the Spring 2025 semester, including slides and handouts. These materials have been shared with key instructors including the instructors for PSY100 (Christine Meyer), PSY204 (Rob Keating), and PSY312 (Arlen Moller). Updates to these materials to be developed over the next 3 years will include best practices for using generative AI for psych science related professional communication.
CI Plan 2. Work with the Dean's office to provide an opportunity for Psych Majors to present research posters developing in PSY204 Research Methods at the Spring semester Lewis College Undergradua	Prior to 2023-2024, the Lewis College Undergraduate Research Day was hosted during the last week of the Spring semester (late April). This allowed students in the Spring semester sections of PSY204 Research Methods to present conference style posters summarizing their work (an artifact used to assess LO3). As a side note, this college-wide UG Research Day event evolved out of a Dept of Psychology specific UG Research Day event that was initiated in 2012 with support from the Dept of Psychology's UG student org, Psi Chi. This past year, for logistical reasons the Lewis College Undergraduate Research Day event was moved up several weeks to late March or early April, and the new timeline didn't allow PSY204 students to	The Lewis College UG Research Day event was held in May 2025 with encouragement from our program faculty and with program faculty representation on the planning committee (Rob Keating and Arlen Moller). To make the event possible on a shoestring budget, Prof. Arlen Moller along with Profs. Lei Li (Biology) and Matthew Shapiro (Social Sciences) carried materials (easels and display boards to and from a storage area in Pritzker Science building's basement and MTCC ballroom. The Lewis College Dean's office purchased a printer able to print posters to reduce costs and ensure continued capacity to host this event in the future.

te Research Day event.	participate (using PSY204 research project posters). The Program Assessment Coordinator, Prof. Arlen Moller (who is also the Dept's Director of Director of UG Programs) will reach out to Lewis College Dean, Jennifer deWinter, now to begin a discussion about the benefits of scheduling this event later in the Spring 2025 semester and also bolstering marketing, and online archiving of student research projects. Current link to Lewis College Undergrad Research Day info on our university website: https://www.iit.edu/science-letters/research/student-research/undergraduate-research-day	We have stored photos and poster files, and have reached out to Marketing and Communications (M&C) to create a online repository for these materials, but that part of the plan has not yet been implemented because of M&C's limited bandwidth.
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We continue to implement Packback “deep dives” tools into multiple psych courses (including PSY310 and PSY312) as was articulated in our 2023-2024 continuous improvement plan. We believe this helped improve Majors communication related skills (LO3), as this tool includes automated prompts to adhere to APA Format and to cite sources synchronously *while students are writing* (vs. days or weeks later). Indeed, the percentage of Majors meeting or exceeding basic competence increased over the last year.

This cycle's CI plan

Describe any CI plans developed from this cycle's data, as well as the strategy developed to implement these plans.

This cycle's data were discussed by the Dept of Psychology's Undergraduate Programs Committee on October 10, 2025. A transcript and notes from that meeting are provided below. Based on this discussion, our CI plans and strategy for implementing these plans between now and the next planned assessment in 3 years (for October 15th, 2028) is provided below.

Two plans for improving Psych Majors' effective communication to scientific and professional audiences (LO3) to implement over the next 3 academic years (2025-2026; 2026-2027, and 2027-2028) are the following:

(1) We will develop in collaboration with the campus Communication Lab materials for UG Psych Majors on academic writing for psych science that includes discussion of appropriate and inappropriate use of new generative AI tools (LLMs). This may include developing a short in-person workshop on academic writing in psychological science that will include discussion of appropriate and ethical use of generative AI tools, an accompanying slide deck that will be shared with psych instructors, and boilerplate syllabus text related to best practices.

(2) We will continue working with the Dean of Lewis College to provide an opportunity for Psych Majors to present research posters developing in PSY204 Research Methods at the Spring semester Lewis College Undergraduate Research Day event. Work to develop an online gallery where these research posters can be archived and shared with future students. If feasible, we will

also seek to include short videos of students summarizing their research completed in PSY204 and presented at Lewis College UG Research Day.

Transcript Dept of Psych Undergrad Programs Committee meeting on Friday Oct 10th, 2025

October 10, 2025, 7:06PM

□ **Arlen Moller** started transcription

AM **Arlen Moller** 0:05

Good. All right. So that's started and now I'm going to reshare the.

Screen.

close.

Hmm.

Oh, OK, here it is.

OK, so now that should be back up.

I'm going to take notes, but for me it's a little easier to use a pen and scribble and then I'll especially while talking and then I'll add stuff to this to the minutes after the after the call.

But main agenda item for today is going to be discussing the program evaluation reports that are due Wednesday of next week. I mentioned the transcription.

We're not doing a program evaluation report for applied analytics. There's one student left. I believe Christina is advising them and they're on.

KB **Kristina Bauer** 1:28

This semester they graduate.

AM **Arlen Moller** 1:29

This semester, so even sooner. OK, so the first of the two degree programs we're going to talk about and submit reports on is the Psych Science degree. There's a link to the degree requirements.

Um.

In the the header subheading, current enrollment is 52 psych science majors and I believe that includes 6 that are dual degree students.

Mostly biology, maybe one biochem. Does that number?

Seem consistent with folks who are doing advising or been following this.

NL **Nicole Legate** 2:19

It seems a little low to me. Um, but.

AM **Arlen Moller** 2:23

It's a little low to me too, yeah.

NL **Nicole Legate** 2:25

I know there's difficulty with sometimes tracking those students because sometimes they're assigned to bio and we don't even see and then we kind of learn informally. So I it does seem low to me, but it also could be the truth.

AM **Arlen Moller** 2:34

Right.

I.

Yeah, I had the same thought. And so I'm going to, after the call, follow up with maybe Katie Spink and Nick Menhart, because I think they're tracking both bio and biochem dual degrees and they would be the ones that.

Find primary advisor often just to make sure that the six is comprehensive. I had thought in I I looked for the two different reports that we're getting and that this was the number that showed up but.

I think that maybe it's not counting some of the dual degrees, so confirm.

NL **Nicole Legate** 3:28

Yeah, because we sometimes get like 6 a year.

AM **Arlen Moller** 3:32

Yeah.

NL **Nicole Legate** 3:36

I would confirm with them, but sometimes they don't even know.

AM **Arlen Moller** 3:40

I know, I know.

NL Nicole Legate 3:40

But they'll they'll know better than us, I think.

AM Arlen Moller 3:44

Yeah, somebody's somebody's been assigned. The other person I could double check with. So I'll check with maybe Katie Spink first, but I'll check with those professional Lewis advisors cause they may, even though they're brand new, they may have a a more comprehensive list.

Next thing, just to get on the same page about is the three learning objectives for the psych science degree. We haven't changed these in a long time, but they emphasize.

Learning objective one has to do with learning and applying basic research methods in psych research design, data analysis, interpretation, the main artifact we've been using.

To assess this learning objective has been the final paper in Psych 204. So Rob did that part of the analysis last year and this year. Learning objective two is be is pretty broad, being able to describe key concepts.

Principles, applications of psychology.

Last year we used artifacts from Alyssa's Psych 465 course, but that's only offered every other year and and she's left the university.

Um, so for this report.

I double dipped and used the final paper for Psych 312 human motivation for both learning objective 2 and learning objective 3. That's about effective communication to.

Scientific and professional organizations.

So before I go a little further, one thing up for discussion is.

Reviewing which artifacts we're using. Does anyone have any thoughts about?

Other courses or course artifacts that might be appropriate for.

Evaluating in particular, maybe learning objectives 2 and three.

NL Nicole Legate 6:19

I know when I was doing this for the years when I was program director, I would use a lot of different classes or like I I felt that a lot of different classes could count for #2. I think I regularly used.

AM Arlen Moller 6:33
Definitely.

NL Nicole Legate 6:37
Adult and child development to to assess that one like any kind of applied paper that they may have had. I may have also used, I think right, I may have used my own courses just.

AM Arlen Moller 6:41
OK.

NL Nicole Legate 6:57
Because it it helps, you know, it reduces the burden to go and ask someone else to do it. And so you could think about those, but I think that I think that motivation and emotion are just fine to use, especially if there's an assignment that covers both of those.

AM Arlen Moller 7:02
Sure.
Yeah.
Mm-hmm.

NL Nicole Legate 7:17
And then I think for learning objective three, I often would also double dip with 204 because sometimes there's a presentation that happens and and I would use.

AM Arlen Moller 7:26
Hmm.

NL Nicole Legate 7:34
I know when I was teaching the class and also doing the learning objective, I would use their present, their poster presentations. I'm not sure if that's been happening in the past. Like I know some years it it has, some years it hasn't.
But that's that's one that I've used as well.

AM Arlen Moller 7:52

Yeah.

Mhm.

RK Robert Keating 7:56

We we we restarted. So the last I think the first two years I taught 204. We didn't do the poster presentation because the planning for research day was very it was kind of haphazard and it was it was too early in the semester to fit with the timeline of the projects.

But last year we last spring we made it work and even if they don't do the poster they have an in class presentation. But I guess technically it's not really a scientific or professional audience like the like research day would be so but but but we're going to continue to do as long as I'm teaching the course we'll continue to.

Utilize research day as long as it's the timeline work so.

Also for objective 2 it.

AM Arlen Moller 8:40

Mhm.

Mm-hmm.

RK Robert Keating 8:45

I guess this depends. That's another discussion if we're gonna do this every year or if we're gonna open it up to multiple years. But when I teach 310, they have thought papers like 4-4 or five thought papers.

AM Arlen Moller 8:52

Yeah, yeah.

RK Robert Keating 9:01

Where they actually have to take a concept, explain the concept and then apply it to something in like a personal experience or a movie or you know, something like that. So that that could fit for that I think. I think Nikki, I think you might have used it in the past maybe.

NL Nicole Legate 9:16
Mhm, mhm. Yeah, yes.

AM Arlen Moller 9:17
That's for LO two, yeah.

RK Robert Keating 9:17
OK.

NL Nicole Legate 9:21
Mhm.

AM Arlen Moller 9:22
Yeah, I think one of the things to consider is that the course be offered frequently and that we have a high number of majors that are captured.
And so 310 is offered at least recently, I think every.
Every semester other other courses offered every semester are 301 and 303. I don't know if there are.
Writing assignments or other assignments in those? Has anyone taught 301 or 303 recently?

RK Robert Keating 10:07
Teaching 301 now.

AM Arlen Moller 10:10
OK.

RK Robert Keating 10:10
I teach that every year, at least one semester a year.

AM Arlen Moller 10:15
Any any assignment that lines up with learning goals two or three?

KB **Kristina Bauer** 10:23

Does it have to be individually assessed? Can it be a team project?

RK **Robert Keating** 10:23

The job.

AM **Arlen Moller** 10:29

It's a little, it's a little messier as a team project because you're trying to filter out non majors and be evaluating just the just the students who are in a particular major that gets trickier, yeah.

KB **Kristina Bauer** 10:30

And it.

So they have. They work on a group project and there's a presentation.

There's a presentation.

AM **Arlen Moller** 10:50

OK.

KB **Kristina Bauer** 10:53

Yeah, well, oh, no. I have them do a presentation on a scientific article they read.

AM **Arlen Moller** 11:01

Individually or as a group?

KB **Kristina Bauer** 11:03

as a group.

AM **Arlen Moller** 11:06

OK.

KB **Kristina Bauer** 11:09

One of my homeworks is apply motivation to yourself. So they're discussing and I

there are probably several homeworks that I guess that that would would get at the very vague describe key concepts.

AM Arlen Moller 11:14

Mhm.

OK.

OK, alright, so.

LS Lindsay Sheehan 11:30

You can't have exams for these, like just like the midterm exam or something. It just seems.

KB Kristina Bauer 11:37

All this.

AM Arlen Moller 11:37

That's a good question for two, I guess. Two, yeah.

LS Lindsay Sheehan 11:39

Or two. It just seems easier, I don't know.

AM Arlen Moller 11:42

Yeah.

KB Kristina Bauer 11:43

Except that they have to describe it, which suggests either verbal or written self generated content.

AM Arlen Moller 11:49

Hmm.

RK Robert Keating 11:52

In my too.

LS Lindsay Sheehan 11:53

Yeah, like fill in the blank.

KB Kristina Bauer 11:53

Not.

RK Robert Keating 11:56

In my 203 class, I have three essay questions on both. There's two exams a semester and there's three essay questions where they actually have to describe like sampling distributions, you know?

AM Arlen Moller 12:11

OK.

RK Robert Keating 12:11

Geez, I can't even think of what my own questions are. Difference between AT test and ANOVA and stuff like that.

So they could potentially be used those those essay questions.

AM Arlen Moller 12:25

OK, we just got to be careful about 204 being the only artifacts used for the for the whole degree. Oh, that's 2033 has.

RK Robert Keating 12:30

That that's 203. That's that, yeah.

AM Arlen Moller 12:37

Essay questions like this, OK.

All right, that's helpful.

KB Kristina Bauer 12:46

I think one thing to think about is like one thing we talk about in IO we have when we have multiple metrics because we have a couple that are like apply IO knowledge. To real world scenarios loosely defined and like we track them. There's ones in their

first semester typically, although with adjuncts were that's less stable and there's one later where they actually have to present their internship to each other.

AM **Arlen Moller** 13:18
Hmm.

KB **Kristina Bauer** 13:18
And we track growth. So is there something else in this two thing that we're trying to that would be informative to us?

AM **Arlen Moller** 13:31
It's a good question. Um.
That that we have would have two time points to to be able to assess change or growth, you mean?

KB **Kristina Bauer** 13:43
Not like, I mean, it happened. I don't think it was an intentional design choice on our part. I think it just naturally happened that way.
That we are able to well, and now we're less consistently able to do this, but we are able to compare them from their first semester to like somewhere near the end.
Yeah.

AM **Arlen Moller** 14:28
Yeah, I can't think of anything we have.
In place, but I think it's a good suggestion to think about.
In terms of continuous improvements, is there is there something?
And and there may be actually, maybe we can return to this idea because in that improvement, continuous improvement, there are some ideas that might make sense to to to to.
Calculate change in some competency. Yeah, so that's a good suggestion. Cool.
OK, so let's look at what we have in terms of numbers.
You know, in all of our classes we get some non majors. So for last spring's 204 there were 21 students who were psych majors one.
Student who's a behavioral health and Wellness major for last fall Psych 31215 psych majors and.

Zero behavioral health and Wellness majors. So that brings up this question about how frequently we want to. We'd be collecting the data every year, but how frequently we want to analyze the data.

And try to.

You know, um.

Say that that there's something meaningful to be said based on the the sample size for. I'm not sure exactly why might have been tradition we we've been analyzing the data even with.

Similarly small sample sizes every single year. And it was news to me in the workshop that lots of other degree programs are doing it once every three years, as as little as once every five years.

They still collect the artifacts and have a a data file that is aggregating over multiple semesters as long as they hopefully the assignments stay consistent.

But I think one discussion for us, I think to start with psychology is do we want to reduce the frequency? And if so, like how frequent would be appropriate?

Any thoughts?

KB

Kristina Bauer 17:29

We do this for the I/O program. We only evaluate. So I report numbers every year and but we only do like a really deep dive discussion and and like we combine everything for the PhD program. I've gotten Scott to go to four years.

And for the master's program, it's every three.

And that gives us with current cohort, that gives us between 30 and 45 masters students. And our PhD numbers are still, I think, too small to make meaningful decisions.

AM

Arlen Moller 17:51

OK.

OK.

Got it. Even waiting four years.

KB

Kristina Bauer 18:09

Yeah.

AM

Arlen Moller 18:12

Yeah.

Well.

NL **Nicole Legate** 18:19

Christina, when you say you still report the numbers, what does that look like? Like what's the difference in your report on on versus off years?

KB **Kristina Bauer** 18:33

So in off years I report well.

So last year I put in those dumb figures that and tables they wanted, but they took me a long time. So I think I'm gonna skip them this time and see if anyone gets mad at me. And historically what we had done is just like list our categories and the number of students that fell in each of our our categories on our our rubric.

When? So that's the off years. I just give the flat number and I write a sentence that says we looked at the data, there were no red flags. Full evaluation will come in the report due whenever it's actually.

Do when we do the full report, I put in whole data tables aggregating across all the years and and we have texts that actually like interprets all of the results in terms of the patterns that that we see. So there's like meaningful.

Um, description and like an interpretation of the numbers.

NL **Nicole Legate** 19:36

Got it. So the numbers definitely you report like NS, but then are you also giving the like number of A's or whatever it is, are you giving those numbers as well or is it just NS basically?

KB **Kristina Bauer** 19:57

So it's no, it's N well cause.

We used to have, right? Last year they made everyone update their rubrics.

So I I would just list the categories of our rubrics and put the number of students that fell in each of those buckets.

But it was a simple like I would just the bullet list. Here's the number next to the the bullet. I could send everyone an example of like one of our full reports and one of our like this is not a full report if that's helpful.

AM Arlen Moller 20:33

Yeah, that'd be helpful.

NL Nicole Legate 20:36

Yeah, I think so. It's just helpful to hear like what our different options are. I think knowing, you know, what is acceptable to them and I definitely am in favor. I had no idea either. I would definitely be in favor of.

Combining across years for, you know, more comprehensive versus less comprehensive report. Or if you want to do it like we assess learning objective one every three years, then the next, you know the second year we do 2 and the third year.

AM Arlen Moller 20:55

Mm-hmm.

NL Nicole Legate 21:12

Well, however you think. Hmm.

AM Arlen Moller 21:14

I think because sometimes artifacts from the same course may be used for more than one learning objective, I'd rather.

Do the full analysis for for multiple learning objectives.

Every few years versus do like a deep dive just for learning objective one, then the next year deep dive for learning objective two. I think that might be more burdensome, but.

But I could, I could see advantages of of staggering it that way too. That that was presented as a option in the workshop that you could have different schedules of frequency for different learning objectives like.

You know, depending on how frequently the artifacts are available.

Based on these numbers of, you know, 21 majors to 15 majors, how many years?

Would we want to be aggregating between?

Dee dive analysis.

I was thinking every two or three years, but.

NL Nicole Legate 22:45

I was thinking three to four, so maybe three might be appropriate.

AM Arlen Moller 22:53

Yeah, that would be like 45 to 60.

Per time point.

Any other any anyone else wanna chime in on frequency?

RK Robert Keating 23:14

I think 3 sounds good.

AM Arlen Moller 23:17

OK.

RK Robert Keating 23:17

So we so in the years leading up to the actual year of the program assessment, we would just basically be compiling in those years and then and then just submitting the assessment report the third year. So we're kind of do the same thing artifact wise and.

AM Arlen Moller 23:31

Yeah.

RK Robert Keating 23:35

And stuff. It's just submitting it a full report every three years.

AM Arlen Moller 23:39

Yeah, we would have these shared folders and you know, the director would maybe prompt the instructor who taught the course with the artifact to put their.

A data file and maybe copies of the papers into that folder, but then they wouldn't get fully analyzed until we were due for a aggregate look.

RK Robert Keating 24:01

OK.

AM Arlen Moller 24:09

OK.

RK Robert Keating 24:29

Be right back. Just have to grab my charger.

AM Arlen Moller 24:31

Yeah, no problem.

OK, all right, so let's take a look at the report for Psych Science before we go into. Behavioral health and Wellness. If I change the tabs, are you guys able to see now a new document program assessment report?

OK, so this uh.

This document I haven't combined. Um.

Rob's report data with mine from 3/12. So first we're going to look at learning objective one with data from Psych 204, OK.

So this is.

This business.

So learning objective one applied basic research methods in psych, including research, design, data analysis, interpretation.

There's some text describing. Um.

The final paper assignment for 204 and then we've copied and pasted the scoring rubric that Rob gives to students.

And there were not all but these sections of the paper, 60 points out of 100 that were most related to learning.

Objective one.

And.

Percentages scores out of 100% translate to five different.

Levels of proficiency.

So the most frequent level of proficiency is achieved advanced competency, so 14 out of 18.

Um.

For psych majors got that highest level, one got exceeding basic competency, one got achieving basic competency, 0 got developing basic competency and two.

Out of a team, we're below expectations.

So I don't know, Rob, do you want to say anything more about we might, we might comment on those two that.

We're below expectations.

Maybe it's already in here, but I.

Oh, one of those two, I guess, didn't turn in the final paper, and so that's why they're getting classified. You're muted, Rob.

RK **Robert Keating** 27:46

Sorry about that. Yeah, I wasn't sure what to do with that because they they technically were still in the class and but they just didn't submit a paper. So there was really nothing to evaluate, but I I don't know if we count that.

AM **Arlen Moller** 27:46

Yeah, no problem.

RK **Robert Keating** 28:01

In the in the evaluation. So they didn't, they were below expectations, but because they just didn't meet any any bit of the expectation. So it's not like they had work to evaluate though if that makes sense.

AM **Arlen Moller** 28:04

Mhm.

Yeah, maybe I'll ask Nick Menhart about how to handle that. The other thing I just noticed that I want to maybe get some clarification on.

We've got an N of 21 imputed here, but then in other places we're we're doing percentages out of 18 and.

We've got total enrollment 18 here.

RK **Robert Keating** 28:44

Yeah, that was probably an oversight by my part.

'Cause I actually saw a 17 there too.

AM **Arlen Moller** 28:52

OK, yeah, so.

RK Robert Keating 28:52

And that might have been that was before I added the person who didn't submit a paper because I wasn't sure to do that, sure what to do there. So that they should all be 18. Sorry about that. I didn't even notice that N equals 21 in the in that column of the table there.

AM Arlen Moller 29:04

OK.

Um, and uh, what was the total enrollment for the class? It wasn't probably wasn't 18.

RK Robert Keating 29:14

No, it was. It was like it was 20. I think two people, one person for sure, did a late withdraw. Actually, I can look real quick.

AM Arlen Moller 29:29

OK, and there were there weren't more like non psych majors who like didn't withdraw but but there's weren't psych majors.

RK Robert Keating 29:37

Yeah, there were. There were none.

There was two. I think there was one. Computer science one.

One something else. One second. I'll pull up the.

AM Arlen Moller 29:51

Yeah, sure.

RK Robert Keating 29:51

A great sheet.

See.

Oh, no. Yeah, I I have 18 total.

AM Arlen Moller 30:19

OK.

RK Robert Keating 30:20

Yeah.

AM Arlen Moller 30:21

And and so all of them were seg majors then.

RK Robert Keating 30:23

It was it was 19 originally and then document to late, late withdrew or or something. I don't know what happened. They were there for most of the semester and then all of a sudden I I never heard from them again.

AM Arlen Moller 30:36

OK, yeah.

RK Robert Keating 30:37

So I'm assuming they they managed some kind of late withdraw. So yeah, I.

AM Arlen Moller 30:40

I think they, yeah.

I think if that's the one student, especially since they don't show up on the list of students from the evaluation office. So Nick Menhard sent us.

RK Robert Keating 30:43

OK.

AM Arlen Moller 30:59

Um.

The course rosters with majors for each student and I think.

Is it Nathan Demento that he he he was removed? Um.

RK Robert Keating 31:11

Yeah, that. Yeah, exactly. Yeah.

KB Kristina Bauer 31:21
He's in psych 409 this term, I think. Is he not a psych major?

AM Arlen Moller 31:22
Just want to do it.
Oh, is he? I don't know. I.

RK Robert Keating 31:28
I think he is.

KB Kristina Bauer 31:28
Yeah, I mean, his attendance is terrible, but um.

AM Arlen Moller 31:31
Right. Yeah. I'm not sure if he's still a site major. We're going to have to.

RK Robert Keating 31:32
Yeah.

AM Arlen Moller 31:38
Look for that. He could be, and he was just given a late withdrawal. I think he had some pretty serious.

KB Kristina Bauer 31:39
Oh, wait.

AM Arlen Moller 31:48
Sickness that semester like he might have had something that was like highly contagious and they like made a university wide announcement. It was like a. I forget what the condition is. It's the symptoms are similar to like an extreme flu, but it's like highly it was like highly contagious.

RK Robert Keating 32:13

He's I got him here as a business administration major. I just went back to the to class list and just looked at the the roster and he.

KB **Kristina Bauer** 32:13
Oh.

AM **Arlen Moller** 32:19
OK.

KB **Kristina Bauer** 32:21
Oops.
Sorry, go ahead, Rob.

RK **Robert Keating** 32:25
Oh no, he and he's just he's listed here as Business Administration.

AM **Arlen Moller** 32:29
OK.

KB **Kristina Bauer** 32:29
His self report this term in my getting to know you Excel file is business administration and psychology.

RK **Robert Keating** 32:36
Oh, OK.

AM **Arlen Moller** 32:37
OK.

KB **Kristina Bauer** 32:41
So.

RK **Robert Keating** 32:42
Oh, there was. We did have one other.

Another business administration manager, Daniel Lavev. So, but yeah, there's 19 total students listed here too, so that matches that matches my record.

AM **Arlen Moller** 32:54
Oh.

KB **Kristina Bauer** 32:54
OK.

AM **Arlen Moller** 32:57
OK, so total enrollment 19. I'm just cleaning this up a little bit.

RK **Robert Keating** 33:03
But psych students is 18. Yeah, Yep.

AM **Arlen Moller** 33:03
18.
Got it. OK. I I think we, I think it's cleaned up now.

RK **Robert Keating** 33:09
Sorry about that.
Hey, there was one other thing I wanted to mention about this, if it's OK. So we're using the methods, results and discussion section of the paper to evaluate this.

AM **Arlen Moller** 33:12
Um.
Yeah.
Mhm.

RK **Robert Keating** 33:23
And I I I kind of been thinking about the the introduction section I think should be included too for just based on looking at the objective, especially for this project which is a replication project and the method and results are really they're just following a road map.

AM Arlen Moller 33:30

Yeah.

Yeah.

RK Robert Keating 33:40

To do those like they're just basically, you know, essentially copy pasting what the materials are and stuff like that and you know, writing it in their own words and stuff and then the results, you know, they they have an example to go off of and they're just inputting the results and have you know, they're basically if you look at their papers, the results are almost identical for a lot of them.

It's the intro and the discussion where they really get to show like what they know about the methodology and how like how they describe the methodology in the front end of the paper and the intro and then how they evaluate it in the discussion. So I would argue that the intro and discussion are probably map on more to the learning objective than the.

AM Arlen Moller 33:59

Yeah.

E.

RK Robert Keating 34:16

The method ends result section, which is just writing, you know what they're already seeing somewhere else, if that makes sense.

AM Arlen Moller 34:18

Oh, wow. OK.

Well, so I don't know that we need to redo the numbers for this report, but I think we could.

Make it clear that instead of those 60 points that we use all 100 points for.

The next report.

Unless folks think think it should be changed for this year, what what if others think about?

NL Nicole Legate 34:57

That sounds like a good plan to me, and that could be one of the continuous improvement things to like make a more robust assessment of L1LO1.

RK Robert Keating 35:05
Yeah.

AM Arlen Moller 35:06
Mhm.
L01, yeah.
Um.

Is there? Do you think there's any value in reporting?
Separate tables or histograms for different sections? Or are they all sort of interrelated and?

RK Robert Keating 35:32
Did different sections of the paper.

AM Arlen Moller 35:32
We would do it out of.
Yeah.

KB Kristina Bauer 35:38
Are the histograms even that helpful?

AM Arlen Moller 35:43
Not to me, but they ask for them.

KB Kristina Bauer 35:46
I think also it's supposed to be flipped. The the bars are supposed to not this. You count the number of students by proficiency level.

AM Arlen Moller 35:56
Uh, I see. Um.

RK Robert Keating 36:00

That's the the there's a graph that does that below. I I kind of just went off of what was there.

AM **Arlen Moller** 36:00

So.

Oh.

KB **Kristina Bauer** 36:03

OK.

AM **Arlen Moller** 36:06

Gotcha.

RK **Robert Keating** 36:08

And just mimic the graphs that were there the year before. Is that what you're talking about, Christina, that one below there?

KB **Kristina Bauer** 36:13

Yeah, the yes, the second one.

AM **Arlen Moller** 36:14

This is better. Yeah, this conveys more information to me. Should we leave this one in or?

RK **Robert Keating** 36:15

Yeah.

Yeah.

KB **Kristina Bauer** 36:23

I I don't think you it's not. I don't think it's terribly informative.

AM **Arlen Moller** 36:24

It's redundant.

Alright.

We have a copy of this, so I'm going to.

Have it just like this.

All right, this is the very detailed scoring rubric that students got.

And then this is um.

Providing text for the different levels of proficiency.

OK, so that's what we have for learning objective one. Any comments or thoughts for?

Changing or improving this one?

OK.

So I'm gonna flip to the artifact from Psych 312 that's for.

Learning objectives two and three.

So it starts the same, just restating the two learning objectives.

Um.

Artifact is a final paper for Psych 312.

There's some justification about what the assignment involved.

And then um.

I had in last year's report done a subset like Rob did this year and I sort of came to the same conclusion that actually.

All of the sections were relevant, so I ended up.

Doing, let's see, scores out of 100 mapping onto these five levels of proficiency.

So then there's the assignment instructions. This is long, so I'm going to scroll through this because they had several three different options for the focus of this paper.

They could apply it to their motivation to learn to video game design or climate change risk mitigation, but they're like using theories from motivation theories to apply to these kinds of challenges.

All right, so I'm going to scroll through this. It's kind of lengthy scoring.

Uh.

This is a different.

Table mapping levels of proficiency on to like specific.

Uh.

Components that were scored in the paper.

Um, and these were the.

Data from from that course. So so about half got achieving advanced competency, 33% got exceeding basic competency and 20% got achieving basic competency.

I haven't done a histogram yet, but that that's that would be pretty easy to add.

None were at developing basic competency or below.

Um, so let me pause.

For this one, for this artifact and these learning objectives, questions or things to add.
Other than a histogram.

NL **Nicole Legate** 41:07

No, looks good.

AM **Arlen Moller** 41:14

All right, so then the last thing, basically I'm going to combine the thing we just looked at with this document and then I have PDFs with de-identified.

Artifacts. So we have the papers from 204, the papers from 312 that student numbers, not their I I T student numbers, but like just codes.

RK **Robert Keating** 41:31

Yes.

AM **Arlen Moller** 41:46

In a table map onto levels of proficiency, and if someone wanted to go back and regrade those papers, they could, you know, start to like check our work. Essentially, that's one of the things they're asking for.

So for continuous improvement plans.

I don't have much right now. One of the things we talked about was.

Um.

Let's see. Oh, this was these were two things from last year.

So a number of courses implemented using pack back deep dives. I listed 310, at least some sections of 310 and 312. Are any other courses that you guys are aware of using?

The Packback Deep Dives tool.

LS **Lindsay Sheehan** 42:49

What is that exactly?

AM **Arlen Moller** 42:51

So Packback is this platform that is like more of a dynamic. It started as like a course

message board system, but there's like a chat bot that.

Nudges the students to go deeper, to cite a primary source, and as they're typing, they're getting like a continuous.

I think they call it like a curiosity score or something that they're getting like sort of instant feedback on what they're writing and they have a newer tool that's like includes this A I.

Writing coach for longer papers and some of what it's doing is nudging them to use APA style formatting to like have transition sentences between paragraphs.

So there's like some of the.

The especially like sort of communication skill feedback that we might be writing on papers is getting automated and getting to the students while they're still working.

And so I I've had some success with the papers that I get back are in a more polished form.

Using the tool.

And it's sort of a compromise between letting students use LLMS unsupervised for writing help, where there's a bigger risk maybe that the LLM is generating substantive.

Content and not just like writing tis.

KB **Kristina Bauer** 44:49

Oh, I use questions, not deep dives. I don't have anything that I felt would fit deep dives yet, so I just didn't do it. But I we could just ask Michelle.

AM **Arlen Moller** 44:55

Yeah, mhm.

KB **Kristina Bauer** 45:02

If we want the specific number of site courses that have implemented deep dive, Michelle is the one of our reps. I'm often in meetings with two people and I can't remember his name, but he's funny.

AM **Arlen Moller** 45:07

OK.

Yeah.

KB **Kristina Bauer** 45:18
Uh, the other guy.

AM **Arlen Moller** 45:19
Oh yeah, the guy from Buffalo, John. Yeah.

NL **Nicole Legate** 45:20
John.

KB **Kristina Bauer** 45:21
OK. Thanks, Nikki.

NL **Nicole Legate** 45:22
Mm-hmm.
umm I think I just got an e-mail today saying that I have a new pack back rep. So they're they're always there's a lot of turnover there, I think.

AM **Arlen Moller** 45:31
Hmm.
OK.

KB **Kristina Bauer** 45:35
Yeah, it didn't used to be that way when they first started and now it the churn is is increased.

AM **Arlen Moller** 45:42
I noticed that, yeah.

KB **Kristina Bauer** 45:44
Yeah.

AM **Arlen Moller** 45:45
OK. Yeah. So maybe I can, you know, go ahead.

RK Robert Keating 45:46

Ireland.

Sorry, I I don't know if it's worth it to mention, but I'm I'm I started using Yellowdig in my 301 class this semester.

AM Arlen Moller 45:55

Yeah.

RK Robert Keating 45:57

It's good. I like it a lot. It it doesn't have, you know, the, you know, the assistance I guess that like deep dives has with like nudging citations and things like that. It really operates like a social media platform.

KB Kristina Bauer 45:57

How's it going?

AM Arlen Moller 46:10

Mhm.

RK Robert Keating 46:14

So the the way I have it set up for now, it's kind of like a trial run is it's just very it's a little bit loose with with what I don't, I don't, I don't provide prompts and say hey respond to this prompt. It's just each week they have a certain amount of posting, commenting.

You know, assigning like accolades and stuff to other people's posts that they have to do that's I/O relevant, relevant to whatever topic we're we're, you know, covering that week. So it takes, it takes more monitoring by by me and and mainly Rima.

That's kind of like Rima's main job. She's my TA.

AM Arlen Moller 46:42

Mhm.

RK Robert Keating 46:50

That's one of the main things she does is kind of she's like the the main overseer of

that, make sure people are posting and and lets me know if you know people aren't or if there's anything going on. But I I like it a lot. The posts are good. The the students are just kind of like they're they're out there finding articles and videos that are I/O relevant.

AM **Arlen Moller** 46:50
Yep.

RK **Robert Keating** 47:10
Relevant and somebody found like a blog that a student wrote a a masters IO student from. I can't even remember where about.
You know what it's like to be a an IO student and stuff like that. So they're they're out there finding stuff and it seems they seem to be using it really well. So, so yeah, so far I like it. It's just kind of a community building platform more or less so.

 47:23
E.

AM **Arlen Moller** 47:25
OK.
Yeah, it does sound like there's some overlap with packed back questions. That's the that's the sort of more discussion board style thing and and you can do the, you know like the equivalent of likes or.

RK **Robert Keating** 47:37
Yeah.
Mhm.

AM **Arlen Moller** 47:49
You know, accolades, I think you you said different ways of sort of engaging with other people's posts in a way that might be lower barrier to entry. So the the yellow dig is being used in 301 you said.

 48:00
Yeah.

RK Robert Keating 48:05
Yeah, section.

AM Arlen Moller 48:07
And what's the?

RK Robert Keating 48:09
Two. Am I Section 2? I think I'm Section 2, yeah.

AM Arlen Moller 48:10
Faster.
OK.
And what's the cost to students for Yellowdig?

RK Robert Keating 48:17
It's \$16.00 per student.

AM Arlen Moller 48:19
OK.
Got it.
All right, and I'm gonna mention I I'm not able to edit this document for whatever reason right now, so I'm writing stuff down, but I'm gonna mention a change in. Um LO1.

RK Robert Keating 48:40
Yes.

AM Arlen Moller 48:42
Considering all 100 points from the 204 paper instead of just 60 points.
OK.
OK.
Um, oh, so here are the two things. I don't know if we want to include this in this. Report could be things that we do internally and are not, you know, part of this, but in particular for the learning objective three with communication.

It seems like the changing landscape around access to LLM and students using them frequently is something we should be responsive to. I had some meetings with. The staff from this new communication lab. It's like the the rebooted writing center about trying to.

You know, be more actively responding to this and I have a slide deck that I used before introducing the long paper assignment in Psych 312 this semester.

Where I talk about sort of like strengths and weaknesses of these tools and ethics. O.

I guess one one question is if folks are interested in that, I'd be interested in getting some feedback from the committee like sharing that slide deck and maybe being more collaborative about.

You know what? What we're telling students in psych courses who are doing paper long papers.

Um.

But a question is it does does this, does that belong in the continuous improvement plan?

NL

Nicole Legate 51:06

I would love to see it and collaborate on that. I wonder if you do put it in, it could be more preliminary stages, you know, just saying like beginning to collaborate.

AM

Arlen Moller 51:21

Mhm.

NL

Nicole Legate 51:25

On teaching strategies to address AI use or just to kind of don't over promise, but I think you could mention it.

AM

Arlen Moller 51:30

So don't don't over, don't over promise.

KB

Kristina Bauer 51:37

I agree and I like that you already highlighted the the idea that this is the next like this is a long term plan, right? I'd I'd go with like 3-4 years, maybe two I don't.

AM **Arlen Moller** 51:46

Mhm.

Mhm.

KB **Kristina Bauer** 51:50

There's a lot of moving pieces right now. I feel like two years will be here very quickly.

AM **Arlen Moller** 51:53

Yeah.

Yeah.

KB **Kristina Bauer** 51:56

I think there's a lot of variability in the way our students do or don't use LLM's.

AM **Arlen Moller** 52:03

Yeah.

KB **Kristina Bauer** 52:05

When they were, I had a one of the teams discussing their construct development.

They were really having an excellent conversation about like a definition of something and they were going back and forth and they had like 2 days to finish this debate and write all their items.

So for the that were due for for class, they were they were having a discussion on Tuesday, Thursday with the items were due. I was like guys you have a time crunch here. It might be worth like pro conning with ChatGPT. It could help you make a decision.

And they looked at me like I had three heads.

AM **Arlen Moller** 52:36

Mhm.

Oh, wow.

KB **Kristina Bauer** 52:40

Yeah.

AM Arlen Moller 52:41

Hmm.

KB Kristina Bauer 52:41

Like, all right. I mean, I wouldn't suggest this if you had all the time in the world, but your items are due in two days.

AM Arlen Moller 52:48

Yeah, yeah, you.

Yeah, it's interesting. Yeah, I'm not sure what contributes to that, but it does seem like different students have more familiarity with these tools and.

KB Kristina Bauer 53:03

There seemed like A and I I had one of our Julia Honda said the same thing to me in a conversation we had we were having a research meeting. It seemed to be this like this like aversion to ChatGPT because it like they could do it and that's like cheating.

AM Arlen Moller 53:22

Mm-hmm.

KB Kristina Bauer 53:23

And the the knowledge that it still hallucinates is like I'm I'm I'm better than this. I I can do it myself.

AM Arlen Moller 53:33

Yeah.

Yeah, um.

NL Nicole Legate 53:38

I I'm so sorry. I have another thing at three. I gotta run. Um.

AM Arlen Moller 53:41

Yeah, I'm, I'm sorry too. I I I think we're gonna put off having a report for behavioral health and Wellness because there's so little data. There's was one student in 204

zero students in.

312 So that's not a there's very little to go on, but I do think we need to have a a conversation or next meeting about like the future of behavioral health and Wellness.

OK. Yep. Bye, Nikki. Yeah. Yeah, I see that we're at time. I'm just gonna flip to the agenda just to see if there's anything else.

NL **Nicole Legate** 54:11

Sounds good. OK, bye guys. Sorry, gotta run.

KB **Kristina Bauer** 54:13

Bye, Nikki.

AM **Arlen Moller** 54:26

Um, I don't think there's anything.

Um.

That is time sensitive in terms of submitting the report on Wednesday, but I do think there's a longer conversation that I'd like us to have about changes to.

The program may be looking at courses that we've historically offered that we've had trouble covering and whether or not we want to keep offering those. I can also just.

Put a pin in something and tell you I just John and I just got an e-mail from Carly Kasurik saying that they had had warned about this and they're coming back to say we they want us to renumber.

Our courses based on, you know, doing specific things or different levels of challenge. So that conversation I think is going to begin in our next meeting, OK. And some of it has to do with aligning us with.

Community colleges and two year schools to make it simpler for folks to transfer in that for me, I'm sympathetic to that as a rationale, so.

OK, so we're at time. Anything. I can stick around for a few minutes. Is there anything pressing that folks want to?

Add to the conversation.

KB **Kristina Bauer** 56:03

When's our next meeting? I'm sorry, I could just scroll up in the agenda. I realize that now.

AM Arlen Moller 56:09

No, it's OK. And um.

KB Kristina Bauer 56:11

October 21.

AM Arlen Moller 56:14

Yeah, we may have one sooner on the calendar. I think there is one coming up on the calendar sooner that's not in this doc. Let me look at my calendar.

Can anyone else do a search of their calendar?

LS Lindsay Sheehan 56:34

Sorry, I need to go to another meeting. I'll see you guys.

AM Arlen Moller 56:36

OK. That's OK. Thanks. Thanks, Lindsey. Undergrad.

KB Kristina Bauer 56:45

3rd 10s. The next one I see is November 18th.

AM Arlen Moller 56:50

OK, I've got October 3rd, October 10th. We did those and I got one. Yeah, so this is not right. November 18th is the next one in the calendar. So I'm glad you brought this up because I got a.

Go into calendar and at least add.

October 21st and November 4th.

Right.

RK Robert Keating 57:18

I have October 21st in my calendar, yeah.

KB Kristina Bauer 57:19

Yes.

AM Arlen Moller 57:20

You do. Do you, Christina?

KB Kristina Bauer 57:25

Uh, OK, hold on, let me come back.

My computer's being dumb. October 21. Oh yeah, I just didn't find it when I searched for it. I don't. I don't know. I do have 1021.

AM Arlen Moller 57:39

OK, yeah, same same for me. I'm searching right now and it jumps like you to the 18th, but I just put in undergrad. OK, so it looks like our next one back to Tuesday, one to two, October 21st.

RK Robert Keating 57:46

Just.

KB Kristina Bauer 57:59

Yes.

RK Robert Keating 57:59

Cool.

AM Arlen Moller 58:00

OK.

All right guys, so I will bundle this into one big PDF and circulate it before uploading, but if you think of anything else that you think should be in there, let me know. Any other parting thoughts? Yeah.

KB Kristina Bauer 58:16

Well, I have two quick schedule things for you. One is there's it's like a I'm confused. Frank's buying out of a course, but he's supposed to teach psychopathology and John is telling me it's 303.

AM **Arlen Moller** 58:21

Yeah, sure.

KB **Kristina Bauer** 58:35

But Kelly didn't put him in the schedule for anything. I think Frank Lane will be teaching the other section of Psych 303, which means we need one fewer adjunct than I thought. And then two is I told students that today was the deadline to apply for adjunct positions because.

AM **Arlen Moller** 58:47

OK.

KB **Kristina Bauer** 58:56

I was Roxanne and I had an e-mail exchange. She said do you have the list of adjuncts? I said no, it's too early. Why? Why? Why? This feels early for that kind of an ask. Short version is that the Dean and Tara are already requesting and.

AM **Arlen Moller** 59:03

Too early? Yeah, yeah.

KB **Kristina Bauer** 59:16

Roxanne thinks if we have a specific person already that we'll be better positioned to justify the adjunct.

AM **Arlen Moller** 59:24

OK. I mean, if you have the bandwidth to to manage that, I guess it's work that's going to have to happen now or later.

Do you want let me know? Yeah, if you wanted me to help help review the applications, I'm happy to take a look.

KB **Kristina Bauer** 59:43

Well, yeah, you. I mean, you ultimately, but I get them and then we, I don't make the decision, right? Like Nikki and I would sit down and we don't have to sit down.

Anyway, if there was a like a clear winner, I would make a recommendation. But sometimes it comes down to like two people who are equal and then.

AM **Arlen Moller** 59:51

OK. Yeah.

Sure.

Right, right. OK. So yeah, I I'm happy when whenever you're ready to share applicant materials and you want to start that conversation, I'm, I'm happy to jump in on that.

KB **Kristina Bauer** 1:00:03

And.

Yeah.

OK, I bandwidth. It's probably not today.

AM **Arlen Moller** 1:00:20

Right.

KB **Kristina Bauer** 1:00:21

Yeah, and then hopefully earlier ish next week, not including Monday.

AM **Arlen Moller** 1:00:28

OK, yeah, it's Friday, so and it's a holiday holiday weekend. So hope you guys have a good weekend and Indigenous Peoples Day on Monday.

KB **Kristina Bauer** 1:00:30

Yeah.

Oh.

It's my birthday. I'm gonna have a great day.

Yeah.

OK. Thanks. Bye.

RK **Robert Keating** 1:00:51

Happy birthday.

KB **Kristina Bauer** 1:00:52
Thanks. Bye.

AM **Arlen Moller** 1:00:53
Bye.

KB **Kristina Bauer** 1:00:54
Thank you.

□ **Arlen Moller** stopped transcription

Awe & Humility: A Replication Study

Student 1

Department of Psychology, Illinois Institute of Technology

PSYC 204: Research Methods in Behavioral Science

Dr. Robert Keating

May 9, 2025

Abstract

Background

The present study aims to replicate the findings of Stellar et. al. (2018). Humility is known for encouraging prosocial behavior, which can contribute to stronger, meaningful relationships, and overall well-being. Awe, as an experience which increases wonder and decreases feelings of self importance, was hypothesized to trigger humility. In operationalizing awe, the original study used a video of the universe to induce awe, and a video of building a fence as a control condition. They asked participants to list their strengths and weaknesses, and took this proportion to assess humility. They found that participants who were in the awe condition listed more weaknesses to their strengths than those who viewed the control video, indicating that experiencing awe influenced their humility.

Methods

Participants were recruited via IIT's SONA system. Once completing a consent form, participants were randomly assigned to either the awe condition or a neutral condition. After viewing the assigned video, participants were asked to list as many strengths and weaknesses as they could in 2 minutes.

Results

No significant results were found. However, there were some limitations to the study which could have contributed to null findings.

Discussion

By conducting this replication, we are aiming to contribute to the growing body of literature regarding the relationship between awe and humility, and what can encourage increased humility.

We also hope to highlight areas for future research, and provide ways to improve future replication studies.

Awe & Humility: A Replication Study

Often associated with epiphanies and mystical experiences, awe results from the feeling of being small, and pushes personal identity out of the forefront of awareness. Awe can be experienced when bearing witness to the vastness of the universe, the glory of nature, or even the wonders of religious artifacts (Bai et. al., 2017). Losing this feeling of personal importance has been previously shown to boost prosocial behavior and humility (Kahn & Cargile, 2021). Humility on its own is able to enhance wellbeing, and even encourage trust, skill building, and reduce psychological distress (Manix & Abston, 2025). By better understanding the connection between these two concepts, encouraging awe could help foster humility, and the benefits that arise from being in a humble state. The overall goal of the present study is to further examine the relationship between awe and humility.

Literature Review

Awe

Awe is a “self-transcendent emotion” which has the capability to inspire prosocial behaviors (Kahn & Carhile, 2021). Awe often leads to a weakening of normally concrete beliefs, and reduces feelings of self importance. It is also able to inspire feelings of “oneness”, which is the idea that all humans are connected, regardless of individuality. This is beneficial to inspiring empathy, care, and compassion towards others (Stancato & Keltner, 2021). The small-self is theorized to be created through experiencing awe. This small-self encourages humility, among other humble behaviors and attitudes in people with strong moral obligations. Reducing feelings of self-importance can help create deeper connections with others, and provide more motivation to work towards the goals of others (Bai et. al., 2017).

Humility

Strong humility has been often linked to good psychological wellbeing, strengthened relationships, and easier recovery periods (Stellar et. al., 2018). Having a strong sense of security in the self allows for an open acceptance of others, and willingness to change (Stellar et. al., 2018). Humility is also seen as a critical personal resource, as it counters selfish and arrogant behaviors which inhibit the development of strong relationships, outward goals, and even the ability to learn new skills (Manix & Abston, 2025). The fostering of these desirable, positive effects of humility is being continuously investigated.

The relationship between awe and humility was previously studied by Stellar et. al. (2018). In this instance, there was a significant observed effect of an awe inspiring stimulus leading to a greater report of humility than when presented with a neutral stimulus. This effect is important for many purposes, and exploring the direct effect of awe on humility holds great promise for future research (Stellar et. al., 2018). A better understanding of the basis of humility could lead to the development of humility-inducing exercises, which could greatly improve the lives of those who may not experience naturally high levels of humility.

Study Overview

The present study aims to replicate the findings of Stellar et. al. (2018). Using a between-subjects design in which participants will be randomly assigned to view one of two conditions: either a video regarding the scale of the universe intended to instill feelings of awe, or a video regarding the construction of a fence intended to instill a neutral emotional response. We hypothesize that participants randomly assigned to the awe condition will report a higher level of humility as indicated by a greater number of weaknesses relative to strengths listed. The

null hypothesis states that there is no relationship between the number of reported traits, and the video stimulus that was presented.

Methods

Design

Data collection was conducted through a between-subjects experiment using an online survey. The independent variable in this study was the video condition the participant was assigned to (awe inducing or neutral). The dependent variable was the amount of strengths and weaknesses self-reported by the participants.

Participants

Ninety current IIT students were recruited via IIT's SONA system, and were compensated up to 0.5 extra credit points for their participation. However, 16 participants were excluded for incomplete responses, 1 participant was excluded for knowing the hypothesis, 7 participants were removed for not viewing the video, and 1 participant was removed for incoherent responses. In total, 65 responses were used in the final analysis.

Materials

Awe

Participants randomly assigned to the “awe” condition were presented with a video regarding the vastness of the universe (http://www.youtube.com/watch?v=ZoXqj-VDhws&feature=player_embedded). Participants randomly assigned to the “neutral” condition were presented with a video describing how to build a fence (<https://www.youtube.com/watch?v=XYGzuPpw50g>).

Humility

After watching the assigned video, participants were first asked to list as many strengths they could think of, followed by any weaknesses in 2 minutes. The proportion of strengths to

weaknesses was used as the scoring for this item. Higher scores indicated a greater proportion of weakness to strengths, implying a greater sense of humility.

Participants were also asked if the video played all the way through, and to describe the content of the video, as quality checks. They were then asked to report what they thought the study was about. The last portion of the survey presented several questions regarding demographic information about their age, race, and gender.

Procedures

Participants were recruited via IIT's SONA system, which showed them a brief description of the study, and a link to the study. If they chose to voluntarily participate, they were led to the online survey to complete on their own. A consent form was presented, and they were then taken to the next stage, or to the end of the study if they consented or not, respectively. Participants were then randomly assigned to view either the awe condition video or the neutral condition video. After viewing the video, they were then asked to list their strengths and weaknesses. Then, they were assessed on whether or not the video actually played, confirmed by their description of the video content. Afterwards, participants were asked what they thought the study might be about, before being presented with the demographic questions. They were then presented with a debriefing message explaining the full purpose of the study and thanked for their involvement.

Results

An independent sample t-test was conducted to test the relationship between the proportion of reported strengths and weaknesses (DV), and the video condition (IV). Scores on the dependent variable, humility, was calculated as the number of strengths / the total number of listed traits. The independent variable was coded as a 1 or 2, for the control and experimental conditions, respectively. The descriptive statistics are reported in Table 1, and results are displayed in Figure 1. The relationship between the control condition and the DV was positive and significant ($M = 0.237$, $SD = 0.200$). The relationship between the experimental condition and the DV was positive and significant ($M = 0.282$, $SD = 0.205$). The model revealed an insignificant effect, $t(63) = -0.891$, $p = 0.376$, $d = -0.222$. This indicates that there was no relationship between listed traits as an indicator of humility, and the video condition participants were assigned to.

Discussion

We hypothesized individuals who were exposed to the awe inducing condition would list more weaknesses in comparison to strengths, indicating that they were experiencing increased humility. However, the results of the independent sample t-test failed to reject the null hypothesis. As the best-fit analysis for the given study, it can be assumed these results are providing an accurate portrayal of any effects that would have occurred in the collected data. The theory that experiencing awe increases humility is generally supported by the literature. While this study failed to replicate the findings of Stellar et. al (2018), there were some flaws in the study which could have contributed to the null findings.

Although participants seemed to have positive reactions to the awe-stimulating video, it is difficult to say to what extent they actually experienced awe. There could also be a certain threshold of awe that must be reached in order to trigger stronger humility. The control video could also be serving as a confound, if it reduces interest from a baseline level. It could also be beneficial to assess how other variables, like baseline personality traits, are impacting the effect. For future investigation, it is important to consider an improved way of stimulating awe, and ensuring the control does not arouse an opposite effect.

In addition, as the measure of humility was not standardized, it is difficult to draw conclusions on how humility was genuinely affected. There were also several participants who did not specify which traits were strengths and which were weaknesses, or specified that they were only going to list one of each, indicating that they did not read or understand the instructions. As the survey was conducted online, it is difficult to control for external factors like instruction confusion or other distractions.

Because this study was conducted using a sample of undergraduate college students, it is difficult to say whether the observed null effect was due to the sample, and whether or not the effect could be generalized to a greater population. In addition, because of the limited number of participants, it is difficult to draw definitive conclusions on this effect.

A past replication study, Bertrand et. al. (2024), reported several limitations which contributed to null findings. First, they acknowledged that it is difficult to evoke awe through the use of a singular, short YouTube video. Likewise, the control video may have decreased feelings of awe or wonder in the control group, and was therefore not entirely neutral. Among other concerns, they also noted that as their sample was limited to undergraduate students, the generalizability of their results is in question. However, despite their failure to reject the null hypothesis and recreate the original findings, they note that future study is required to further understanding of the relationship between awe and humility (Bertrand et. al., 2024). While the present study may suffer from similar weaknesses, expanding upon the current literature holds a great importance, as does the search for effects similar to the original study in other populations.

For future research, it would be beneficial to create new operationalized versions of the variables. Finding a method of creating a stronger sense of awe could make any effect that occurs easier to identify. In addition, finding a way of standardizing the measurement of humility will make comparisons between an awe and neutral condition more accurate. In doing so, it is possible that a stronger effect could be observed, as experiences will be strengthened and easier to interpret. It may also be worth considering changing the form the experiment is conducted in to ensure that participants fully understand all instructions. Introducing a measure of baseline humility and related personality traits will also ensure that any observed effect is indeed caused by the awe inducing stimuli, rather than baseline characteristics.

By understanding emotions and traits that cause greater senses of humility, it is possible to develop methods of encouraging humility. The current research suggests many similarities between awe and humility (Stellar, et. al., 2018). As this relationship continues to be explored, more light can be shed on the basis of humility (Bertrand et. al., 2024). Encouraging humility has the potential for a large positive effect on individuals who may not experience naturally high levels of humility. Giving them access to this important aspect of the human experience may allow them the capabilities to develop stronger relationships and create a sense of open-mindedness that comes with a slew of other positive impacts on life (Manix & Abston, 2025).

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Appendix

Table 1.

Descriptive statistics

Independent Samples T-Test

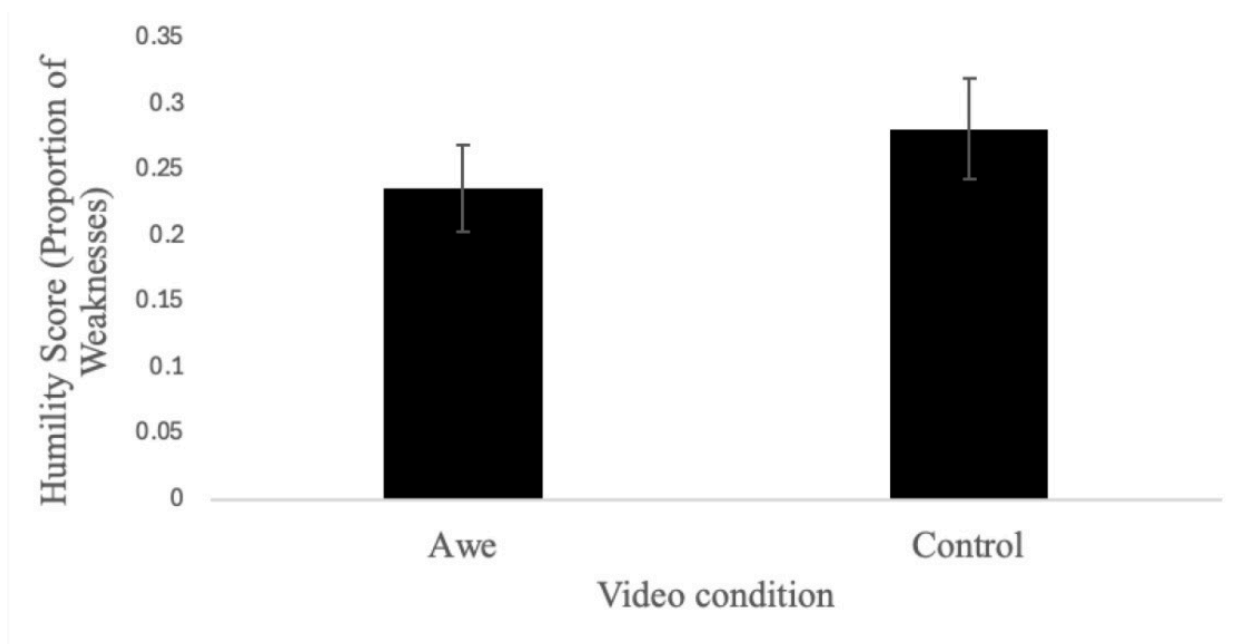
		Statistic	df	p		Effect Size
DV- weakness_ratio	Student's t	-0.891	63.0	0.376	Cohen's d	-0.222

Note. $H_a: \mu_{IV1} \neq \mu_{IV2}$

Group Descriptives

	Group	N	Mean	Median	SD	SE
DV- weakness_ratio	IV1	36	0.237	0.250	0.200	0.0333
	IV2	29	0.282	0.333	0.205	0.0381

Figure 1.

Results

Note. Bars indicate mean humility scores by video condition. Humility scores were calculated by the proportion of weaknesses to strengths (higher scores indicate more weaknesses, and higher reported humility). Error bars indicate the standard errors (0.03 for awe, 0.04 for the control).

A Replication Study of the Effect of Awe on Humility

Student 2

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PSYC 204: Research Methods in Behavioral Science

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March 12, 2025

Abstract

Awe is an emotion that can decrease a sense of self and bring a new perspective to usual thinking patterns. This study aimed to replicate study 3 of Stellar et al. (2018), which describes awe as a transcendent emotion that can catalyze greater humility. Participants (N = 65) were randomly assigned to two different video conditions: an awe-eliciting video or a neutral stimulus video. The behavioral measure of humility was calculated based on the strengths and weaknesses they listed. We hypothesized that in our replication study, watching the awe-eliciting video would have participants list more weaknesses based on the previous findings in the original study. The results showed no significant differences between the awe condition and control condition with humility score. These findings suggest awe and humility may have more complex measures than previously understood and can be affected by many limitations like mood, distractions, and sample characteristics.

Keywords: awe, humility, self-focus, behavioral measure, replication

Introduction

Humility is a realistic psychological view that someone has of themselves. It involves having a balanced but open view of their strengths and weaknesses (Tangney, 2000). In contrast, Awe is an emotional response to a stimulus that the person perceives as powerful and even changes their perceptions towards certain aspects (Stancato & Keltner, 2021). Since awe is able to keep someone grounded and open to new concepts, it is assumed there is a smaller amount of self-centering. This would make humility more apparent, giving people a more balanced perspective, especially about themselves (Bai et al., 2017). Emerging research has looked into the connection between awe and humility and has explored how they may interact with each other.

A study that directly examined the relationship between awe and humility was Stellar et al., (2018). The researchers organized several experiments to analyze if awe led to humbler views within participants. In study 3, this idea was tested by having participants randomly view an awe-eliciting or neutral instructional video, then proceeding to measure behavioral humility. The results found that the awe group listed fewer strengths than the control group. These results were perceived to be due to a higher sense of humility from experiencing awe. The idea that awe seems to lessen the focus someone has on themselves and that they even view their flaws was supported in this case.

Studies build on these findings, like those of Bai et al., (2017), who found that awe promoted a “small self,” acting as a mediator for prosocial behaviors like generosity and ethical decision-making. Supporting the view of awe as not only affecting someone’s own perception of themselves but also how they may view others. New approaches have also been presented in studies to elicit awe. Song et al., (2022) looks at awe and moral considerations where study 4

used Virtual Reality to induce a more prominent sense of awe. They found that there was a greater effect of eliciting awe than the control group and a significant relationship with moral expansion. These findings further the concept of the minimized sense of self where it can direct views from oneself to broader awareness of others and one's environment. Another study that took the approach of eliciting a stronger sense of awe through Virtual reality was Kahn and Cargile (2021) which recognized that virtual reality was effective in inducing awe but it may also impact prosocial behavior. In this study, awe was positively correlated with volunteering which shows a specific example of how it leads to behavior that connects with others. Stellar et al. (2017), focused on transcendent emotions such as compassion, gratitude, and awe to highlight these emotions may have emerged in the human cognition to address issues involved with group cooperation and taking care of others. These studies propose that awe can ground someone's perception of themselves, which can bring them closer to qualities of humility.

The goal of this study is to look into these findings and see if the exposure to an awe eliciting video increasing humility is still in effect when replicated. We followed a similar structure to study three of stellar et al., (2018) where participants were assigned randomly to either the awe eliciting video group or the control group with a neutral video. After watching the video, they were asked to do an open-ended self-description task. The number of strengths and weaknesses will then be coded to see their score on humility and compared for both groups. We hypothesized that participants who view an awe-eliciting video will exhibit more humility compared to participants in the control condition.

Methodology

Participants

A total of 90 participants were recruited for this study. Many of them were undergraduate students enrolled at The Illinois Institute of Technology and received course credit. They were recruited through the university system SONA which also provided where they could directly take the survey. Data from participants who didn't consent or did not complete the writing task were excluded from this study. 25 of the participants' data was removed as they either knew the data, included incomplete responses, did not view the video or provided responses that were not clear. In total, 65 participants remained in the final sample.

Design

A between-subjects experimental design was used for this study to analyze the effect of awe on humility. Random assignment was utilized to put participants in either the control or experimental group. This allowed there to be more control for factors like individual variability. The independent variable has two different video conditions that would be viewed by participants prior to answering the questions in the survey. This variable was manipulated to elicit either a neutral stimulus in the control group, or awe in the experimental group. The dependent variable was the humility scores obtained from the traits listed by participants. A higher ratio of weaknesses to the total amount of traits listed would indicate greater humility.

Materials and Measures

Qualtrics was used by participants to access the survey through any technological device. Both videos used in the study's survey were obtained through youtube. The awe condition involved participants watching an awe eliciting videos of the universe at a larger scale than

humans would be able to see. The control group was shown a video that was meant to elicit a neutral emotional response with instructions on how to build a patio wall.

Once exposed to the videos, participant's will take a self description task where the number of strengths and weaknesses they write will be recorded. The humility score was calculated by dividing the amount of weaknesses by the number of strengths and weaknesses. Awe was measured through how much they perceived emotions watching the video using a likert scale from 1 being not at all, to 7 being very much of the following emotions: awe, fear, wonder, Happy, Amazement. To see if our video condition was delivered effectively to participants, the survey contained a series of questions asking, "What was the video you watched about?", "What did you think the study was about?", "Did the video load properly and play all the way through?".

Procedure

Participants were first asked for their informed consent to proceed with the study. Once given their consent, participants would be randomly assigned through the Qualtrics survey to either the control or experimental video condition. After the participants watched the assigned videos, a writing task was given where people had to list both their strengths and weaknesses. The responses were then analyzed to count how many strengths and weaknesses were listed. The humility score would be calculated and compared for both groups to see the effects of each condition.

Results

To test the hypothesis that the awe condition would have higher rates of humility, we conducted an independent sample t-test to compare the humility scores between the two conditions. Results indicated that participants in the awe condition ($n = 36$) there was a mean

obtained of $M = 0.24$ ($SD = 0.20$) for the humility score. However, the participants in the neutral condition had a mean of $M = 0.28$ ($SD = 0.21$). The difference in these groups was not found to be statistically significant, $t(63) = -0.89$, $p = .376$, $d = -0.22$. There was also a small negative effect size, which showed a negative trend in the opposite direction. Overall, the results obtained did not support our hypothesis that awe increases humility.

Discussion

The study that was conducted focused on replicating Study 3 of Stellar et al., (2018) and looking at the original findings of the increase of awe caused by people reflecting on their awareness of their weaknesses. We hypothesized that those exposed to the awe-eliciting condition would have a greater portion of weaknesses than those in the neutral condition. This did not occur for our study as our hypothesis was not supported. When looking at the control group, participants listed more weaknesses than those in the awe condition. This difference was not statistically significant, and its effect size was small.

There are several factors that could explain the results obtained for this study. A limitation involved could be with construct validity, such that the way of measuring humility by counting weaknesses could not capture the full effects of humility. For internal validity, there could have been uncontrolled factors that could have influenced their responses to the survey and how many weaknesses they listed. For instance, their mood or if they were having a hard day while answering the survey could cause more weaknesses to be listed. Looking at external validity, our sample included mainly college students, which was also the same as in the original study. However, looking at external validity, this is difficult to generalize to a full population because of the use of college students.

Measuring awe and its effects on psychological traits such as humility seems to be broader than initially thought. A recent study by Li et al., (2024) found that awe was also associated with empathy. Specifically looking at cognitive empathy and affective empathy which found that a person's tendency to perceive awe was positively correlated with their self reported levels of empathy. This could explain some reasoning why our study did not show the exact humility findings as in Stellar et al., (2018) when awe was elicited. Participants in the study are assumed to all perceive awe equally the same, but that might not have been the case when viewing the videos..

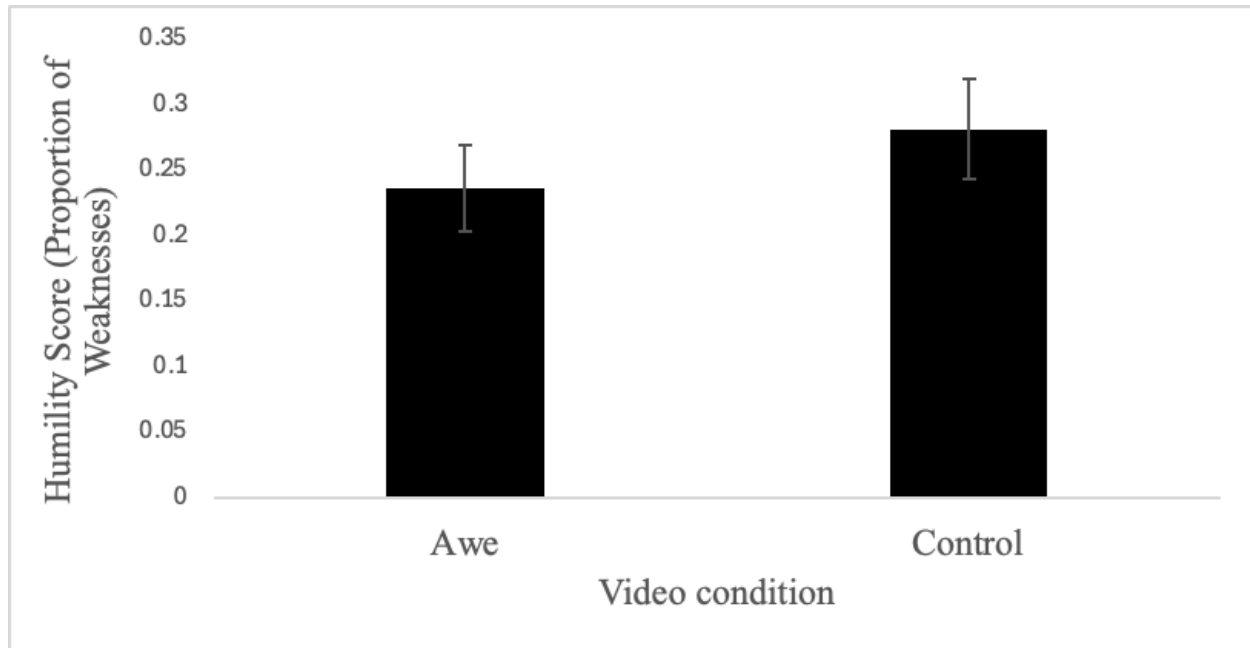
Furthermore, while awe may correlate with humility in past studies, it is also important to analyze and consider other psychological factors that are affected by this emotion, like empathy. Mainly when the emotion is elicited and a measure of humility can't be found. Then, it could be possible that other traits could be activated through awe, which may also need to be measured to see its effect.

Directions for future research could be improving the awe-eliciting experience. Instead of having people take the survey in their own time, there could be an arrangement where participants can come into an actual lab and finish the study there so we can limit other factors. Environmental distractions, like if participants were having a hard day, or emotional factors like the person's mood could also be recorded prior to the study to see if these factors affect responses. Participants could be asked more about their humility through questionnaires to add more details to their view of themselves and even the world around them to widen the analysis of their perceptions. The measure of awe is also something to look into as to whether participants have a similar tendency to perceive awe. A questionnaire similar to the Dispositional Awe Scale (Shiota et al., 2006) can be used to measure an individual's tendency to experience an emotion

like awe. Overall, without supporting the original findings of this study, we can continue to look at the psychological complexity of awe as an emotion and how it reflects the way we view ourselves, others, and our environment.

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Appendix**Figure 1.**

Mean humility scores (proportion of weaknesses listed) by video condition. Error bars represent standard errors (0.0333 for Awe, 0.0381 for Control).

A Replication Study on the Effect of Awe and Humility

Student 3

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PSYC 204: Research Methods in Behavioral Science

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March 12th, 2025

Abstract

AWE & HUMILITY REPLICATION STUDY

This study replicates Stellar et al.'s (2018) research on the relationship between awe and humility. Awe, as an emotion known for its vast impact on perception and behavior, was hypothesized to increase humility. The original research suggested that experiences of awe lead to greater humility. In the original study, participants watched either an awe-inducing video or a neutral video. After this, they were then asked to write about their personal strengths and weaknesses, simulating a self-introduction to a stranger. Participants in the awe condition described themselves in a more balanced way, listing fewer strengths before moving on to their weaknesses, suggesting an expression of humility. Our goal is to replicate this study and see if exposure to an awe eliciting video still increases humility. We hypothesize that participants who view an awe-eliciting video will exhibit more humility, operationalized as a more balanced presentation of personal strengths and weaknesses, compared to participants in the control condition. By replicating this study, we hope to contribute to growing psychological research on how emotion shapes self-concept and interpersonal behavior. We aimed to contribute to a larger ongoing dialogue and body of research on the replicability of psychological phenomena such as awe and humility and the complexities of human emotion, feelings, and experience.

A Replication Study on the relationship between Awe and Humility

AWE & HUMILITY REPLICATION STUDY

Awe and humility are two distinct emotions that have gained attention in psychological research due to their profound effects on human behavior and functioning. Awe is often described as a positive response to stimuli that is vast and creates a sense of wonder for the world, leading to an altering of perspective and emotion. (Keltner & Haidt, 2003). Humility, on the other hand, involves a realistic self-assessment that leads to an acknowledgment of one's limitations and an openness to receive information from others. (Kruse et al., 2019). Both emotions, despite their differing qualities, share a common ability to both acknowledge and challenge perspectives about the self, inspiring a deeper understanding of one's relationship with the world and others. Stellar et al. Sought to further the depth of research regarding these concepts, considering linkages between the two and adding to the overall literature.

Stellar et al. (2017) investigated the relationship between awe and humility, finding that creating experiences of awe, such as viewing awe-inducing videos that dealt with the vastness of the world, led to greater humility in comparison to neutral experiences. These findings suggest that awe can foster humility. The current study seeks to replicate Stellar et al.'s (2017) experimental design by exposing participants to awe-inducing and neutral video stimuli. After these videos are viewed, levels of humility are measured to understand how awe might impact self-reflection and humble behavior. While these findings are essential for understanding these topics, replication studies are crucial for confirming reliability. These replication efforts are essential for ensuring generalizability across different populations and contexts. This replication heightens the understanding of the relationship between awe and humility and further solidifies the growing body of research surrounding these concepts.

This link between awe and humility found in Stellar et al.'s study aligns with previous findings by Piff et al. (2015), who showcased that both awe and humility promote increases in generosity and putting other first, while also reducing narcissism. The paper recognizes, "awe

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shifts people away from being the center of their own individual worlds, toward a focus on the broader social context and their place within it (Piff et al., 2015, p. 1833). Additionally, research by Van Cappellen et al. (2013) supports this connection by demonstrating that experiences of awe contribute to feelings of oneness with others and activated further emotion and action. Both of these findings suggest that awe, as it reduces egocentrism and provides perspective on the world, has an important role in promoting humility, which can affect personal growth and social harmony.

Method

Design

This design replicates Stellar et Al's study #3, which was an online between-subjects experimental design. In this experiment, awe was manipulated variable and operationalized through the viewing of a video (awe-inducing video and neutral video). The dependent variable is humility, which was determined through a self-report measure of participants' weaknesses and strengths.

Participants

A total of ninety-two participants were recruited through an open Qualtrics survey link distributed online. Data from 27 individuals were excluded from the final analysis due to one or more of the following reasons: failure to provide informed consent, indications of prior knowledge of the study's purpose, or incomplete responses to the strengths and weaknesses task. The final sample consisted of 65 undergraduate students from the Illinois Institute of Technology, recruited through departmental outreach and the IIT SONA system.

AWE & HUMILITY REPLICATION STUDY

The sample included 24 male and 41 female participants, ranging in age from 18 to 26 years old, with the largest age group being 21-year-olds ($n = 19$). The racial composition of the sample was as follows: 14 Hispanic or Latino, 10 non-Hispanic White or Euro-American, 9 Black, Afro-Caribbean, or African American, and 8 East Asian or Asian American participants. Additionally, the sample included 3 South Asian or Indian American participants, 2 Middle Eastern or Arab American participants, 3 who self-described their racial identity, and 2 who reported more than one race.

Participants were randomly assigned to one of two conditions. 36 participants were assigned to the awe condition and 29 were assigned to the neutral condition. All participants provided informed consent and received up to 0.5 extra credit points in an eligible course as compensation for their participation.

Materials

The study was administered entirely online using Qualtrics. Participants were randomly assigned to view one of two YouTube videos: an awe-inducing video depicting the vastness of space and planets, or a neutral instructional video on how to build a fence. Following the video, participants completed a timed, open-ended self-report task in which they listed their personal strengths and weaknesses. The task was displayed with a two-minute countdown timer. The proportion of weaknesses listed relative to the total number of strengths and weaknesses combined was used as the operational measure of humility.

Procedure

After the option for providing consent, participants were first presented with a brief introduction to the study's purpose, followed by a set of demographic questions. Following the demographic information participants were randomly assigned to either the awe condition or the

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neutral condition. The awe condition was a video about the planets and space, while the neutral condition was an instructional video on building a fence. They completed a self-report measure of listing their strengths and weaknesses for two minutes, beginning with their strengths. After this data was collected the strengths and weaknesses were counted and a proportion of the DV was calculated by averaging the total weaknesses listed over the total number of strengths and weaknesses listed by a participant.

Results

We tested the hypothesis that participants who viewed an awe-inducing video would report greater humility, measured as a higher proportion of self-reported weaknesses, in comparison to participants who watched a neutral video. Humility was operationalized through a weakness ratio, calculated by the number of weaknesses listed divided by the total number of traits (total strengths and weaknesses) each participant reported. The data was used in an independent samples *t*-test to compare the weakness ratio between different conditions, showcased in Table 1 and Table 2. The 36 participants in the awe condition reported a mean weakness ratio of 0.24 ($SD = 0.20$), while the 29 participants in the neutral condition had a slightly higher mean of ($M = 0.28$, $SD = 0.21$). The difference between these two groups was not statistically significant, $t(63) = -0.89$, $p = .376$, $d = -0.22$.

Table 3 presents the distribution of weakness ratios by condition. Although the neutral condition showed a marginally higher average weakness ratio, the large overlap in variability between groups and the small effect size suggest minimal difference in the way participants described themselves after watching the respective videos. In addition to analyzing the weakness ratio, we also explored the average number of traits listed in each condition. On average, participants in both groups listed more strengths than weaknesses overall. There was no notable

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difference in the total number of traits generated between groups. These findings do not support our hypothesis and differ from the original results reported by Stellar et al. (2017), who found that awe increased participants' willingness to report personal weaknesses.

Discussion

Our hypothesis paralleled Stellar et al.'s, that participants in the awe condition would demonstrate greater humility, operationalized as a higher weakness ratio, compared to those in the control condition. However, our results did not support this hypothesis. The observed difference in weakness ratios between groups was not statistically significant. This finding contrasts the final results of Stellar et al.'s (2017) original study, which found that awe increased humility.

Although our study was a direct replication of Stellar et al.'s original study, there were multiple potential limitations. Construct validity in this study may be limited due to the open-ended format used to measure humility. While this method aligned with the one in the original study, self-reporting personal strengths and weaknesses may introduce individual differences in interpretation or possibly social desirability bias out of fear of judgment. A validated humility scale could improve measurement consistency in future replications. Internal validity was reasonably strong and random assignment to video conditions further reduced the influence of confounding variables. However, subtle factors such as individual differences in mood or attention prior to participation may still have influenced responses. External validity is limited by the narrow participant pool, undergraduate students from Illinois Institute of Technology, and by the brief data collection period. These factors reduce the generalizability of our findings to a larger context and population. Statistical validity was also limited due to the relatively small sample size. Our non-significant result and minimal effect size indicate that future studies could

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benefit from larger sample sizes to better and more impactfully evaluate the hypothesized relationship.

Future research should replicate this study with more participants and consider using multiple measures of humility, including validated self-report scales. Improving construct and external validity will also enhance the reliability and generalizability of findings.

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Table 1 and Table 2

Independent Samples T-Test						
		Statistic	df	p	Effect Size	
DV- weakness_ratio	Student's t	-0.891	63.0	0.376	Cohen's d	-0.222

Note. $H_0: \mu_{IV1} = \mu_{IV2}$

Group Descriptives						
	Group	N	Mean	Median	SD	SE
DV- weakness_ratio	IV1	36	0.237	0.250	0.200	0.0333
	IV2	29	0.282	0.333	0.205	0.0381

Figure 1

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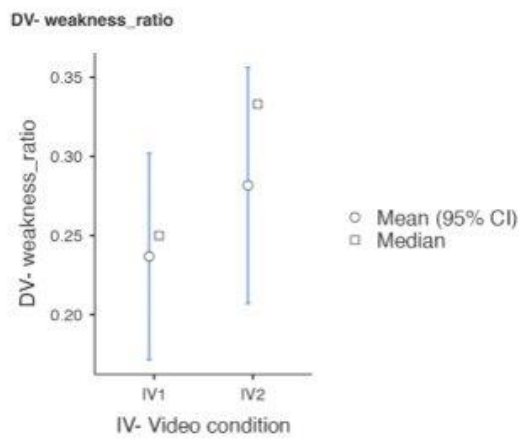


Table 2

Descriptives		
	IV- Video condition	DV- weakness_ratio
N	IV1	36
	IV2	29
Missing	IV1	0
	IV2	0
Mean	IV1	0.237
	IV2	0.282
Median	IV1	0.250
	IV2	0.333
Standard deviation	IV1	0.200
	IV2	0.205
Minimum	IV1	0.00
	IV2	0.00
Maximum	IV1	0.500
	IV2	0.500

Dynamic Norms Promote Sustainable Behavior, Even if It is Counter normative

Student 4

Illinois Institute of Technology PSYC 204

Research Methods in Behavioral Science

Dr. Robert Keating

4/25/2025

Abstract

In this study it is investigated whether informing that people are increasingly reducing meat consumption, could induce people to reduce meat consumption themselves. The respondents were presented with two messages, one informing them that meat consumption is restricted by a certain percentage of the people, and the other stating that this percentage is growing through years. They wanted to find out if the notion of shifting momentum in one's dietary habits would encourage more people to rethink their own choices. Notably, we did not find that there was any significant difference in responses between the two groups, which raised the possibility that dynamic norms may not be as effective as it was supposed to be. These results have important implications for how social messages influence actions,

Introduction

Social norms in psychology, are believed to influence social behavior within various groups. Cultural Context and Contingent Norms These norms are shaped by culture, tradition, and social expectation to inform individuals how to act in different contexts. For instance, the Filipino culture valued the ideal of “utang na loob,” which has strong implications in interpersonal relationships. Filipinos feel a sense of obligation to give back because they want to live up to the kindness or generosity that you have shown their way. It’s not about contracting a debt. Rather, they believe that the “favor” or “payment” you give them is a measure of love, gratitude, and loyalty, and it’s something they will consider when it is their turn to give back to you. Norms of this kind help people find their way in the world, and fit in with their communities. In essence, they serve as the bond that brings others together. In sociological terms, norms are informal understandings that govern the behavior of members of a society. In social psychology, norms shape everything from large-scale cultural expectations to small-group dynamics. Studies have found that messages that encourage or discourage behavior can influence people’s decisions. These messages, referred to as the dynamic normative messages, point to current shifts in society which should be emphasized to change behavior.

The Sparkman and Walton study investigated the effect dynamic norms could have on meat consumption behaviors. The study, investigated whether reading about more and more people cutting back their meat consumption would inspire more typical eaters to question their own meat consumption. The researchers also measured the weekly meat consumption of participants to see if exposure to a passage about changing norms affected their personal habits. This research raises vital questions: How are norms established, and why do they have such influence over behavior? Social norms play a role in how people interact with society and they influence decisions on a level that people might not even realize. Over time, once a behavior becomes accepted, it becomes normalized, but not all norms are good. Some are benign and even helpful to the social system, like etiquette or dietary preferences, and others, like materialism and racial or gender-related stereotypes, can lead to unhealthy behaviors. These communal standards also differ between cultures and traditions. The Filipino food culture is a partner in promoting communal dining and sharing with one’s family and friends being the order of the day. But in some Western traditions, individual portions and preferences are the norm. Norms change, and often they change in response to the repetition of behaviors that defy the expected. At the end of the day, social norms are the culture we live in, and they mold personal decisions and community interactions. When people move or join unfamiliar groups, though, they tend to take behavioral norms with them, adjusting in time to their new context. Norms take the form of collective decisions within groups, and common practices or traditions that slowly become established. In examining how people react to passing social trends, psychologists are attempting to determine what causes people to embrace or eschew particular conduct. This knowledge illuminates human behavior and provides tools that can be used to help make the world a healthier, safer, and more harmonious place.

Methods

We used a Qualtrics-based survey for this study that was near-exact replication of the original protocol by Sparkman and Walton (2017). About 100 individuals, who were all undergraduate students in the Department of Psychology at Illinois Institute of Technology, participated in the experiment. The study population was mostly composed of young adults in their early 20s with varied backgrounds reflecting an urban university environment. Participants were solicited from psychology courses and participants in relevant classes had the opportunity to earn 0.5 SONA credit for their participation in the survey. Although we tried to create an almost identical replica of the original work, some demographic differences, political beliefs, cultural influences, may have affected the results. Furthermore, according to previous work, those with more liberal attitudes are likely to be more willing to change their meat consumption, which we took into account in the present study. Materials and Procedure Recruitment was through the psychology department at Illinois Institute of Technology with students receiving e-mailed invitations to complete the survey. The Qualtrics survey exactly mirrored the original questionnaire with few alterations. Participants joined one of two conditions (static norm or control; dynamic norm experimental) where they were asked the same survey questions, except for a critical message treatment difference:

1. *The statement in the control condition (static norm) was: “According to recent studies, 30% of Americans are trying to eat less meat. That’s because 3 in 10 people eat less meat than they normally would.”*
2. *The experimental group (dynamic norm) was given the following information: “We have only just begun to see a huge shift in attitudes toward how we eat and produce our food.*

Recent studies have shown that over the past 5 years 30% of Americans now are making a conscious effort to reduce their meat intake. In the past few years, 3 out of 10 have adjusted their behavior and are eating less meat than they used to. Participants were randomly assigned to either group. Following the exposure to the message, subjects were requested to report their political ideology, gender, and eating habits, namely whether they were vegetarian or vegan. This information was useful for investigating the correlation between social norms and personal decision-making on meat.

Results

The results of this study were not aligned with that obtained in the previous results of Sparkman and Walton (2017). There was no significant difference in intention to reduce meat consumption between those exposed to the static norm message and those exposed to the dynamic norm message ($p = 0.716$). Effect size (Cohen’s d) suggested that there was a very small difference in the influence of dynamic norm messaging on dietary behaviors. Based on these findings, we could not reject the null hypothesis, which challenges the efficacy of dynamic norms as a device for social influence. Demographic differences, such as

who the psychology student population at the Illinois Institute of Technology is composed of, may have affected the results, as past studies indicate that political ideology, cultural background, and already established dietary practices can influence reaction to normative messages. These results highlight the need for replication studies in various populations to examine the generalizability and the robustness of dynamic norm interventions.

Discussion

Social norms exert a powerful influence on human behavior, guiding the decisions made and interactions had, often without our knowledge. We sought to investigate whether dynamic norms, which emphasize that social behavior is continuously on the move, could be more successful in promoting change than static norms. One of the underlying assumptions was that static norms do play a role in behavior change, yet dynamic norms may have a greater impact. By re-performing a previous study, we wanted to provide congruent evidence that the social norm does drive people's attitudes more so than that people do engage in attitudinal adjustments. There were differences in the demography of our cohort and the population of the original experiment. The undergraduates on which this study is based and who were invited to participate are students in psychology courses at the Illinois Institute of Technology, whose backgrounds are different from those of the sample in the original study, which focused on students at a religious, very conservative university. Political ideology, cultural diversity, and urbanicity were possibly influencing different results. Although we don't think these are reasons enough to account for why we find no substantive results, they are worth pursuing as possible determinants. The observations made in our study leave us with some challenging open questions:

- How powerful indeed are dynamic norms for behavioral change?

Previous research had indicated that making changes in behaviors within a population could prompt people to change their own behaviors accordingly. Yet our results did not find support for this claim, and we go on to ask if dynamic norms matter at all in decision-making. - Would other populations have different results? Our participants were undergraduate students in a young urban technology-oriented university. It may be that you can test this theory in a different social context, in say, a rural or politically liberal area, and get different results. Future research is needed to test how age, cultural values and socio-economic factors moderate responsiveness to norm-based interventions. How might cultural attitudes toward norms bias the results of the study? Cultural context greatly influences beliefs and behaviors related to food and eating. In Filipino culture, communal sharing of meals also promote close social relationships, and some dietary practices are influenced by religious and familial values. If this study were replicated in a more collectivist culture, or a culture with dissimilar dietary restrictions (eg, Hindu or Islamic traditions), this could also help to clarify the interplay between social norms and cultural expectations. Several studies have indicated a link between static norms and behavior change, leading us to expect equally

dynamic norms to shape behavior in the same way. Yet, the lack of replicability of this effect means it might not be generalizable across contexts. This is especially important in the field of psychology because findings from psychological studies need to be replicated to ascertain whether or not a particular theory is applicable to different populations and contexts. If findings such as these are not upheld across further studies, researchers need to readjust their thinking to make certain that inferences are based on sound scientists. This reflects a general problem in the field of behavior research: many studies find effects that are difficult to repeat. Failure to replicate puts the field at risk of progressing on shaky foundations and applying psychological principles wrongly in policy, marketing and social interventions. More research is required to understand who norms determine behaviors, and how they work, and to understand who is likely to follow norms. That through further investigation of these questions we may develop a more nuanced understanding of why people decide the way they do, what leads behavior to change and how society can more effectively harness social influence to promote positive results. All told, this knowledge enables us to better develop our insight into human behavior, improve social interactions, and increase quality of life for people and populations.

References

A Replication Study of the Effect of Awe on Humility

Student 5

Illinois Institute of Technology

PSYCH 204: Research Methods in Behavioral Science

Dr. Robert Keating

May 5, 2025

Abstract

This study aimed to replicate the finding of Stellar et al. (2018), which suggested that the feeling of awe can increase a person's humility. Humility is characterized as a self-transcendent state involving self-awareness and recognition of personal limitations. The 59 participants in this study were randomly assigned to watch either a two minute awe inducing video or a neutral video. They then had to list their strengths and weaknesses within a two minute time frame. The study operationalized humility by calculating a score (value between 0 and 1) which was the proportion of weaknesses to total traits listed. After running an independent samples t-test on the humility scores, no significant difference in the awe condition and the neutral condition were found. These results did not support the original study's findings nor our initial hypothesis. Possible explanations for the null results include construct validity issues with unclear survey instruction, internal validity issues with uncontrolled testing environment, and external validity issues with low diversity within the sample.

Keywords: awe, humility, self-awareness, self-transcendent emotions

Introduction: Awe and Humility

Awe is a profound emotional response to something that is extremely vast, abstract, or beyond someone's understanding. This can lead to individuals having a need to explain the extremity of the experience by finding a way to rationalize the idea, or cognitive accommodation (Keltner & Haidt, 2003). Awe typically consists of wonder, amazement, and occasionally fear. Because it is such a complex emotion, it can cause individuals to feel more vulnerable or fearful due to the vastness of the concept. This can also contribute to self growth (Kashdan & McKnight, 2013). It is also associated with self-transcendence, or the shift of attention away from the self and towards something greater. (Gordon et al., 2017).

Humility is characterized by self-awareness, openness, and recognition of personal limitations. (Tangney, 2000). However, recent research suggests that humility can function as a transient state influenced by external stimuli such as emotional experiences (Chancellor & Lyubomirsky, 2013). It causes individuals to focus on others as opposed to the self. Humility has been associated with many positive outcomes, including improved well-being and interpersonal functioning due to the recognition of one's smallness (Ivtzan et al., 2016).

Recent psychological studies have been suggesting a link between the feeling of awe and humility. Awe is a feeling that can be invoked in an individual, while humility is a personality trait. The affect of awe that causes a diminished perception of the self, might increase a person's capacity for humility (Piff et al., 2015). In the Stellar et al's study (2018), it was found that those who are introduced to awe-inducing stimuli are more likely to describe themselves with more modesty and attribute their successes to external factors.

This study was done in hopes of replicating the findings of the Stellar et al. (2018). The goal was to be able to find a relationship between awe and humility by exposing participants to

awe inspiring stimuli and then rating their levels of humility. Participants in this study were either part of the awe condition or the control condition. Those in the awe condition watched a video with the purpose of inspiring awe, while those in the control condition watched a neutral video. Both groups were then asked to list their strengths and weaknesses. The independent variable was awe, manipulated with the videos. The dependent variable was humility, measured by the strength and weaknesses listed. We hypothesized that those in the awe inspired group would rank higher on humility.

Method

Design

The replication of this study that we did was a posttest-only experiment that had a between-subjects design. The manipulated, independent variable was the induced feeling of awe and the measured, dependent variable was the humility a participant had.

Participants

In this study, there were 92 participants. The participants were students recruited from Illinois Institute of Technology through the SONA research system. The participants were volunteers that gained course credit for their participation in the study. Of the original 92 participants, 33 of them were excluded from the final analysis due to incomplete responses, incorrectly following instructions, or failing manipulation checks. The final sample size was 59. The participants range of age was 18-26 years ($M = 20.04$, $SD = 1.78$). 47.4% of the participants identified as male, 49.1% as female, and 3.5% as non-binary. As for racial and ethnic identity information, 35.7% were Hispanic or Latino, 19.6% were non-Hispanic White or Euro-American, 16.1% were East Asian or Asian, 16.1% were Black, Afro-Caribbean, or

African American, 5.4% were South Asian or Indian American, 1.8% were Middle Eastern or Arab American, and 5.4% were of other races that were not listed.

Materials

The study was accessed online through a Qualtrics survey. Participants were randomly assigned to with the awe condition or the control condition. Participants in both conditions were required to watch a 2 minute YouTube video. Those in the awe condition watched a video that depicted the scale of the universe by zooming out more and more to show the planets and stars. This video was the stimulus that elicited awe in participants. The individuals in the control condition watched a video on how to increase the strength of a courtyard wall. This video was intended to provoke minimal feelings in those who watched it.

To assess the feeling of awe in each participant, each person completed a series of Likert scale questions, rating how much they felt certain emotions such as awe, fear, wonder, happiness, and amazement. Responses ranged from 1 ("not at all") to 7 ("very much"), with higher scores indicating stronger emotional reactions.

Humility was measured by having the participants list their own strengths and weaknesses within two minutes. Coders that were blind to the conditions in the study counted the number of strengths and weaknesses given by each participant. The coders found the average number of weaknesses listed and divided that value by the total product of strengths and weaknesses, to get a score between 0 and 1. This score represented the ratio of listed weaknesses to total listed traits. A lower score indicated less weaknesses, while a higher score indicated more listed weaknesses and therefore, greater humility.

There were also a series of demographic questions to collect data on age, gender, and ethnicity. When asked about their gender and race, participants were allowed to select multiple answers, and/or choose to self describe their identity.

Procedure

This study was conducted online using Qualtrics. Participants were immediately required to review and sign an informed consent form. Those who completed the consent form were then randomly assigned to the awe or neutral condition. The respective 2 minute videos for each condition were embedded into the Qualtrics survey so that the assigned participants could watch it. After watching the video, the participants were given two minutes on a timer to list their strengths and weaknesses. Next, all of the participants completed the five Likert scale questions assessing their various emotions. They were also required to answer a series of questions about the content of the video they saw and the survey they were completing to make sure that they had followed instructions correctly. The end of the survey included the demographic questions. We made sure to debrief the participants once the survey was completed.

Results

We conducted an independent samples t-test in order to test our hypothesis that inducing awe would cause greater humility in an individual. The mean humility score (ratio of weaknesses to total listed traits) in the awe conditions was $M = 0.252$, $SD = 0.203$. Which was slightly lower than the mean humility score in the control condition ($M = 0.261$, $SD = 0.203$). This difference was not statistically significant, as $t(57) = -0.169$, $p = .567$, $d = -0.0442$.

Discussion

We had initially hypothesized that those in the awe condition would have a greater sense of humility, measured by a greater proportion of weaknesses listed than strengths. The data from this replication study did not support the hypothesis. Those in the awe condition did not significantly differ from those in the neutral condition when considering the proportion of weaknesses to strengths. This contradicts the findings of the original Stellar et al. (2018) study, where they observed a significant increase in humility after an awe inducing stimuli.

There are several possible reasons why our replication did not match the findings of the initial study. First, the construct validity might not have been the strongest. Even though the sample size was initially 92 participants, many of them had to be excluded from the analysis in the end due to failure to follow instructions or pass manipulation checks. This could have been due to misunderstanding of instructions when completing the survey. If participants weren't fully aware of how to complete the task, then the survey might not have been an accurate measure of their humility.

Additionally, the study faced challenges with internal validity. Because the study was conducted entirely online and asynchronously, there wasn't a way to control for environmental factors. If students completed the survey with a friend, this could have affected the results. A snowball sampling effect is also possible if students recommended the study to their friends.

Due to the sample coming entirely from Illinois Institute of Technology, the external validity was very limited in this study. Since all of the participants came from the same school, the results aren't very generalizable to the whole population. Students who participated were also given course credit, which could have created an incentive to sign up, and as a result a self-selection bias. In the original study (Stellar et al., 2018), there was a wider range of where the participants were selected from which strengthened the external validity.

If this study was done again, the construct, internal, and external validity could be strengthened. To improve the study, the instructions given to participants could be rewritten to make them clearer. The survey could also be taken in a more controlled environment to account for distractions or outside input. To increase external validity, we could consider using volunteers from other school campuses to broaden our range of participants. Unfortunately, this replication study did not yield significant results. However, it gave us good insight into how we might improve the design in the future.

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A Replication Study of the Effect of Awe on Humility

Student 6

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PSYC 204: Research Methods in Behavioral Science

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May 3, 2025

AWE AND HUMILITY REPLICATION STUDY

Abstract

Stellar et al. (2018) examined how awe influences humility, using videos to depict the relationship between these concepts. Awe, characterized by wonder at something vast, may foster humility by making individuals feel smaller compared to the world. To replicate their findings, we hypothesized that participants viewing awe-eliciting videos would report higher humility, defined by their self-reported strengths and weaknesses. Ninety-two participants were recruited from the Illinois Institute of Technology's online portal, assigning them to either an awe-inspiring or neutral video condition. The ratio of weaknesses to strengths listed was used to measure humble behaviors and thus humility. Results showed no significant difference in humility levels between groups, contradicting our hypothesis and earlier findings. Possible reasons for this include a restricted sample size and a lack of environmental control. While data collection is ongoing, these insights could deepen our understanding of how emotions like awe influence self-perception. Harnessing awe might benefit fields such as education and therapy by promoting humility. Future research could explore varying types of awe and the duration of their effects on self-perception.

Keywords: Awe, Humility, Self-Perception, Replication.

AWE AND HUMILITY REPLICATION STUDY

Introduction

Does experiencing something greater than the human condition broaden our awareness of the physical universe and expand our moral outlook? One significant psychological process through which awe may facilitate this transformation is its profound impact on self-concept. Specifically, awe has been shown to evoke a sense of a "small self," where individuals feel humble and diminutive within the vastness of existence. This feeling of self-diminishment often leads to an enhanced sense of connection to a larger whole, emphasizing the enormity of the universe compared to one's individual experiences, thereby inducing greater humility. By reshaping individuals' perceptions of themselves in relation to their surroundings and fostering a sense of interconnectedness, the experience of awe may inspire a broader moral consideration for others. This expansion of greater humility can manifest as increased empathy, compassion, and altruistic behavior towards a diverse range of individuals, ultimately reinforcing social bonds and community cohesion. This study aims to replicate Stellar et al.'s work to examine the relationship between awe and humility.

The concepts of awe and humility have been studied independently since the early to mid-20th century, but both have recently gained renewed attention within psychological research. While extensive studies have explored these individual concepts, the intricate relationship between awe and humility remains inadequately documented and understood. Awe is an emotional response that combines wonder and respect, typically arising from experiences that transcend ordinary understanding (Gordon et al., 2017). Humility, related to awe, involves recognizing one's strengths and weaknesses, being open to new perspectives, prioritizing the needs of others, and acknowledging the inherent worth of all individuals (Ruberton et al., 2017).

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Research shows a strong connection between experiencing awe and engaging in prosocial behaviors, such as increased generosity and a willingness to volunteer for charitable initiatives (Guan et al., 2019). Awe is a temporary emotional state triggered by extraordinary experiences (Liu et al., 2023) and is associated with improved mental and physical health (Stellar et al., 2015). Studies have indicated that awe enhances environmental responsibility and fosters humility, both of which are essential for altruism (Piff et al., 2015; Wang & Lyu, 2019; Stellar et al., 2018). Furthermore, research has found that awe influences moral concern by changing our self-perception, leading to feelings of insignificance and a sense of connection to the universe (Piff et al., 2015). A recent example is the awe-inspiring stories from the COVID-19 pandemic, which motivated prosocial actions, such as helping others and donating blood (Luo et al., 2022). Psychological well-being is linked to positive emotions and the promotion of humility (Tong et al., 2019). While humility is often viewed as a stable trait (Nielsen & Marrone, 2018), some researchers suggest it can be developed through practice (Wright et al., 2017). Since both humility and awe can enhance well-being, exploring their relationship further may reveal how awe can foster humility over time. Thereby, awe can inspire individuals to contribute positively to society.

In this study, a replication study of the seminal work by Stellar et al. (2018), we aim to test the hypothesis that experiences of awe will cultivate a greater humility. Stellar and colleagues aimed to examine the connection between awe and humility, assessing whether momentary experiences of awe could promote humility by using an appraisal-tendency framework to clarify the underlying processes. This paper focuses on replicating Stellar and colleagues' study, with awe as our manipulated variable and humility as our measured variable and aims to deepen our understanding of humility as a dynamic, fluctuating state influenced by

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encounters with awe-inspiring experiences. We aim to observe the effect of awe on humble behaviors by showing participants awe-inspiring or neutral videos and asking them to list their strengths and weaknesses using a questionnaire format. By examining how awe can transform our moral landscape, we can better appreciate its potential for fostering greater empathy and ethical considerations in our interactions with others.

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Method

Design

The study employed a posttest-only experiment with a between-subjects design. The key variables examined were the independent variable awe, which was manipulated via YouTube videos intended to induce awe or a neutral experience, and the dependent variable, humility.

Participants

Ninety-two students from the Illinois Institute of Technology participated in this study. The sample size reflects the number of individuals who completed the survey while it was open. Participants were recruited through the university's SONA system for course credit. An online link was provided to access the survey. Those with incomplete data, who did not follow instructions, or who failed manipulation checks were excluded from the study (33 participants, leaving a sample size of 59).

The participants' ages ranged from 18 to 29 years old, with a mean age of ($M=20.3$) and a standard deviation of ($SD = 1.81$). Of the participants, 47.4% reported being male, 49.1% female, and 3.5% non-binary. The ethnic composition of the sample was as follows: 35.7% identified as Hispanic or Latino, 19.6% as non-Hispanic White or Euro-American, 16.1% as East Asian or Asian, 16.1% as Black, Afro-Caribbean, or African American, 5.4% as South Asian or Indian American, 1.8% as Middle Eastern or Arab American, and 5.4% preferred to self-describe their race.

Materials and Measures

Participants accessed the survey via Qualtrics on their own electronic devices, such as smartphones, computers, or tablets. The experimental manipulation consisted of two 2-minute

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YouTube videos. Those in the awe condition of the study watched an embedded video designed to evoke feelings of awe, wonder, and interest, as demonstrated in a pilot study (Stellar et al., 2018). During the survey, this video displayed the universe on a grand scale, gradually zooming out from planet Earth to illustrate the sizes of increasingly larger celestial bodies, thereby conveying the vastness of the universe (Appendix 1). Participants in the neutral condition watched an instructional video on enhancing the structural integrity of a garden wall, which elicited low levels of nearly all emotions.(Stellar et al., 2012). (Appendix 2). The rest of the study was identical between conditions.

To assess humility, participants were given two minutes to list their strengths and weaknesses, a method previously employed in various studies (Tangney, 2000). They were instructed to frame their responses as if they were sharing this information with someone they had just met. Unaware of the study's conditions, four independent coders counted the number of strengths and weaknesses each participant listed, and the proportion of weaknesses was then calculated to derive a score ranging from 0 to 1, representing the proportion of a participant's responses that were weaknesses. A higher score indicated a greater prevalence of weaknesses relative to strengths; thus, lower scores suggested less humility, while higher scores indicated a greater degree of humility.

Participants were instructed to assess the extent to which they experienced various emotions—specifically awe, fear, wonder, happiness, and amazement—following their viewing of the video associated with their designated condition. A manipulation check was added to assess whether the videos effectively manipulated awe, with scores ranging from 1 (not at all) to

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7 (very much). Lower scores signified a lack of emotion or minimal emotional response, whereas higher scores indicated a strong emotional experience.

Participants were presented with a series of demographic questions focusing on age, gender, and race/ethnicity. These questions were designed to allow participants to select multiple options and/or provide a self-descriptive response.

Procedure

Informed consent was obtained from all individuals before they participated in the study. Participants engaged in an online survey conducted through Qualtrics and were randomly assigned to either the awe or neutral condition videos. Following the video, participants completed the dependent measures of listing their strengths and weaknesses, within a two-minute timeframe monitored by a countdown timer.

The subsequent section included three manipulation check questions to ascertain participants' comprehension of the video content and the study's objectives, as well as to confirm that the video loaded and played correctly. This measure ensured that participants completed the survey to the best of their abilities and with full attention.

The survey concluded with three demographic questions that inquired about age, gender, and race/ethnicity. Upon completion, participants were thanked for their involvement and provided with a debriefing.

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Results

We conducted an investigation to test the hypothesis that individuals who viewed an awe-inspiring video would exhibit more humble behaviors than those who viewed a neutral video, employing an independent samples t-test for analysis. As shown in Figure 1, results revealed that the mean proportion of weaknesses compared to strengths listed was lower for participants in the awe condition ($M = 0.25$, $SD = 0.20$) than for those in the neutral condition ($M = 0.26$, $SD = 0.20$). However, this difference did not reach statistical significance, $t(57) = -0.17$, $p = 0.56$, $d = -0.04$. The observed mean differences indicate an insignificant effect that runs counter to the initial hypothesis, with participants in the awe condition listing weaknesses at a level 0.04 standard deviations below that of their neutral counterparts.

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Discussion

Our hypothesis was that participants who viewed the awe-inducing video would present a higher ratio of weaknesses to strengths, reflecting an enhanced sense of humility compared to those who watched the neutral video. However, the results indicated that the average ratio of weaknesses to strengths did not reach statistical significance, with participants in the neutral condition yielding a slightly higher mean. This outcome facilitates important discussions regarding our replication of the study, particularly in the context of the significant results reported by Stellar et al. (2018). The findings of this research may underscore the intricacies involved in study replication rather than undermining the theoretical framework.

The results of this study did not support our hypothesis and were not consistent with the findings of Stellar et al. (2018). This may have been due to several limitations. First, issues affecting construct validity may have influenced our results. Although our study directly replicated the design, materials, and procedures of Stellar et al. (2018), we employed a smaller sample of participants. A significant proportion of the original sample had to be excluded due to non-compliance with instructions or failure to pass manipulation checks, which may suggest potential shortcomings in the wording or design of the questionnaire. In contrast, Stellar et al. (2018) observed a significant effect with the same materials, indicating that a larger usable sample in our study might have mitigated these concerns.

Furthermore, internal validity was compromised as participants completed the study asynchronously and in an online environment lacking environmental controls. Additionally, the external validity of our findings may be limited, as the survey was administered solely to Illinois Institute of Technology students seeking course credit on SONA, restricting the generalizability

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of our results to a broader population, while Stellar et al. (2018) accessed a more diverse global audience. Despite these potential weaknesses in the study's design and implementation, it provides a solid foundation for future research. The experimental conditions were well-controlled, similar to those employed by Stellar et al. (2018). Additionally, the manipulation of the independent variable had been pretested in previous research and was executed in the same manner as in Stellar et al. (2018), where it yielded significant effects. Therefore, further replication of this study is essential to validate its methods and findings, as well as to elucidate why our study did not yield significant results.

Future research on the relationship between awe and humility could improve upon the current study's design by providing clearer instructions for participants when listing strengths and weaknesses and by considering alternative operationalizations of humility (Song et al., 2023). The use of a timed listing may not effectively measure humility in online surveys, prompting the exploration of other methodologies.

Additionally, the reliance on online videos to elicit awe may not produce the strongest effects; thus, alternative approaches, such as vignettes or immersive experiences, should be examined. Conducting the survey in a controlled, synchronous, in-person setting could minimize potential confounding variables. Future studies should also aim for larger, more representative samples to enhance the validity of findings and investigate the influence of moderators like connectedness and other self-transcendent emotions (Pizzaro et al., 2021).

This study did not replicate the significant effects identified in the research conducted by Stellar et al. (2018). Specifically, participants who viewed an awe-inspiring video did not demonstrate a significant tendency to list more weaknesses than strengths when compared to

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those who watched a neutral video. These findings underscore the importance of conducting further replications and additional research to understand the relationship between awe and humility.

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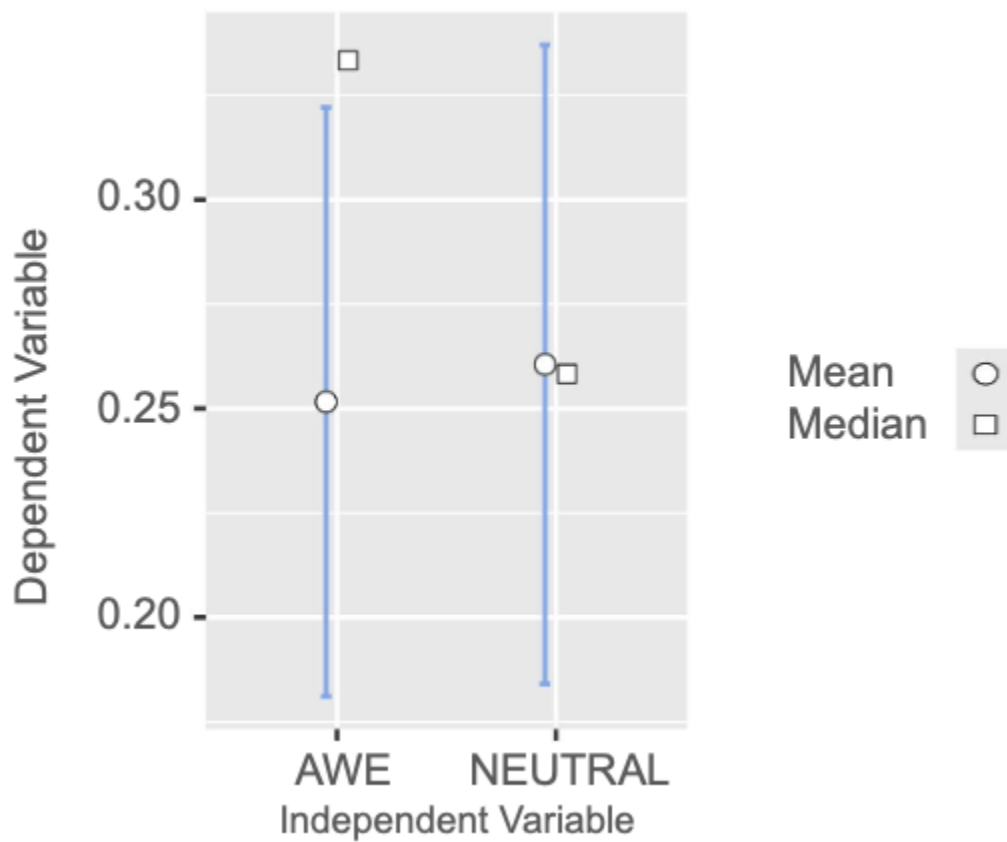
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Appendix

1. Video link: <https://youtu.be/ZoXqj-VDhws?si=ZhbDnOxx5mHOdBT7>
2. Video link: <https://youtu.be/XYGzuPpw50g?si=DiatRdgAwGu0XVZo>

Figure 1

Mean Proportion of Weaknesses to Strengths



A Replication Study of the Effect of Awe on Humility

Student 7

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PSYC 204: Research Methods in Behavioral Science

Dr. Robert Keating

May 9, 2025

Abstract

This study aimed to replicate Stellar et al. (2018) by investigating whether experiences of awe can influence humility. Humility, often seen as a stable personality trait, has more recently been conceptualized as a dynamic emotional state that can be shaped by external experiences such as awe. In this study, participants ($N = 67$) from the Illinois Institute of Technology were randomly assigned to watch either an awe-inspiring video depicting the vastness of the universe or a neutral video about fence-building. Humility was measured by calculating the proportion of self-reported weaknesses to the total listed traits. Contrary to our hypothesis and the original study's findings, participants in the awe condition did not demonstrate a greater degree of humility. In fact, those in the neutral condition had a slightly higher average proportion of weaknesses, although the difference was not statistically significant ($p = .893$). These results highlight challenges in measuring humility as a state and act as a stepping stone for further work to build upon. Limitations related to sample characteristics, experimental setting, and measurement strategies are discussed, along with directions for future research to improve methodological approaches and enhance the accuracy of measuring humility.

Keywords: *Awe, Humility*

A Replication Study of the Effect of Awe on Humility

Humility is a personality characteristic defined by an accurate self-view that acknowledges both strengths and limitations, an appreciation of others' contributions, and an openness to feedback and growth (Nielsen & Marrone, 2018). It is grounded in the recognition that something greater than the self exists (Nielsen & Marrone, 2018). Despite its significance, humility remains poorly understood, particularly regarding the factors that shape it and whether it can be influenced by external forces beyond one's internal beliefs. To address this, Stellar et al. (2018) suggested that experiences of awe may shape humility. According to Guan et al. (2017), awe is an emotional response to vast and novel stimuli that challenge one's existing mental frameworks, often leading to a diminished sense of self in relation to a greater whole. This study seeks to replicate the work of Stellar et al. (2018) and paint a more complete picture of the relationship between awe and humility.

Humility has been widely recognized as highly important across various disciplines. Despite this, there has been slow progress in the empirical understanding of humility, which has been primarily attributed to the absence of a coherent measurement strategy (Davis et al., 2010). Traditionally, humility has been viewed as a stable personality trait that is largely resistant to change through experimental manipulation. This perspective suggests that humility is an enduring characteristic not easily influenced by situational factors. However, more recent research suggests it can also be experienced as an emotional state (Kruse et al., 2017). This shift allows for exploring how humility evolves and whether individuals could develop humility in response to significant events that humble their self-perception (Kruse et al., 2017). Building on this idea, researchers began examining the impact of different emotional states, such as awe, on humility. Piff et al. (2015) found that feelings of awe increased ethical decision-making and

prosocial behavior, suggesting a diminished sense of entitlement and increased concern for others. Across five studies, they demonstrated that awe not only enhanced generosity and helping behavior but also reduced self-focus, with these effects mediated by a heightened sense of the “small self,” highlighting awe’s role in promoting collective social concern (Piff et al., 2015). Further examining the impact of awe on the self, Gordon et al. (2017) distinguished between positive and threat-based awe, noting that only the positive variant was associated with well-being and humility-related outcomes. Additionally, both humility and awe are considered self-transcendent emotions, which shift an individual's focus from self-interest to the needs and well-being of others, fostering social behavior and collective action (Pizarro et al., 2021). Given these parallels, exploring the relationship between awe and humility offers valuable insights into how those emotions interact with each other as well as their influence on human behavior and social dynamics.

This research aims to solidify the findings of Stellar et al. (2018) and build a stronger foundation of research on the connection between awe and humility. In our study, we operationalized experiences of awe through a video illustrating the vastness of the universe, contrasting it against a neutral condition video depicting the process of building a fence. Humility was the measured variable, operationalized through the balance between an individual's perception of their strengths and weaknesses. We hypothesized that exposing participants to momentary experiences of awe would lead them to demonstrate more humble behavior compared to those in the neutral condition.

Method

Design

This study utilized a posttest-only, between-subjects experimental design. The independent variable was awe, which was manipulated through a standardized video induction. The dependent variable, humility, was assessed following the manipulation by evaluating the balance in participants' self-presentation of strengths and weaknesses.

Participants

The study included 92 Illinois Institute of Technology students who completed the online survey while it was available. Participants were recruited via the university's SONA system in exchange for course credit. Upon clicking the link, students were directed to an online survey comprising our study. Of the initial participants, 25 were excluded due to incomplete responses, failure to follow instructions, or not passing manipulation checks. The final sample comprised 67 participants aged 18 to 29 ($M = 20.23$, $SD = 2.08$). Gender identity was distributed as follows: 46.97% identified as men, 46.97% as women, and 6.06% as non-binary. Regarding ethnicity, 36.51% identified as Hispanic or Latino, 19.05% as Non-Hispanic White or Euro-American, 11.11% as Black, Afro-Caribbean, or African American, 17.46% as East Asian or Asian American, 6.35% as South Asian or Indian American, and 9.52% reported other racial identities.

Materials

The experimental manipulation was performed using identical videos to those employed in the original research conducted by Stellar et al. (2018). The participants, in awe condition, watched a recording depicting the vastness of the universe, which showed the distance between planets and stars by slowly zooming out from the Earth (http://www.youtube.com/watch?v=ZoXqj-VDhws&feature=player_embedded). This video evoked high levels of awe, wonder, and interest (Stellar et al., 2018). In the neutral condition,

participants were shown a nonemotional instructional video about how to build a fence (<https://www.youtube.com/watch?v=XYGzuPpw50g>). The video in the neutral condition was found to provoke little emotional reaction except relaxation and calmness (Stellar et al., 2011). After viewing the video, both groups of participants responded to an identical set of questions.

Humility was operationalized by asking participants to list their strengths and weaknesses, beginning with their strengths. The prompt instructed them to imagine describing these traits to someone they had just met. Participants were given two minutes to complete the task and were provided with an on-screen timer to help manage their time. Once data collection was complete, two independent coders—blind to participants' assigned conditions—reviewed all responses and counted the strengths and weaknesses each participant listed. The final counts were determined by averaging the two coders' assessments. The number of weaknesses was then divided by the total number of traits listed. This produced a coefficient ranging from 0 to 1 that represented the proportion of an individual's responses that were weaknesses. A higher score reflected more weaknesses relative to strengths, indicating a heightened state of humility.

Following this task, participants were presented with a Likert scale prompting them to report how much awe, fear, wonder, happiness, and amazement they felt, ranging from 1 (*Not at all*) to 7 (*Very much*). Lower scores indicated that participants did not feel the emotion, and higher scores indicated that the participants felt the emotion greatly.

Participants also answered three questions about their understanding of the study's hypothesis and whether or not the video had loaded properly. Finally, participants provided demographic information, including age, gender, and race. For gender and race, they selected from predefined options with the opportunity to self-describe.

Procedure

To participate in the study, participants completed an online survey via Qualtrics using a device of their choice, which was not specified as part of the study. After reviewing and providing consent, participants were randomly assigned to either the awe or neutral condition. They then watched the corresponding video embedded in the survey and moved to the next page, where they had two minutes to write about their strengths and weaknesses, starting with their strengths. They were instructed to imagine sharing these traits with someone they had just met, with a countdown timer displayed on the screen. Next, participants assessed the emotions they experienced during the video. These items were identical for all participants, regardless of their assigned condition. Following this, participants answered three manipulation check questions. Finally, they responded to demographic questions. Upon completion, participants were thanked for participating and provided with additional material to read about the study's hypothesis and its significance.

Results

Using an independent samples t-test, we tested the hypothesis that individuals exposed to the awe-inspiring video will demonstrate a greater degree of humility than those in the neutral condition. We found that the mean proportion of weaknesses was greater for those in the neutral condition ($M = 0.3$, $SD = 0.21$) rather than for those in the awe-inspiring condition ($M = 0.24$, $SD = 0.2$). This result can be seen in Figure 1. The difference between groups was not found to be statistically significant, $t(65) = -1.26$, $p = 0.893$, $d = -0.309$. The results of our study represent an effect in the opposite direction of the hypothesis, which is not statistically significant. Despite this finding the results of our manipulation check showed that the awe condition video provoked greater feelings of awe ($M = 4.32$, $SD = 1.9$) and happiness ($M = 3.43$, $SD = 1.62$) relative to the

neutral condition (awe ratings: $M = 2.13$, $SD = 1.52$; happy ratings: $M = 2.9$, $SD = 1.66$), which was in line with our desired effect.

Discussion

We hypothesized that individuals randomly assigned to the awe-inspiring video would list a greater number of weaknesses than those in the neutral condition. Our findings did not support this hypothesis and represented results going in the opposite direction. The mean proportion of weaknesses to total traits was greater for the people assigned to the control condition than those exposed to the awe-inspiring video. Although the mean was greater for those in the neutral condition, the difference between the two groups was not statistically significant. Our results also did not replicate the findings of Stellar et al. (2018), who found a significant difference in the proportion of weaknesses between the two conditions. This discrepancy does not necessarily dispute the validity of Stellar et al. (2018) but could be attributed to our replication.

Despite our study not finding statistically significant results, it serves as a valuable stepping stone for future research by contributing to the ongoing effort to refine how humility is measured and understood. Given the longstanding challenge of developing a coherent measurement strategy for humility (Davis et al., 2010), our replication highlights the complexities of operationalizing humility and the importance of continued exploration into how emotional states like awe may influence it. Even without significant effects, studies like ours help identify methodological challenges and offer a foundation for future work to build upon and improve.

Numerous factors could have been attributed to the fact that we could not establish results that support our hypothesis, as well as the findings by Stellar et al. (2018). Most notably, several

issues posed significant threats to our construct validity. Although our initial sample was similar to the one used by Stellar et al. (2018), in the process of cleaning the data, we removed a substantial number of participants due to failure to follow instructions. Even after the data was cleaned, there were 17 entries that listed only strengths and no weaknesses, which we also could have considered a failure to follow instructions, given that the prompt asked for both strengths and weaknesses. Unfortunately, removing those 17 additional participants would have resulted in an insufficiently small sample size. Nonetheless, we reran the analysis excluding these entries and reached the same conclusion. However, the reduced sample size would not have supported generalizable results. Given that we used materials identical to those of Stellar et al. (2018), having a larger final sample could have allowed us to draw a different conclusion.

The internal validity of our replication was also a factor that might have hindered our results. The fact that the sample of participants was exclusively Illinois Institute of Technology students was systematically different from Stellar et al. (2018), who gathered a broader sample from Mechanical Turk. Additionally, our study was exclusively executed online, limiting our ability to control for external factors that might have impacted the way students filled out the survey. Distractions, skipping through the videos, and rushing through questions are serious concerns related to internal validity. Finally, the external validity of our study could also be questioned, given that our sample was solely comprised of IIT students seeking course credit, which is too narrow to generalize to a broader population. All these are considerations that, in combination with each other, could have hindered our ability to obtain the same statistically significant result as Stellar et al. (2018).

Future research could improve upon our study in several key ways. First, increasing the sample size would enhance statistical power and allow researchers to confidently exclude

participants who fail to follow instructions without compromising generalizability. Recruiting participants from more diverse populations, rather than relying solely on a single university's student population, would also improve external validity and better reflect broader demographics. Additionally, conducting the study in a controlled in-person setting could minimize distractions and ensure participants fully engaged with the video stimuli, reducing the likelihood of rushed or careless responses. Refining the humility measure is another important issue that could be improved upon. For example, altering the strengths-and-weaknesses task to prompt the participant for several single-word responses instead of the currently open-ended format. Incorporating a Likert scale for the participants to rate themselves on characteristics such as intelligence, physical ability, patience, and others could offer a more nuanced understanding of humility. Finally, future studies might explore alternative awe inductions, such as immersive virtual reality, to see whether different formats elicit stronger emotional responses and, thus, stronger effects on humility.

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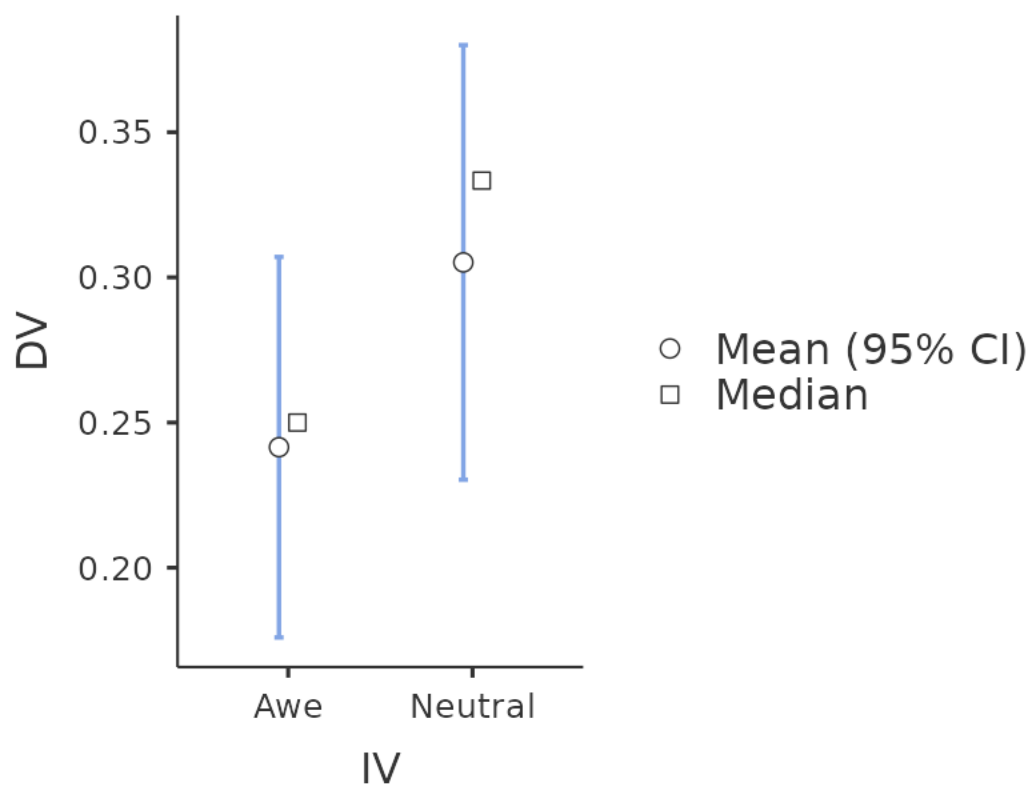
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<https://doi.org/10.1037/a0026508>

Figure 1

DV



DYNAMIC NORMS REPLICATION

DYNAMIC NORMS REPLICATION STUDY

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May 9th, 2024

Abstract

This replication study examined the effects of dynamic norm statements versus static norm statements on individuals' intentions to reduce their meat consumption. Previous research by Sparkman and Walton (2017) demonstrated that dynamic norms, messages emphasizing how individuals' behaviors are changing over time are far more effective than static norms in promoting sustainable behaviors. We conducted a between-subjects design study, 89 participants were randomly assigned to read either a dynamic or static norm statement regarding meat consumption. Participants then rated their interest in eating less meat on a 7-point Likert scale. While mean interest was higher in the dynamic norm group ($M = 3.95$, $SD = 2.13$) than in the static norm group ($M = 3.44$, $SD = 2.20$), the difference was not statistically significant, $t(87) = 1.11$, $p = .27$, $d = 0.235$. These results suggest that there is a potential trend consistent with previous findings, but fail to provide significantly conclusive evidence of dynamic norms' superiority in this context. Limitations can be seen related to sample size, survey setting, and external validity. Future research is greatly encouraged and perhaps even looking at changing different types of behavior.

DYNAMIC NORMS REPLICATION

Introduction

Humans are heavily influenced by our environment, specifically the evolving norms of society. People are continuously changing our behavior and beliefs in response to other individuals around us. Gregg Sparkman and Gregory M. Walton researched the effect of dynamic versus static norms on individuals' sustainable behavior. Dynamic norms are defined as social norms about how other individuals' behavior and attitudes are changing over time (Sparkman and Walton, 2017, 2019). Different from static norms in the way that they explain the current state of individuals behavior. Their results demonstrated that dynamic norms were significantly more influential in changing individuals behavior when they learned that other individuals behavior was actively changing.

There has been further research done around the effects of dynamic norms on various different behaviors. Dynamic norms have been used to see if they could encourage young adults to vape less. The results from that test demonstrated that “dynamic norms messages led to higher pre conformity, self -efficacy and lowered freedom threat and reactance”(Ma, H., & Reynolds-Tylus, T., 2024). Dynamic norms have been found to also help lower individuals' use of water when it was emphasized that there was an increase in water consumption behaviors across time(Mortensen et al., 2019). Furthermore, dynamic norms have also been found to better influence individuals actions than static norms in intentionally increasing the use of reusable cups(Loschelder et al., 2019). Many studies have shown that dynamic norms can significantly influence behavioral change by highlighting how societal behaviors are currently actively changing.

Dynamic norms in some cases may be seen as influential in changing individuals behavior in a positive way. Sparkman and Walton specifically looked at meat consumption.

DYNAMIC NORMS REPLICATION

Individuals that read the dynamic norms statements that stated that people are actively shifting their behaviors and eating less meat were more inclined to eat less meat. Various other studies have looked at other health and environmental behavior changes. Dynamic norms may be able to influence individuals to live healthier lifestyles if they hear that others are also changing the way in which they live. Dynamic norm statements seem to motivate individuals to align with emerging trends. Many young adults are heavily influenced by the attitudes and behaviors of peers since they are motivated and inclined to gain peer approval and to “fit in”(Ma & Reynolds-Tylus, 2024). Due to this I hypothesize that our experiment too will demonstrate similar effects as those in Sparkman and Waltons (2017) as supported by many others.

Design

The study compares the different norm statement groups using a between-subjects design. Participants were randomly assigned either a dynamic or static norm to read. There is only 1 independent variable in this study with two levels being the dynamic and static norm statements, with 1 dependent variable being measured. The static norm statement group read that 30% of Americans try to limit their meat consumption. The Dynamic nor statement group read that 30% of Americans are actively eating less meat, demonstrating behavior changed over time. This study measured how interested participants were to want to consume less meat after reading the norm statements. We gathered this data by using the likert scale and asking from 1 (not at all) to 7 (extremely), how interested they were to eat less meat.

Participants

The total participant submissions we received from our survey was 103. However, only 89 participants were used for our analysis due to not consenting, incomplete data, and unsure responses. The participants in this study were Illinois Institute of Technology (IIT) Students and

DYNAMIC NORMS REPLICATION

other individuals over the age of 18 on SONA. Of those participants in our analysis 48.1% (43) identified as males, 48.1% (43) as female, .02% (2) as non-binary and .01 (1) chose to self-describe their gender. Ages of those who participated ranged from 18 to 29, average age being 20.4 years. Looking at political views, 33.7% (30) identified as moderate, 23.6% (21) as somewhat liberal, 22.5% (20) as liberal, 10.1% (9) as very liberal, 0.06% (5) as somewhat conservative and 0.04% (4) as conservative. The racial composition included 32.6% (29) identifying as Hispanic or Latino, 20.2% (18) as East Asian or Asian American, 14.6% (13) Non-Hispanic White or Euro-American, 12.4% (11) as Black, Afro-Caribbean, or African American, 0.08% (7) as South Asian or Indian American. The remaining individuals identified as either mixed and or prefer to self identify individuals. Only 0.06% (5) of participants identified as vegan or vegetarian. As compensation to our participants all of those that completed the survey through SONA received 0.5 course credit for participation.

Materials

Our essential tool for this study was Qualtrics, this software was used in the design and development of the survey. Most importantly out of two norms statements, Static: Recent research has shown that 30% of Americans make an effort to limit their meat consumption. That means that 3 in 10 people eat less meat than they otherwise would. Dynamic: Recent research has shown that, in the last 5 years, 30% of Americans have now started to make an effort to limit their meat consumption. That means that, in recent years, 3 in 10 people have begun to eat less meat than they otherwise would. Our questionnaire also included questions about age, race, political ideologies and vegetarian and veganism.

Procedure

DYNAMIC NORMS REPLICATION

The study was accessible and taken through SONA systems. Initially, they were presented with a consent form in which they were informed of the study's purpose, risk and benefits, payments and reminded that this was voluntary and that they were allowed to withdraw at any time. After consenting, participants were then told some instructions on what the study was going to be about and how we were defining meat. Participants were then taken to either the static or dynamic norm statement. After reading, they were asked about their interest in consuming less meat as well as their perceptions on how likely they believed other individuals made an effort to consume less meat. Continuing this, participants were asked about their political ideology, gender, whether they were vegetarian or vegan and concludingly their race. To finish off the study participants read the debriefing form in which we thanked them for participating as well as disclosed our study purpose and hypothesis.

Results

We tested the hypothesis that participants that read the dynamic norm statement would be more likely to be influenced and would want to consume less meat. We did this using an independent sample *t*-test to determine whether there was an interest in reducing meat consumption between the static norms statements and dynamic norm statements. Our results did not support the hypothesis, participants were more likely to want to consume less meat in the dynamic norm statement group($M = 3.95$, $SD = 2.13$) than in the static norm statement group ($M = 3.44$, $SD = 2.20$). The difference is not statistically significant, $t(87) = 1.11$, $p = 0.27$, $d = 0.235$. The observed mean difference represents a low effect; mean meat consumption influence for those in the dynamic norm statement group was 0.235 standard deviations higher than mean meat consumption influence for those in the static norm group.

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Discussion

Our study aimed to see the exposure of dynamic norms and their effects on influencing individuals' thinking and behavior, compared to static norms. However, our results failed to reject the null hypothesis, indicating that we did not find sufficient evidence that dynamic norms had a significantly different effect. Though our findings do not support Sparkman and Waltons (2017) findings we did have some strengths and some improvements may be done in future studies.

Looking at our study's construct validity, how well our variables were manipulated and measured, both our dynamic and static norm statements were taken directly from Sparkman and Walton (2017). We measured our variables using a likert scale from 1 to 7 which may not completely represent participants intent in sustaining this behavior. As for internal validity we did use random assignment in our online study to assign participants randomly to either the static or dynamic norm statement. There were no systematic differences between the groups having equal participants going into those 2 conditions. Weaknesses are seen in our internal validity, as it was an online study. We had no control over where and when and what participants used to take the study. Perhaps in future research this weakness can be better controlled by having the study done in a lab setting. To make sure that all participants are in the same type of environment and using the same electronics to take the study with minimal to no distractions. As for external validity our results cannot be generalized. Individuals that took the study were primary Illinois Institute of technology students. Looking at our sample size of around 90 which is smaller than the original study, participants in our study are pretty split between men and women. This does not represent IITs demographic, IIT is composed primarily of a male demographic so our results cannot be generalized to the university or real world populations.

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Future research is very much supported and encouraged, perhaps looking at other variables aside from meat consumption. There has been lots of previous research but much more can be looked at in the future to see if trends are still visible. Perhaps our study may have had better results if conducted in a lab setting to help it be more controlled as well as having a larger sample size. Our study was only out and available for about 2 weeks till we gathered our data, maybe having the study out for longer would provide different results. Future search can help build on this by looking at dynamic norms and other behavioral aspects and perhaps focusing on specific populations.

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Appendix

Figure 1

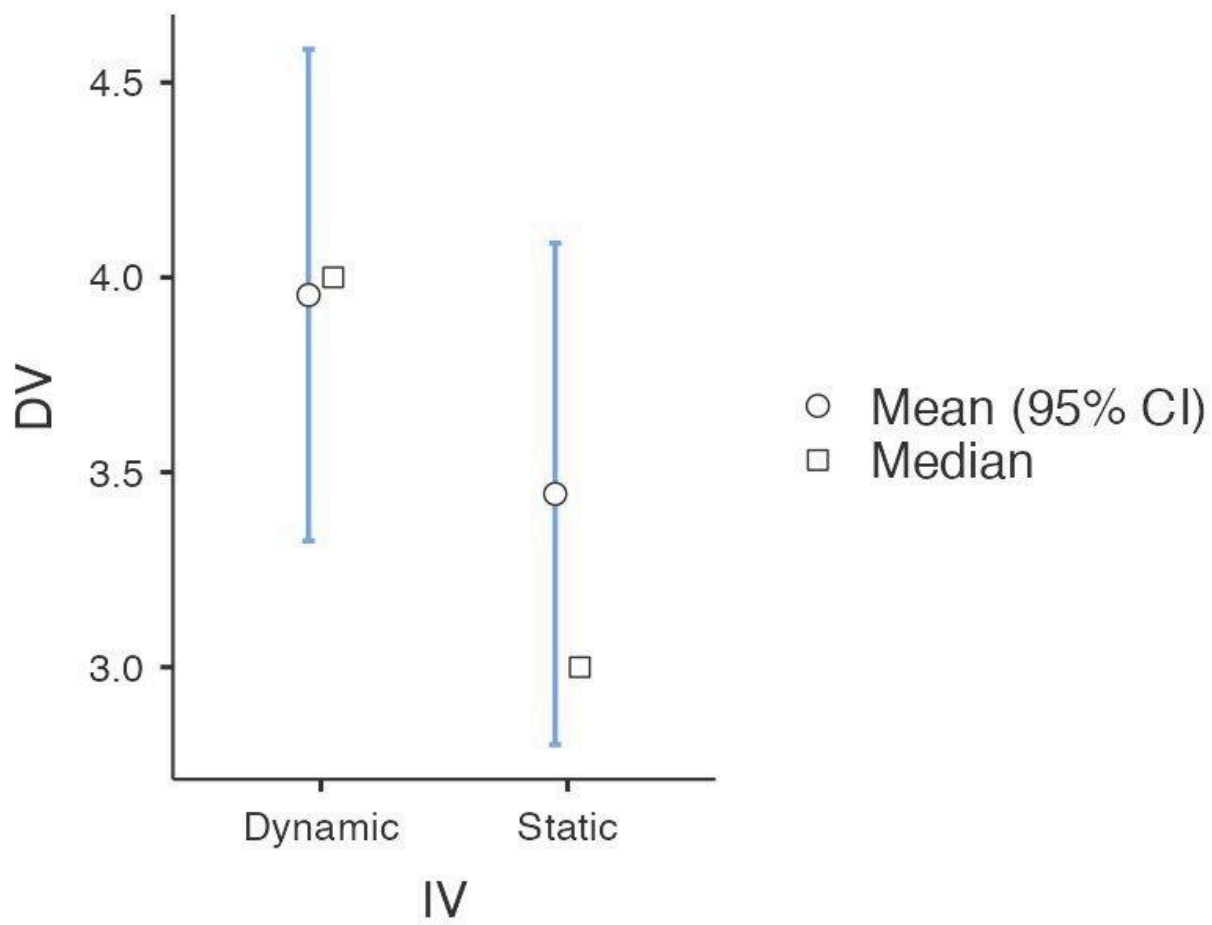


Figure 2

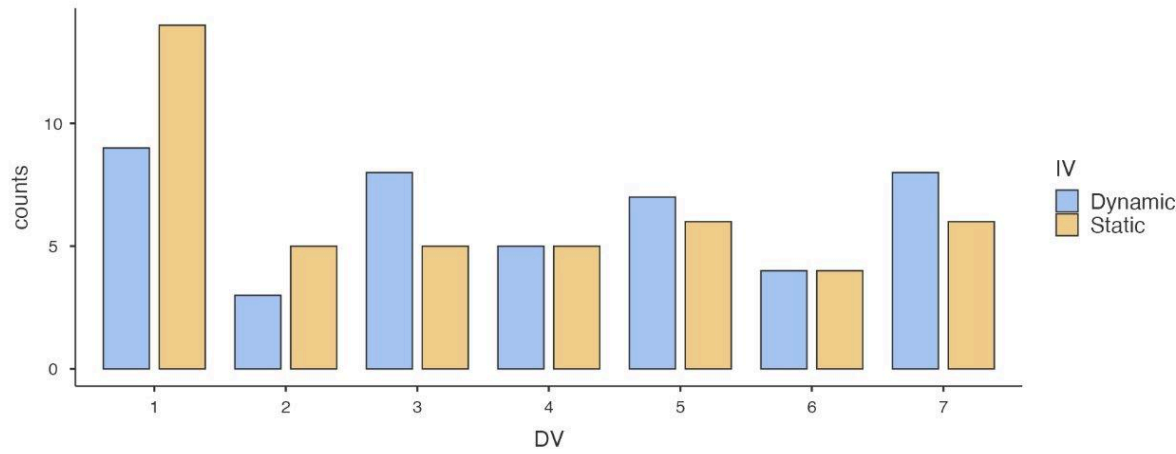
DYNAMIC NORMS REPLICATION

Descriptives		
	IV	DV
Mean	Dynamic	3.95
	Static	3.44

Plots

DV

>



**A Replication Attempt: The Impact of Dynamic and Static Norms on
Sustainable Behavior**

Student 9

Illinois Institute of Technology

PSYC204 – Research Methods in Behavioral Science

Dr. Robert Keating

May 9, 2025

Abstract

Social norms play a central role in guiding individual behavior, particularly in environmentally significant domains such as food consumption. This study investigates how the framing of normative messages—either as static norms that reflect current behaviors or dynamic norms that emphasize change over time—influences people’s willingness to reduce meat intake. We hypothesized that dynamic norms would be more effective than static norms in encouraging interest in reducing meat consumption. To test this, an online experimental study was conducted in which adult participants were randomly assigned to read either a static or dynamic norm message before reporting their interest in changing their dietary behavior. Contrary to our hypothesis, the findings revealed no meaningful difference in responses between the two groups. These results suggest that dynamic norms may not always be more persuasive than static norms in promoting sustainable behavior. The study highlights the importance of considering situational and psychological factors when designing norm-based interventions and suggests directions for future research to better understand when and why such messages are effective.

Introduction

Human behavior is often influenced by the actions and expectations of those around us. Social norms—shared beliefs about what is typical or appropriate in a given context—play a crucial role in shaping decision-making and behavior (Cialdini, Kallgren, & Reno, 1991). These norms can be categorized into static norms, which describe how many people currently engage in a particular behavior, and dynamic norms, which emphasize that a behavior is becoming more or less common over time, suggesting a shift in what people view as acceptable or typical (Capps et

al., 2022). The framing of these norms can significantly impact whether individuals adopt new behaviors or maintain existing ones. For instance, dynamic norms can be especially persuasive because they signal social change in progress, encouraging individuals to align with emerging trends. In contrast, static norms may reinforce the status quo by highlighting what is currently common. Understanding how these two types of norms function differently provides insight into how social influence shapes behavior.

This paper examines how the framing of dynamic versus static norms affects individuals' openness to reducing meat consumption, thereby contributing to the broader literature on social norm interventions and behavior change.

One key study examining the influence of dynamic norms is Sparkman & Walton (2017), which investigated whether learning about a shifting social trend could encourage behavior change. Their study found that participants who were told that an increasing number of Americans had begun reducing their meat consumption over the past five years were significantly more likely to express interest in cutting down their own meat intake. This suggests that highlighting a growing movement toward sustainable behavior may be more persuasive than simply stating how many people currently engage in it.

The relevance of these findings extends beyond individual dietary choices. In an era of increasing consumerism and environmental concerns, promoting sustainable behaviors is more important than ever. By replicating Sparkman & Walton (2017), this study aims to further explore whether dynamic norms can serve as an effective tool for encouraging pro-environmental behavior. While some research suggests that dynamic norms are more influential

than static norms, other studies have reported mixed results. For example, Mortensen et al. (2019) found that dynamic norms significantly increased participants' willingness to engage in sustainable actions, suggesting that emphasizing change over time can motivate behavioral shifts. In contrast, Göckeritz et al. (2010) observed no significant difference between static and dynamic norm conditions in promoting behavior change, raising questions about when and why these effects emerge. Similarly, Tylus et al. (2024) reported that both types of norm statements were more effective than no norm at all but showed limited differences between each other.

Despite the growing body of work in this area, it remains unclear under what conditions dynamic norms are more effective than static norms, particularly in motivating dietary behavior change. Given these conflicting findings, this study seeks to clarify the role of dynamic norms in shaping behavior and assess their potential for fostering long-term societal change.

The present research replicates Sparkman & Walton's (2017) work by examining whether participants exposed to dynamic norm messages—emphasizing a growing trend of people limiting meat consumption—will express greater interest in reducing their own meat intake compared to those exposed to static norm messages, which describe the number of people who currently limit meat consumption without suggesting change. In this online experimental study, participants were randomly assigned to view either a dynamic or static norm message about meat consumption and then completed a measure assessing their interest in reducing their meat intake. It is hypothesized that exposure to dynamic norms will lead to higher levels of interest in reducing meat consumption compared to static norms.

Method

Design

This study employed a between-subjects design, where participants were randomly assigned to one of two conditions to prevent cross-contamination between conditions. The independent variable had two levels: Static Norm and Dynamic Norm. The dependent variable was participants' self-reported interest in reducing their own meat consumption.

Participants

The study included students from the Illinois Institute of Technology who participated in exchange for course credit. They were recruited via SONA. Several participant responses were excluded during data cleaning. The initial sample size was 103 participants, and after elimination, it was 89, with 14 omitted. Some participants were removed due to incomplete survey responses or due to evidence of repeated participation. Additionally, some responses were recategorized for demographic reporting purposes, and outliers based on survey completion time were also excluded.

Of the participants, 51.42% identified as male and 43.7% identified as female, 3.86% as non-binary, and 1.02% chose to self-ascribe gender. The age range was 18 to 29, with an average age of 20.2 years. In terms of racial composition, 32% identified as Hispanic/Latino, 22% as East Asian/Asian American, 18% as Non-Hispanic White, 10% as South Asian, 13% as Black/Afro-Caribbean/African American, and 4% identified as multiracial. As for political views, 34.3% identified as moderate, 21.4% as liberal, 20.9% as somewhat liberal, 10.8% as very liberal, 6.1% as somewhat conservative, and 6.5% as conservative.

Materials and Measures

The study was conducted using an online Qualtrics survey, which participants accessed via personal electronic devices (e.g., laptop, smartphone, or tablet). The primary independent variable was norm type, with two conditions: *Static Norm* and *Dynamic Norm*. Participants were randomly assigned to either the *Static Norm* or *Dynamic Norm* group. In the static norm condition, participants read a message stating that 30% of Americans currently limit their meat consumption. In the dynamic norm condition, participants read that in the past five years, 30% of Americans have started reducing their meat intake, implying an ongoing shift in behavior. These norm statements were adapted from Sparkman and Walton (2017).

The dependent variable was measured using a single item: “*How interested are you in reducing your meat consumption?*” Responses were recorded on a 7-point scale ranging from 1 (*Not at all interested*) to 7 (*Extremely interested*), with higher scores indicating greater interest in reducing meat intake.

Additional survey items asked participants about their dietary identity (e.g., vegan, vegetarian, or neither) and typical eating habits. Demographic questions included age, gender identity, racial/ethnic background, and political ideology to enable exploratory analyses of potential moderating variables.

Procedure

Participants accessed the study through an online Qualtrics survey using their personal devices. Before beginning, they reviewed an informed consent form outlining the study’s purpose, potential risks, and their right to withdraw at any time without consequence. After

providing consent, participants were randomly assigned to one of two conditions: Static Norm or Dynamic Norm. Participants then completed the dependent measure of willingness to reduce meat consumption. Following the dependent measure, participants completed follow-up questions assessing their perceptions of how common meat reduction is among others, as well as their own meal habits. Demographic information, including age, gender, and dietary preferences, was also collected. At the conclusion of the study, participants were provided with a debriefing that explained the study's objectives and thanked them for their participation.

Result

An independent samples *t*-test was conducted to examine whether there was a difference in participants' interest in reducing meat consumption between those exposed to dynamic norms and those exposed to static norms. Participants in the dynamic norm condition ($M = 3.89$, $SD = 2.16$) reported a slightly higher interest than those in the static norm condition ($M = 3.60$, $SD = 2.19$). However, this difference was not statistically significant, $t(90) = 0.646$, $p = 0.520$. The *d*-value (effect size) was -0.133 . These results suggest that exposure to dynamic versus static norm messages did not lead to a significant change in participants' reported interest in reducing meat consumption. A figure displaying the data for both conditions is provided in the Appendix to illustrate the lack of significant difference.

Discussion

This study aimed to examine whether dynamic norms could significantly influence interest in reducing meat consumption compared to static norms. Our results showed no meaningful difference between the two groups. Participants exposed to dynamic norm messages

reported slightly more interest in reducing their meat consumption than those exposed to static norm messages, but this difference was too small to be considered reliable. These findings do not align with prior research suggesting that dynamic norms can effectively promote pro-environmental behavior, and raise questions about the broader applicability of these results.

One potential explanation for the lack of significance is that the dynamic norm messages in this study may not have been compelling enough to influence attitudes. Future research could test more engaging methods of presenting dynamic norms, such as incorporating narratives, testimonials, or visual storytelling. Additionally, the demographic homogeneity of our sample, which consisted primarily of university students, limits the generalizability of our findings. A more diverse sample, including participants of different ages, cultural backgrounds, and educational levels, could provide a more comprehensive understanding of how dynamic norms influence different populations.

Although random assignment to conditions helped control for individual differences, the relatively small sample size may have limited the statistical power of our study. Unmeasured factors, such as prior dietary habits, environmental awareness, or attitudes toward sustainability, could still have influenced participants' responses. A larger sample size could potentially detect more subtle differences between the conditions and increase the generalizability of the results.

In conclusion, while dynamic norms did not significantly influence interest in reducing meat consumption in our study, the findings suggest that normative messaging needs refinement. Future research should explore alternative methods of presenting dynamic norms, consider

additional confounding factors, and use larger and more diverse sample sizes to better assess the effectiveness of dynamic norms in promoting sustainable behaviors.

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Appendix

Figure 1

Interest In Reducing Meat Consumption by Norm Type – Static vs Dynamic

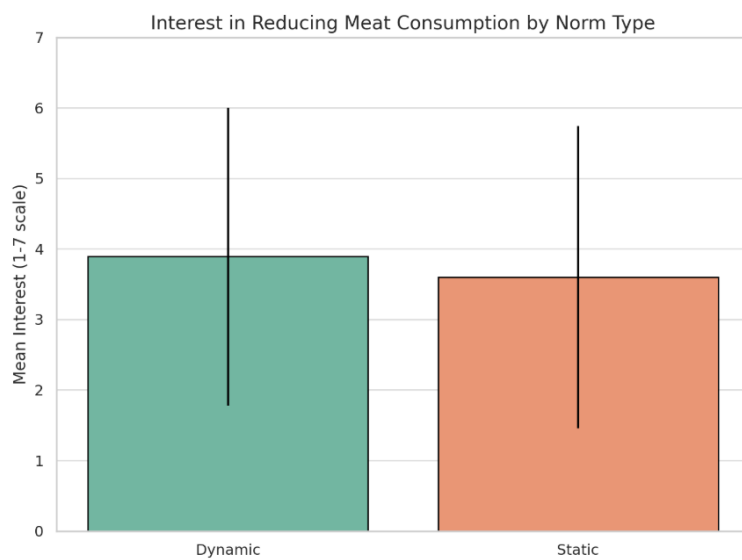


Figure 2

Independent Samples t-test

Independent Samples T-Test

Independent Samples T-Test

		Statistic	df	p
DV	Student's t	0.646	90.0	0.520

Note. $H_0: \mu_{\text{Dynamic}} = \mu_{\text{Static}}$

Group Descriptives

	Group	N	Mean	Median	SD	SE
DV	Dynamic	45	3.89	4.00	2.16	0.321
	Static	47	3.60	4.00	2.19	0.320

A Replication Study of the Effects of Awe on Humility

Student 10

Department of Psychology, Illinois Institute of Technology

PSYC 204: Research Methods in Behavioral Science

Dr. Robert Keating

May 10, 2025

Abstract

While humility is often seen as a fixed personality trait, recent research raises the question of whether it can be temporarily influenced by specific emotional experiences. Can feeling awe, an emotion that shifts attention away from the self, lead to more humble behavior? This study aimed to replicate a study conducted by Stellar et al. (2018) to investigate whether brief exposure to awe could influence an individual's humility, defined as the tendency to recognize one's own limitations relative to one's own personal strengths. This study was conducted using a posttest-only, between-subjects experimental design. Ninety-two participants were recruited through the Illinois Institute of Technology's SONA system. After data cleaning, the final sample included 68 participants. Participants were randomly assigned to view either an awe-inducing video or a neutral video. After they finished watching the video, participants had two minutes to list as many personal strengths and weaknesses as possible. Humility was measured as the ratio of weaknesses to strengths provided by the participants. We hypothesized that participants in the awe condition would display greater levels of humility than participants in the neutral condition. Our results, which showed no statistically significant difference between the two groups, were not able to confirm our hypothesis and the findings of Stellar et al. (2018). These results may have been influenced by several factors, including the uncontrolled nature of the online environment, limited diversity among participants, or potentially unclear instructions. Overall, no significant group differences were found.

Keywords: awe, humility, humble behavior

A Replication Study of the Effect of Awe on Humility

Awe is commonly defined as an emotional response to something vast, powerful, or beyond our usual level of understanding—something that makes us feel small in a meaningful way (Monroy, 2022). Humility, on the other hand, involves a recognition of one's limitations, valuing the strengths of others, and maintaining a view of the self that is accurate (Tongeren et al., 2019). While there have been numerous attempts to study awe and humility individually in the field of psychology, the direct connection between awe and humility hasn't been explored in-depth. Stellar et al. (2018) suggested that awe might actually increase humility by shifting attention away from the self and toward something greater. This study aims to replicate their research and contribute to the limited but growing body of work on how awe might influence humble behavior.

Traditionally, humility has been thought of as a stable personality trait—something individuals either possess or don't possess. This suggests that humility is categorical, but individual differences more realistically vary on a continuum. There has been a shift in this view as more recent research is starting to frame humility as a state that can change based on context (Kruse et al., 2017). Awe might be one of potential experience which can influence humility. Awe tends to promote feelings of connection while reducing self-focus, both of which could set the tone for more humble thinking and behavior (Piff et al., 2015). Experiencing awe has been shown to diminish self-focus and promote prosocial behaviors, which are associated with enhanced psychological well-being. For instance, awe can lead to a "small self," where individuals feel less self-important and more connected to others, which helps to foster collective engagement and moral expansiveness. (Song et al., 2023). This boost in well-being may be linked to a mindset of humility, which increases as individuals become more aware of their place

in a larger context and more receptive to others' viewpoints (Rowatt et al., 2006). All of this leads to the question of whether experiencing awe—even for a brief period of time—could lead someone to act or think more humbly, at least in the short term.

In this study, we test if exposure only to awe-inducing stimuli results in more humble behavior than exposure to neutral stimuli. Participants viewed either an awe-inducing or neutral video. Afterwards, they were asked to describe their personal strengths and weaknesses, which were measured afterwards to determine their level of humility. With awe as the independent variable and humility as the dependent variable, we aim to determine whether awe can influence humility. We predict that participants who are in the awe condition will exhibit greater levels of humility than those in the neutral condition, which aligns with the findings of Stellar et al. (2018). In doing so, we hope to fill a potential gap of information about the relationship between awe and humility and meaningfully contribute to ongoing research about the relationship.

Method

Design

A posttest-only, between-subjects experimental design was utilized in this study. The experience of awe was the independent variable of the study and was manipulated by assigning participants to watch either an awe-inducing video or a neutral video. Participant humility was the dependent variable of the study and assessed as participants' self-reported strengths and weaknesses.

Participants

The Illinois Institute of Technology's SONA system was used to recruit 92 students, who were then paid with course credit for their participation in the study. Sixty-eight participants were included in the final dataset after 24 people were eliminated for not finishing the study, not following instructions, or failing the manipulation checks. The final sample was composed of individuals aged 18 to 29 ($M = 20.13$, $SD = 1.98$). Among these individuals, 49.25% identified as male, 47.76% identified as female, and 4.48% identified as non-binary. Regarding racial and ethnic background, 34.33% of participants identified as Hispanic or Latino, 22.39% identified as non-Hispanic White or Euro-American, 19.40% identified as East Asian or Asian American, 13.43% identified as Black, Afro-Caribbean, or African American, 5.97% identified as South Asian or Indian American, 4.48% self-identified with another racial identity, and 2.99% as Middle Eastern or Arab American.

Materials and Measures

The study was completed through Qualtrics, which is accessible using common electronic devices such as a computer, phone, or tablet. Manipulation of the awe variable involved participants viewing one of two two-minute-long YouTube videos. Participants assigned to the awe condition watched an awe-inducing video intended to showcase the vastness of the universe, slowly zooming out from Earth to reveal the planets and stars of the universe (<https://www.youtube.com/watch?v=ZoXqi-VDhws>). Participants assigned to the neutral condition watched a video outlining the instructions for building a fence, a task that typically evokes minimal emotional response (<https://www.youtube.com/watch?v=XYGzuPpw50g>).

Humility was operationalized by giving participants 2 minutes for the participants in which they were asked to list their strengths and weaknesses as if they were telling them to someone they had just met. After the participants had listed their strengths and weaknesses, two

independent coders counted the number of strengths and weaknesses listed. Then, the total number of weaknesses was divided by the total number of strengths and weaknesses to generate a proportion score between 0 and 1. A higher score was an indication that there was a greater proportion of weaknesses listed by the participant relative to strengths, which was a sign of greater levels of humility. Additionally, the participants used a 7-point scale, ranging from 1=*not at all* to 7=*very much*, to measure how strongly they felt awe, fear, wonder, happiness, and amazement in response to the videos they watched. Asking the participants questions about their racial and ethnic identity, age, and gender, as well as giving them the option to select multiple categories or self-identify, was done as a means of obtaining demographic information about the participants.

Procedure

An online Qualtrics survey was filled out by participants to complete this study. Once participants agreed to the informed consent form, they were randomly assigned to either the awe or neutral video condition. After being randomly assigned to either condition, they watched the video corresponding to their condition and were asked to write down their strengths and weaknesses for two minutes afterwards, with a 2-minute timer being present on the screen. Once they had written down their strengths and weaknesses, the participants rated their emotional responses to the video on the five scale items. The participants then completed a set of manipulation check questions, ensuring they had understood the content of the study and the video and that the video played correctly. The survey concluded with a section to collect demographic data about the participants, including their age, gender, and race/ethnicity. After they had completed the survey participants were thanked for their participation and debriefed at the end of the study.

Results

We hypothesized that participants who watched the awe-inspiring video would demonstrate greater levels of humility—operationalized as listing a higher proportion of weaknesses compared to strengths—than participants who watched the neutral video. To test our hypothesis, we conducted an independent sample *t*-test comparing the proportion of weaknesses to strengths in each condition. As demonstrated in Figure 1, the mean proportion in the awe condition ($M = 0.38$, 95% CI [0.25, 0.51], Median = 0.25) was lower than the mean proportion in the neutral condition ($M = 0.666$, 95% CI [0.46, 0.87], Median = 0.67). However, this difference was not statistically significant, $t(66) = -1.59$, $p = .116$, $d = -0.395$.

Discussion

This study aimed to replicate Stellar et al. (2018) and test whether exposure to an awe-inducing video would lead participants to display more humility than a neutral video, as measured by the proportion of weaknesses to strengths listed. We hypothesized that individuals exposed to an awe-evoking experience would be more likely to acknowledge their own weaknesses, resulting in a higher proportion of weaknesses to strengths compared to those who viewed a neutral video. However, participants in the awe condition actually reported fewer weaknesses on average than those in the neutral condition. While the numerical difference suggests a trend, the result was not statistically significant.

These results contradict the findings of Stellar et al. (2018), which did find a statistically-significant effect. One possible explanation for this difference in results could be related to how we implemented our procedure. Several participants had to be removed due to not following the instructions and only listing strengths, which could be a potential indicator of

possible confusion about what was being asked. This could have compromised the effectiveness of our humility measure and reduced the accuracy of our data, as well as our construct validity. Because we conducted our study online, we also lacked control over participants' environments. Potential distractions, variation in viewing conditions between the participants, or technical issues are all potential confounding variables which may have influenced how engaged the participants were with the video they were assigned to watch. Given this, we could have improved the internal validity of our experiment by controlling for the participants' environment to make sure that it does not influence their responses in any way. Additionally, since our sample consisted entirely of college students seeking credit, the generalizability of these findings is limited. This demographic could be expanded to improve the external validity of our study, as it is representative of only one segment of the population which makes it hard to draw conclusions about broader populations.

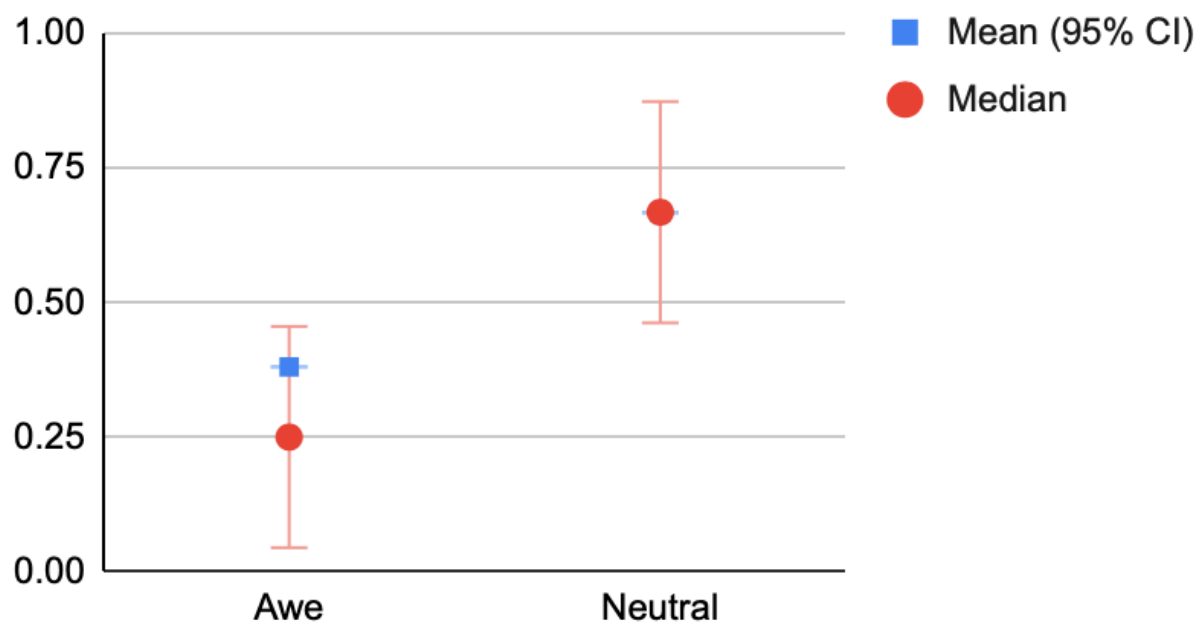
Potential ways to improve future research in this area may be conducting sessions in a controlled environment, providing greater clarity to participants about the task instructions corresponding to their condition, and using a more comprehensive measure to assess levels of humility. While our current measure—comparing the number of weaknesses listed by participants to the number of strengths—offers a basic indicator, humility is a complex construct that may be better understood and measured with additional qualitative and psychometric tools. Making these adjustments can set the tone for further clarification on whether a relationship exists between awe and humility but undetected in our design, or whether this relationship is genuinely absent in this context.

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Figure 1.

Mean Proportion of Weaknesses to Strengths



The Influence of Dynamic Norm Messaging on Willingness to Reduce Meat Consumption: A Replication Study

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PSYC 204: Research Methods

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May 9th, 2025

Abstract

This project examines the initial experiments from Sparkman and Walton's (2017) study on dynamic norms and their influence on sustainable behavior. Experiment 1 demonstrated that exposing individuals to information about increasing trends in meatless eating led to greater interest in reducing meat consumption, even when static norms favored meat consumption. These findings suggest that dynamic norms can effectively promote sustainable behaviors by leveraging perceptions of social change. We hypothesize that our results will also support Sparkman and Walton's findings. We have currently started our data collection process and are going to start reviewing it.

Introduction

Social norms, or the unspoken rules about what is normal and acceptable in a given group, exert a powerful influence on attitudes and behavior (Cialdini, Kallgren, & Reno, 1991). Norms are scripts for behavior, simplifying the negotiation of complex social worlds by signaling what other individuals do (descriptive norms) and what other individuals approve of (injunctive norms). They affect decision in diverse areas, from conservation to food selection (Goldstein, Cialdini, & Griskevicius, 2008; Robinson et al., 2014).

Traditional normative messaging generally emphasizes static norms—messages that convey what most people currently do. More recent research, however, shows that dynamic norms, which express how behaviors are progressing over time, can be even more successful at promoting change, especially when a new behavior has not yet become common (Sparkman & Walton, 2017). Rather than highlighting what is normative, dynamic norms focus on trends in behavior adoption, for instance, that "more and more people are starting to." This can generate a sense of momentum, social change, and future conformity that motivates individuals to conform to emerging trends (Mortensen et al., 2019).

Sparkman and Walton (2017) found that dynamic norm messages were more effective at increasing intentions to reduce meat consumption than static or control messages. In their study, participants who read that increasingly more Americans were reducing meat consumption said they were more interested in trying a vegetarian diet. Their findings suggest that dynamic norms may involve different psychological processes—such as anticipated future majority status or personal identity alignment—that make them particularly persuasive.

Follow-up studies have built on this research, extending the application of dynamic norm messaging to areas such as water conservation (Sparkman et al., 2021), pro-environmental

behaviors (Schultz et al., 2018), and diversity and inclusion (Chang et al., 2020). This work consistently shows that dynamic norms can facilitate behavior change, even in groups where the new behavior is not yet normative. However, some of the effects are mixed, and there are questions about the broad applicability of these effects to various populations and message contexts.

The current study was a replication attempt of Sparkman and Walton's (2017) study to examine the effectiveness of dynamic versus static normative messages on attitudes towards meat consumption. Participants were randomly assigned to read either a dynamic norm, static norm, or control message and then completed a series of questions about their intentions to reduce meat consumption. The current study aimed to establish whether the effect of dynamic norm messaging in the initial study could be replicated with a new sample.

We hypothesized that participants exposed to the dynamic norm message would report greater intentions to reduce meat consumption than participants in the static norm or control conditions.

Methods

Design

The present study used a between-subjects experimental design to examine whether dynamic norm messages influenced attitudes and intentions to reduce meat consumption. Participants were randomly allocated to receive either a control message about general dietary behavior or a dynamic norm message that indicated that increasingly more people are eating less meat. Self-reported intention to reduce meat consumption was the primary dependent variable and was assessed via Likert-scale ratings.

Participants

96 participants were recruited from the Illinois Institute of Technology (IIT) through convenience sampling. Participation was anonymous and voluntary, and informed consent was received from all participants prior to the study. Five participants were excluded because they gave incomplete responses or missing data on one or more key variables (e.g., condition assignment or dependent measure), resulting in a final sample of 91 participants. The sample was representative and diverse in demographics of the student population. Mean age was 20.5 years (range: 18–29 years). For gender, 47.9% male, 46.8% female, 4.2% nonbinary, and 1% preferred not to respond. For political orientation, answers were distributed across the political spectrum, with the majority marking themselves as moderate (scale midpoint). A vast majority (94.8%) of participants were non-vegetarian, and 5.2% were vegetarian.

Materials and Measures

The major experimental manipulation was exposing participants to a short paragraph at the start of the survey. Control participants were given a neutral statement about recent diet tendencies, whereas dynamic norm participants were exposed to the fact that "more and more people are starting to cut down on meat consumption." The wording in this case was borrowed from Sparkman and Walton (2017) and designed to emphasize a social tendency toward behavior reduction.

After participants read the required message, they completed a self-report attitude and behavioral intention scale measuring attitudes and intentions to minimize meat consumption as a dependent variable. The scale consisted of Likert-type items (e.g., "I am willing to decrease my meat consumption"), responded to on a 7-point scale (1 = strongly disagree, 7 = strongly agree). Participants were also asked demographic questions, including age, sex, political belief,

vegetarian, and how often they consume meat. The questionnaire was hosted and completed via Qualtrics.

Procedure

Participants entered the study through a secure Qualtrics link distributed via class communication. After informed consent, they were assigned randomly to one of the two message conditions. Participants then completed the attitude and intention questions followed by demographic items. The survey took around 5–7 minutes. Participants were then debriefed at the end. All procedures conform to ethical standards for student research.

Results

We examined the hypothesis that subjects who received a dynamic norm message would indicate greater willingness to reduce meat intake than subjects who received a static norm message. An independent samples *t*-test was conducted with willingness to reduce meat intake as the dependent variable. Participants in the dynamic norm condition (IV2) indicated a greater mean willingness to reduce meat consumption ($M = 4.73$, $SD = 1.92$) compared to the participants in the static norm condition (IV1; $M = 4.47$, $SD = 1.91$). However, the difference was not significant, $t(89) = -0.63$, $p = .530$, $d = -0.13$.

The relatively small negative effect size (Cohen's $d = -0.13$) only suggests the direction of calculation but not the actual pattern of means. On average, as per the hypothesis, the dynamic norm group showed higher willingness, but an insignificant and effect magnitude-wise insignificant.

Two of the experimental variables were also examined: weekly dinners that included meat and the number of individuals in the participant's social environment who customarily consume meat. On neither did conditions differ." The subjects in the static norm condition

yielded a minimal set of more meat-containing meals ($M = 10.40$, $SD = 5.08$) as opposed to that of the dynamic norm condition ($M = 10.17$, $SD = 4.77$), $t(89) = 0.22$, $p = .825$. In the same way, the static norm participants estimated fewer meat-eaters in their groups ($M = 22.33$, $SD = 10.31$) than those in the dynamic norm group ($M = 26.90$, $SD = 14.75$), but the difference was again nonsignificant, $t(89) = -1.33$, $p = .188$.

Discussion

This study queried whether exposure to a dynamic normative message would enhance motivation to reduce meat consumption in comparison to a static normative message. We anticipated that participants in the dynamic norm condition—those who read that "more and more people are now cutting down on their meat"—would show higher willingness to reduce meat intake. While the dynamic norm group did score higher on average than the static norm group, the difference was small and not statistically significant. The findings therefore were unable to replicate the core effect demonstrated by Sparkman and Walton (2017).

These null findings are intriguing inasmuch as they run contrary to prior research suggesting that dynamic norms are especially effective where a behavior is still uncommon but rising in popularity. Sparkman and Walton (2017) demonstrated that dynamic norms could lead to positive change in intentions to reduce meat consumption, suggesting that the messages would trigger social progress, anticipated majority action, or identity-congruent alignment with evolving norms. As predicted, our results failed to find statistical support for such an effect. Perhaps participants failed to process or attend to the message. The brevity of the manipulation, combined with the self-administered internet format, may have reduced psychological

engagement. Participants might have been distracted, fatigued, or less motivated to consider the content of the messages, particularly in the low-stakes study situation.

Several methodological limitations may have influenced the results. Firstly, construct validity was reduced since there was no manipulation check. Without a guarantee that participants will subjectively experience the message to be dynamic or static, one cannot consistently assign observed effects—or failures to effect—to the deliberate manipulation. Moreover, the dependent measure was restricted to attitudinal intent, rather than behavior per se. Attitudes are sometimes less sensitive to subtle influences, and earlier research (such as Sparkman and Walton's second experiment) showed more robust effects when using behavioral indicators, such as food choices, as measures.

Internal validity concerns also should be addressed. Although participants were randomly assigned to conditions, some individual difference variables—such as eating behavior, moral identity, environmental concern, or health motivation—were not assessed. These variables potentially might moderate the norm-based message effect, in that their uncontrolled variance would have obscured an underlying effect if present. Future replications need to assess these moderators to clarify under what conditions and for whom dynamic norms work.

In addition, there was limited external validity. Our sample of 91 students from one university is not generalizable to the wider population. College students could vary in baseline meat consumption attitudes, social conformity, or exposure to climate change-related dietary messaging. Furthermore, as more public conversation has taken place about plant-based diets, most participants might already have strong opinions, making them less open to normative influence. Replications within more politically varied, older, or more heterogeneous samples may produce different results.

The statistical design of the study may also have constrained our ability to detect an actual effect. While an independent samples t-test was appropriate for comparing two groups, the unequal sample sizes within conditions (static norm: $n = 65$; dynamic norm: $n = 26$) may have attenuated statistical power. In addition, the study was likely underpowered to detect small effects, which are common in social psychological interventions. Future studies should have larger, balanced samples and investigate other statistical approaches (e.g., regression with covariates or ANOVA) that can accommodate potential confounds.

Beyond statistical concerns, framing and content of the normative messages may matter. Sparkman and Walton's first dynamic message emphasized that people were "beginning to" change, rather than stating that many already were. If the language of our message did not approximate this nuance—or if participants interpreted the trend as trivial—it could have watered down message credibility or effect. Recent work on norm salience shows that combining descriptive and injunctive components (e.g., "more and more people are beginning to change, and most people think that's a good thing") can enhance message effectiveness.

Future research needs to overcome these limitations in the following specific ways. First, dynamic norm manipulations should include manipulation checks to confirm message receipt. Second, behavior and attitudes need to be both assessed in research—such as by allowing respondents to receive a food option or donation option after seeing the message. Third, researchers must explore moderators such as political ideology, moral identity, or environmentalism to determine variations in response across people. Finally, variation in message presentation (e.g., video, peer messages, or interactive sites) can maximize salience and effectiveness.

Despite null findings, the study is one of a growing body of research that explores social influence and sustainability messages. Replications, even those with non-significant findings, serve to refine theoretical models and establish boundaries of psychological effects. Our findings suggest that dynamic norms do not always alter attitudes across all contexts and demonstrate the importance of tailoring message content, delivery modality, and sample population for future interventions.

Overall, while we were unable to match the strong effects of Sparkman and Walton (2017), our study gives key evidence for when dynamic norms can or cannot work effectively. Further research into this technique is still beneficial to informing public health and environmental behavior change interventions

Appendix

Table 1: summarizes group means, standard deviations, and statistical tests.

Measure	Static Norm Mean (IV1)	Dynamic Norm Mean (IV2)	Static <i>SD</i>	Dynamic <i>SD</i>	<i>t</i> -value	<i>p</i> -value
Willingness to Reduce Meat Consumption (DV)	4.47	4.73	1.91	1.92	-0.63	0.53
Estimated Weekly Meat-Contain ing Meals	10.4	10.17	5.08	4.77	0.22	0.825
Estimated Number of Meat-Eating People in Social Circle	22.33	26.9	10.31	14.75	-1.33	0.188

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EXPLORING THE
RELATIONSHIPS BETWEEN
AWE & HUMILITY - A
REPLICATION STUDY

Exploring the Relationships Between Awe & Humility - A Replication Study

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Abstract

Within the range of human emotion, awe and humility represent deeper feelings in regards to a person's sense of self. Finding the dynamic that exists between these two vital emotions can help us to better understand them. This paper is a replication of the Stellar et al. (2018) study that aims to use video viewing as an element to study the relationship between awe and humility. By watching a video that induces a sense of awe, a hypothesis emerged: Increased feelings of awe may cause participants to show an increase in humility and report fewer of their strengths. 92 participants from IIT were recruited via the SONA platform to participate in the study. Ultimately, the replication failed, and the null hypothesis could not be rejected; a small sample size and open-ended questions may have contributed to this. However, this study still contributes to the discourse surrounding the relationship between the two emotions by serving as another data point. Further research and improved replications are needed to get a clearer picture.

Keywords: Awe, humility, replication

Exploring the Relationships Between Awe & Humility - A Replication Study

Living in day-to-day life can make us as humans lose sight of the vastness of the world around us and what lies beyond the scope of our daily thinking. Our frame of reference can become much smaller as we define ourselves by the immediate environments that surround us. However, it is important to remember that what surrounds us on a daily basis is just that, our frame of reference. When reminded of how minute we actually are as individuals in comparison to the world and greater universe, the feeling of awe can manifest (Stellar, et al., 2018). Our self-concept can begin to be put into question when looking at ourselves from above versus from ground-level as is typical. It is then possible that the perception of self may change depending on how much awe someone is experiencing, which translates into measurable humility¹. Viewing oneself from a greater context may automatically bring the tendency to feel more humble, as it may become harder to brag about or show off personal accomplishments that can stand out in a more grand frame of reference. The scope of this paper is to explore the relationship between awe and humility, with the goal of deepening our understanding of how these two emotions may work hand-in-hand.

Literature Review

Previous replications of Stellar et al., 2018 and similar studies looking into the relationship between awe and humility have been performed by other researchers with mixed results. Bertrand et al.'s 2024 replication stayed true to the original study by attempting to measure awe and humility based on self-report questions after watching a video clip. The questions were designed to measure the amounts of strengths and weaknesses that a participant reports. Bertrand et al. were not able to replicate the results of Stellar et al.'s study, and as such, a significant correlation between the two emotions was not found. Their replication notes that participants in the awe condition reported a greater number of both strengths and weaknesses. This differs from the results detailed in the original study², which revealed that the participants who watched the awe-inducing videos reported far less strengths, while reporting a similar amount of weaknesses compared to the neutral control group. This discrepancy has led to some uncertainty regarding ways in which awe and humility correlate.

Another study involving awe is Stancato et al., 2021. This study focuses more on the possible relationship between awe and the sense of conviction, and it too used Stellar et al.'s study as a springboard for the possibility of conviction being an emotion linked to awe. This study used writing prompts to measure awe, pride and neutral inductions. Ultimately, this study was unable to prove a significant connection between awe and conviction. One potential issue that may be at play is that awe as an emotion can be facilitated by a very wide range of triggers³, and these triggers may vary for different people. It cannot be assumed that one type of stimulus will be enough to increase awe across all members of the population. Given the variance in how people interpret and process emotion, visual or written stimuli may not be all-encompassing

facilitators of awe for everyone. This may prove to be a limitation when it comes to testing the relationships between these kinds of complex and deeply personal emotions.

Method and Procedure

Our replication study follows the procedure set by Stellar et al., and as such, we sought to answer the same basic hypothesis: An increase in awe experienced by a person can lead to greater feelings of humility and self-diminishment¹. In order to study this hypothesis, a thorough method was followed. Using the SONA platform, 92 students between the ages of 18-29 from IIT were recruited. Of these 92 participants, 24 were removed from the pool due to incomplete/incoherent responses or insufficient participation. One participant was removed for knowing the hypothesis before the replication began. The final selection included 67 participants who were compensated for their participation with SONA credits. The participants were sent an online Qualtrics survey link to complete in an environment of their choosing. First, the participants were randomly assigned by the platform to either watch a neutral video about how to build a brick wall (control group), or one featuring imagery of Earth and the surrounding universe, inspiring awe (experimental group). After watching the video (independent variable), the participants were served a series of questions regarding the magnitude of emotions they felt from a list after watching the video (dependent variable). The numerical range for acceptable responses was 1 (none) to 7 (high). In addition, written questions asking about the participant's personal strengths and weaknesses were also asked. A two-minute timer was made visible on the screen during the duration of this section. After the surveys were completed and the participation

window closed, the collected data was downloaded, converted and run through Jamovi for detailed data analysis.

Results

Of the 92 participants originally recruited for the replication, 67 participants were ultimately selected for the data analysis. This is lower than the amount of participants that were selected in Stellar et al.'s study (85)¹, and also lower than the amount of participants selected in Bertrand et al.'s replication (94)². After processing the data through Jamovi, the relationship that is at the focus of the hypothesis was put into question, as the replication had failed.

Taking a look at the descriptive analysis results in **Figure 1** allows us to lay the foundation for understanding the situation at hand. IV1 is the neutral video group, and has $N = 36$ participants. The mean of IV1's reported weakness ratio is 0.237, with the median equaling 0.250. This indicates a nearly symmetrical distribution. The standard deviation and standard error values were calculated to be $SD = 0.200$ and $SE = 0.0333$ respectively. IV2 is the awe-inducing video group, and has $N = 29$ participants. The mean of IV2's reported weakness ratio is 0.282, with the median equaling 0.333. This indicates a slight skew in the distribution. The standard deviation and standard error values were calculated to be $SD = 0.205$ and $SE = 0.0381$ respectively. This data can be seen pictographically in **Figure 2**, which also shows a 95% confidence interval about both means. Returning to **Figure 1**, it also includes a section containing an independent samples t-test conducted on the collected data. It shows that there was no significant difference across the two video conditions. Upon seeing that *Student's t* = -0.891, it was determined that the IV1 group experienced a lower weakness ratio than the IV2 group. The degrees of freedom were calculated by first counting the number of participants in each

group, then using the formula $df = N_1 + N_2 - 2$. In this case, $df = 63$. For the p -value, we consider $p < 0.05$ to be an indicator of statistical significance. However, this replication yielded $p = 0.376$, which far exceeds our limit for statistical significance. Finally, we observed that *Cohen's d* = -0.222, indicating a small effect size, with IV1 having a lower mean than IV2. In essence, the data presented in this figure shows that the attempted awe induction did not have a measured significant effect on the humility metric.

Discussion

The goal of this replication was to repeat the procedures observed in Stellar et al.'s study as a means to determine whether their results could be achieved in subsequent trials. The original study hypothesized that increased feelings of awe would be associated with increased humility. The results of their study showed a connection between these two emotions. However, our replication did not produce the same results, and has led us to backtrack on the hypothesis that started the study. Bertrand, et al. also failed to replicate the results of the original study. Our data showed a high p -value, indicating low statistical significance. The observed overlap in the confidence interval also puts the validity of the results into question. The bottom line when it comes to the data we collected is that no clear correlation could be established between awe and humility in this sample.

The limited sample size may be partially to blame, as could the limitations of selecting a purely academic crowd and potentially open-ended survey questions. There is, however, another element that should be considered. Awe is a highly personal sensation that can vary from one individual to another. It operates by shifting parts of your mental schema past its usual bounds,

and the accommodation that follows is what gets defined as a sense of awe⁴. The problem with measuring awe, then, is that everyone's bounds for what is considered normal cannot reasonably be the same. In addition, the types of stimuli that can trigger a person's sense of awe can vary as well. What these statements ultimately boil down to is the possibility that a single video being used as an all-encompassing awe induction method may not be fully effective. Some individuals may find awe through spoken or written word, or from performative arts instead of the space visualization video presented in the study. If further replications or related studies are to take place, adding additional pieces of media to spark awe may help to make the emotion more accessible to the researchers. Accounting for innate human individuality may lead to gains in terms of statistical significance down the road, and perhaps bring research to a stronger conclusion as to whether or not awe and humility work hand-in-hand.

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Figures

Independent Samples T-Test

		Statistic	df	p		Effect Size
DV- weakness_ratio	Student's t	-0.891	63.0	0.376	Cohen's d	-0.222

Note. $H_a: \mu_{IV1} \neq \mu_{IV2}$

Group Descriptives

	Group	N	Mean	Median	SD	SE
DV- weakness_ratio	IV1	36	0.237	0.250	0.200	0.0333
	IV2	29	0.282	0.333	0.205	0.0381

Figure 1 - Independent Samples T-Test and Group Descriptives

DV- weakness_ratio

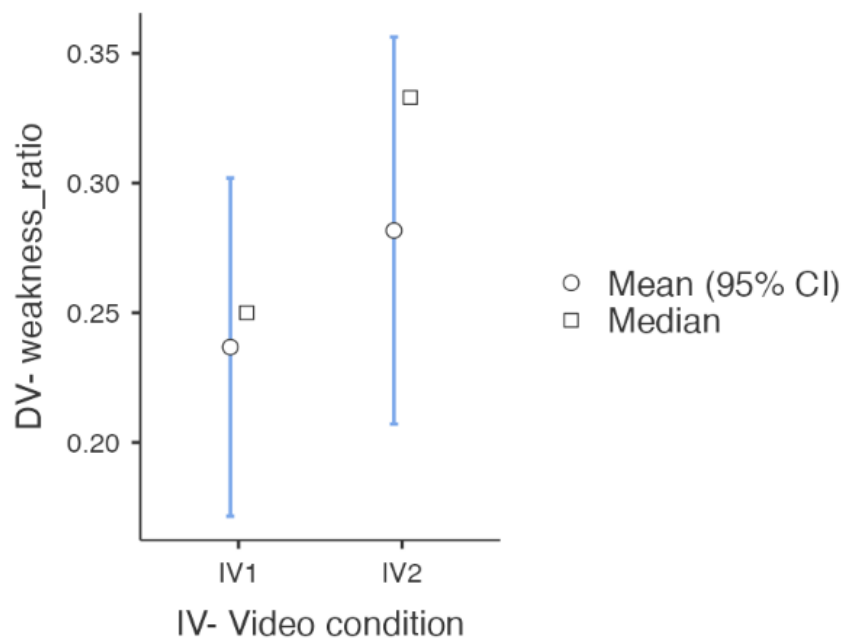


Figure 2 - Dependent Variable Graph

Awe and Humility: An *Awesome* Replication Study

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PSYC 204: Research Methods of Behavioral Science

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Abstract

Awe is a complex emotion that has been a part of many studies. One prominent study has shown that awe may be able to be affected by something such as humility. In the current study, we attempt to replicate the results of this study. The study is by Stellar et al. (2018). We hypothesized that participants who were exposed to feelings of awe would demonstrate more humble behavior than those in a neutral condition. 76 students were randomly assigned either an awe-inspiring video or a neutral video to watch. They were then asked to list their strengths and weaknesses, which were tested to determine humility. In the current study, participants in the awe condition did not exhibit significantly higher levels of humility than those in the neutral condition. This did not confirm our own hypothesis, or the findings of the original study, Stellar et al. (2018). This may have had many confounding issues, including limited sample size and lack of outside control.

Introduction

In psychology, the complex emotion of awe is characterized by feelings of wonder and surprise (Monroy & Keltner, 2023). It has had many definitions, including the idea that awe happens when a person encounters something that is beyond one's current frame of reference (Keltner & Haidt, 2003). Humility, which has been considered a virtue that has a positive effect on people's development as it creates the acceptance that a person is no better or worse than others (Tsatisis, 2023). It also has been defined as being a realistic assessment of a person's contribution versus the contribution of others and any other additional things that helped them (Hendijani et al., 2019). Few studies have examined how awe and humility are related. One recent exception by Stellar et al. (2018) found that momentary experiences of awe induced by a video stimulus were effective at creating instances of humility in the participants (Keating, 2025). The current study aims to replicate the work of Stellar et al. (2018) to potentially learn more about this not very researched connection.

In research, interest in the concept of awe increased following the work of Keltner and Haidt (2003). This research presented ways to approach awe as a concept. One was the idea of perceived vastness. The other was a need for accommodation. This is defined as the inability to perceive the experience while in a specific mental state. An example of a study of awe and another experience is seen in Arcangeli et al. (2020). This article is about the relationship between awe and the experience of the sublime. Another article by Ruberton et al. (2016), does discuss the link between humility and awe. However, they discuss boosting the state of humility using not only awe, but gratitude and self-affirmation as well. However, very few known studies examined the direct effect of awe on humility. Stellar et al. (2018) attempted to address this gap. So, by replicating the work of Stellar et al. (2018), we hope to fill in some more of the missing

gaps or at least reinforce the past knowledge. When looking for literature establishing a connection, Stellar et al. (2018) is the main article that shows up. Stellar et al. (2018) tested to find if introducing people to experiences of awe would induce humility. To do this, they provided a survey where some participants were selected to watch an awe-inducing video. After, they had two minutes to write down strengths and weaknesses. After, they were asked a few base-line questions before the survey ended. Based on the survey responses, they determined that the awe-inducing video led to more humility as represented by a higher number of weaknesses listed compared to strengths. To find other articles testing the link, one must search through articles discussing only one of the variables that we wanted to test. Humility has recently, however, been shown to influence the effort of self-reflection (Hendijani et al. 2019). Awe has also been shown to influence self-reflection. This goes to show that there should theoretically be some link between the two.

In this study, we replicated the study of Stellar et al. (2018) to determine whether participants who viewed the videos had a higher level of humility afterwards. Each participant was asked to self-report their strengths and weaknesses after watching a video provided to them. The independent variable, awe, was manipulated by either showing the participant an awe-eliciting video or a neutral video. The dependent variable then became humility. We hypothesized that the participants who were exposed to the awe-eliciting video would demonstrate higher levels of humility.

Method

Design

A simple posttest-only experiment was conducted with a between-subjects design. The manipulated variable was the experience of awe, created by showing one of two videos. The dependent, or measured, variable was the participant's humility.

Participants

Participants were recruited through Illinois Institute of Technology's SONA system, where they received course credit. There were 92 students who participated in the study, which was obtained through a SONA-provided link that gave them access to the online survey for the study. There were 33 participants whose data was excluded out of the 92. This was due to failing manipulation checks, failing to follow instructions, or not completing the survey. This left 59 individuals. They ranged in age from 18 to 26 ($M = 20.04$, $SD = 1.78$), with the majority identifying as women (49.1%), followed by those identifying as men (47.4%), followed by those identifying as non-binary (3.5%). 35.7% of participants identified as Hispanic or Latino, 19.6% of participants identified as Non-Hispanic White or Euro-American, 16.1% of participants identified as Black, Afro-Caribbean, or African American, 16.1% of participants identified as East Asian or Asian American, and 12.5% of participants identified as other racial identities.

Materials

Awe was the manipulated variable in this study. It was conducted by using two different 2-minute-long videos from YouTube. Participants in neutral conditions were assigned a video that provided instructions on building a fence. This video was shown to produce low levels of all emotions in the original study (Steller et al., 2018). Participants in the awe condition were assigned a video that zoomed out from Earth to show the planets and the stars. In the original study, it was shown to elicit high feelings of awe, wonder and interest (Steller et al., 2018).

Our dependent variable, humility, was measured by having the participants write about their strengths and weaknesses with a time limit. The strengths and weaknesses were counted and then averaged. The average number of weaknesses were divided by the total number of both strengths and weaknesses to get a score between zero and one. This represented the proportion of their response that was weakness. The higher the score, the more weaknesses listed compared to strengths, and therefore more humility.

Participants were also asked to report how much fear, awe, happiness, wonder, and amazement they felt during the video using a Likert scale. This rating started at 1, or “not at all”, and ended at 7, or “very much”. The higher the score, the more of that emotion they felt.

All participants were also asked demographic questions. These were about the content of the video, what they thought the study was about, and whether the video loaded and played for them. The survey then sent the participants to a page with three demographic questions about age, gender, and race/ethnicity.

Procedure

This study was conducted by having each participant fill out an online survey provided by Qualtrics. They first were asked to complete a consent form. If they agreed to it, they were randomly assigned to either the awe condition or the neutral condition. After watching the assigned video, they were sent to a page with a two-minute timer, which counted down to zero. On this page, they were asked to write about their strengths and weaknesses, beginning with their strengths. Once the time was up, they moved to the next page, where they had to respond on a Likert scale about the five emotions they felt during the video. After, they were given three

manipulation check questions. After, all participants were thanked, debriefed, and given a SONA participation credit.

Results

The hypothesis we assessed was that those who were shown the awe-inspiring video would show more humble behavior over those who watched a neutral video. This was tested using an independent samples *t*-test. In Figure 1, the results showed that the mean proportion of weaknesses to strengths was lower for participants in the awe condition, labeled 1 ($M=0.258$, $SD=0.204$) than those of the participants in the neutral condition, labeled 0 ($M=0.313$, $SD=0.195$). This difference was found to not be statistically significant, $t(76) = -1.20$, $p=0.236$, $d=-0.276$. The results represent an insignificant effect that is in opposition to the hypothesis.

Discussion

For this replication study, we predicted that the participants who watched the awe-inducing video would list more weaknesses than the participants who watched the neutral video. This would have been indicative of greater humility. However, the results did not support the hypothesis. There was no significant difference in the number of weaknesses reported compared to strengths based on whether they participated in the awe condition or the neutral condition. This means the results do not match the original study either. The original study by Stellar et al. (2018), did find that a significant difference in the proportion of weaknesses between the awe and neutral condition. The original study had more participants, as well as a different sample of participants. For our study, there were only participants from Illinois Institute of Technology, which could potentially affect results. So, while the study conducted was like the original study, it was different enough to produce different results.

An additional issue may have been internal validity. With the collection of participants coming from the Illinois Institute of Technology through an online survey, there was no way to control the conditions when taking the study, nor were we able to control how much different representation there was. There also may have been issues with external validity. Only students at Illinois Institute of Technology seeking SONA credits took the survey, meaning the sample is infinitesimal compared to the global reach of the original study.

Overall, this study did not find the same results that Stellar et al. (2018) was able to find. In our study, participants who watched an awe-inspiring video were not found to list more weaknesses over strengths compared to those who watched the neutral video. This demonstrates that there should be more replications of the study and more research on how awe and humility relate. It also shows how something such as the type of sample can affect the outcome of a study.

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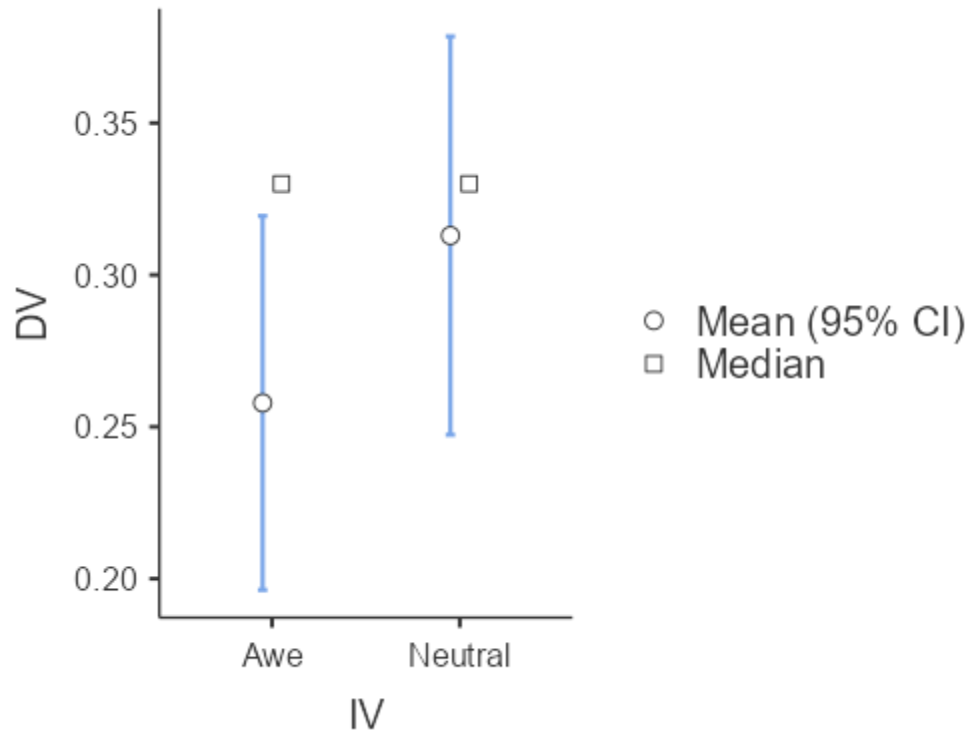
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Figure 1

Proportion of Weaknesses to Strengths Listed by Participant



A Replication Study of the Effect of Awe on Humility

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Psyc 204: Research Method in Behavioral Science

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Apr 4, 2025

Abstract

In this study, we attempted to replicate the Awe and Humility Study by Stellar et al. (2018) to examine how experiencing awe influences humility. The study used videos to research the relationship between awe and humility. Awe can be defined as a feeling of amazement or wonder when encountering something vast or beyond our usual understanding and humility as recognizing both our strengths and weaknesses in a balanced way without over or underestimating ourselves. Previous research suggests that awe makes people feel smaller in comparison to the world around them, which could lead to a more humble perspective.

We hypothesized that participants who viewed the awe-eliciting videos will have higher levels of humility, which will be operationalized in a self-report of personal strengths and weaknesses. In the study, a number of students were randomly assigned to watch either an awe-inspiring video or a neutral video. After this, they were asked to list their strengths and weaknesses in order to measure their humility.

Although data collection is still ongoing and no conclusions can be drawn yet, this research could help us understand how emotions like awe shape our self-perception. If awe increases humility, it could be used in education, leadership training, or even therapy to encourage self-awareness and openness to new perspectives. Future studies could explore whether different types of awe, like social or spiritual awe, have similar effects and how long these effects last. Understanding this connection may offer new ways to foster humility in daily life.

A Replication Study of the Effect of Awe on Humility

Awe is an emotion that happens when we see something so big or powerful that it feels greater than us and makes us stop and think (Keltner & Haidt, 2003). One idea that researchers are interested in is how awe might increase humility. Humility means being honest about both your strengths and weaknesses without thinking you're better or worse than others (Tangney, 2000). Feeling awe might help people become more humble by making them feel like a small part of something bigger (Piff et al., 2015). While some studies have looked at this link, it hasn't been tested enough to know if it's the same in different groups or situations which our study will further aid in. We're replicating a study by Stellar et al. (2018) that looked at whether awe makes people more humble where people who watched an awe-inspiring video were more likely to describe themselves in a balanced way.

In our study, the main focus is on humility, our dependent variable. Humility doesn't mean thinking less of yourself, it means being honest, open-minded, and staying grounded. It matters because it's connected to healthier relationships, less selfishness, being more forgiving, and showing stronger leadership abilities (Exline & Geyer, 2004). Humble people are also more willing to learn from their mistakes and take advice from others, essential for personal growth.

A psychological theory called self-transcendent emotion theory helps explain this idea. According to this theory, some emotions like awe, compassion, or gratitude can move people beyond thinking only about themselves, making them feel connected to others or to something greater (Stellar et al., 2017). Awe, the independent variable in our study, is one of those emotions. One of the main ideas in awe research is the feeling of the "small self." This means that when people feel awe, they start to see themselves as smaller and less important in the grand

scheme of things (Piff et al., 2015). Making people behave less selfishly and focus more on others and the bigger picture, a critical part in what it means to be humble.

In the original study by Stellar et al. (2018), participants watched either awe-inducing or neutral videos and afterward, they were asked to list their strengths and weaknesses. Those who saw the awe videos tended to describe themselves in a more balanced and honest way, giving a fair view of themselves. The researchers believed that experiencing awe helped people shift their perspective, making them feel smaller in a positive sense and encouraging honest self-reflection. Supporting this, Piff et al. (2015) also found that awe made people feel less self-focused and more generous toward others, showing that awe can have social benefits beyond personal reflection. This link between awe and humility matters because it shows that awe could be used in real-life situations, like in schools or therapy, to help people think more deeply about themselves and grow in self-awareness.

Although the original study showed a link between awe and humility, more research is needed to test if those findings are consistent. By repeating Stellar et al.'s (2018) study, we can check if we get similar results. If we do, it will support the idea that awe is truly connected to humility. But if the results are different, it could mean that the effect of awe on humility changes depending on the situation or the type of people being studied. Overall, both humility and awe are important traits to understand and by studying how these two ideas are connected we can learn more about how emotions shape who we are.

In our study, we replicated Stellar et al. (2018) to test whether awe, the independent variable increases humility, the dependent variable. We believe that people who watch an awe-inspiring video will be more likely to describe themselves in a humble way than people who watch a neutral video. We had participants watch an awe inspiring video or a neutral video and

afterward asked participants to write down a few of their personal strengths and weaknesses. Just like in the original study, we used these responses to measure humility. We used the same setup and measures as the original study because we want to see if the results can be repeated in order to help us know if the research findings are reliable.

Method

Design

We conducted a simple experiment in which awe was the manipulated variable, using brief videos designed to elicit emotional responses. The dependent variable or measured variable was the participant's level of humility. The study used a between-subjects design, with participants randomly assigned to either the awe or the neutral condition.

Participants

In this study, 89 students from Illinois Institute of Technology took part in the experiment. From those 89 students, only 72 responses were recorded properly. The responses that weren't recorded properly had insufficient data, technical difficulties with watching the video or failing to comply with instructions. Meaning, the sample size came out to be a total of 72 students. Participants were recruited using IIT's Sona's system and received course credit for their involvement. The study was administered via Sona where participants could access the link for the study. The sample size of 72 participants consisted of an age range from 18 to 26 ($M = 20.30$, $SD = 2.10$) with 50% identifying as men, 44.4% identifying as woman, and 5.6% identifying as non-binary. The demographics of the study sample were, 14.3% Black, Afro-Caribbean, or African American, 32.9% Hispanic or Latino, 21.4% Non-Hispanic White or Euro American, 22.9% East Asian or Asian American, and 8.6% identified as other racial identities.

Materials

The materials used to manipulate the variables were two Youtube videos, each approximately 2 minutes long. The first one was a video that shows the scale of the universe. It visually expands from familiar objects on Earth to the farthest reaches of the observable universe, highlighting the vastness of space. It is designed to evoke a sense of awe by showing just how small we are in the grand scale of existence. Participants shown this video were in the awe condition. The second video was about how to build a fence. The video is designed to be instructional and practical, so it generally elicits low emotional arousal. Viewers are likely to feel neutral, focused, or mildly interested, but it typically does not evoke strong emotions like awe, joy, or sadness. This makes it suitable as a neutral control condition in psychological experiments (Schaefer, Nils, Sanchez, & Philippot, 2010).

In order to measure humility, participants had two minutes to write about their strengths and weaknesses. Participants were instructed to list them as if they were describing them to a person they just met. The number of strengths and weaknesses were added up and then averaged, which was also done for each participant. In order to find the proportions of weaknesses to strengths listed, the total number of weaknesses were divided by the total number of strengths and weaknesses listed for each participant. This was done for calculation and analysis purposes in order to create a humility score ranging from 0 to 1. Scores closer to 0 signified lower humility (less weaknesses listed) while scores closer to 1 signified higher humility (more weaknesses listed).

Awe was measured using a 7 point Likert Scale. Participants rated how strongly they experienced five emotions, awe, fear, wonder, happiness, and amazement, after watching the

video in their condition. The scale ranged from 1 = not at all to 7 = very much, with higher numbers meaning they felt that emotion more intensely.

Participants were also asked basic demographics questions like their age, gender, and race. Both gender and race/ethnicity sections included the option for the participant to self-describe or to select more than one answer.

Procedure

This study was completed through an online survey. Participants were first shown a consent form that described the general purpose of the study. After agreeing to the consent form and participation, participants were randomly assigned in the awe or neutral condition. Those in the awe condition had to watch a 2 minute video about space and while those in the neutral condition watched a video on how to build a fence. Afterwards, they were directed to the next page in which they had 2 minutes to list out their strengths and weaknesses as if they were explaining them to a person they had just met. Following this, participants had to rate how much they felt certain emotions while viewing their respective video. Participants then had to answer a set of questions regarding what the video was about, their thoughts on the content of the study, and video quality. Lastly, participants were asked basic demographic questions like age, gender, and race and on the last page they were thanked and shown a brief summary of the study's purpose.

Results

We tested our hypothesis that viewing an awe elicited video increases humility using an independent sample t-test comparing the proportion of weaknesses scores for the two groups. As illustrated by Figure 1, the mean proportion of weaknesses to strengths in the awe condition ($M = 0.257$, $SD = 0.215$) was lower than that in the neutral condition ($M = 0.295$, $SD = 0.191$). This

difference was not statistically significant, $t(70) = -0.778$, $p = .439$, $d = -0.185$. The effect size was small, $d = -0.185$, indicating a slight, non-significant tendency for participants in the awe condition to report fewer weaknesses relative to strengths compared to those in the neutral condition.

Discussion

Contrary to our hypothesis and prior findings, viewing the awe-eliciting video did not lead to greater humility compared to the neutral video. In fact, the neutral condition reported a higher mean in humility scores than the awe condition. Additionally, our results don't match the findings of Stellar et al (2018). The small and non-significant effect suggests that, in this sample, awe did not reliably shift participants' self assessments of weaknesses versus strengths. It's possible that contextual factors, such as the online testing environment, varying engagement with the video, or differences in baseline humility, dampened the expected effect.

The lack of a significant difference between conditions stands in contrast to Stellar et al. (2018), who found that awe reliably increased humility ($M = 0.32$ vs. 0.25 in control group $d = 0.61$) in a lab setting. One possible reason is that our online format allowed distractions, participants might have been multitasking, which can weaken emotional engagement (Oppenheimer, Meyvis, & Davidenko, 2009). In addition, we did not screen for baseline humility or trait openness, both of which can influence how strongly someone responds to awe (Stellar et al., 2018). Without these controls, individual differences may have obscured any true effect of the awe video.

Our sample size, though consistent with the original study, may still have been underpowered to detect a small effect. Psychological research frequently suffers from low power when samples are small or effects are subtle (Button et al., 2013). Moreover, our reliance on a convenience

sample drawn from a Western, educated, industrialized, rich, and democratic (WEIRD) population limits how well these findings generalize to broader groups (Henrich, Heine, & Norenzayan, 2010). It is therefore unclear whether null effects reflect a genuine boundary condition for awe's impact on humility or simply these methodological constraints.

Looking forward, future studies should combine online convenience with tighter engagement checks, such as possibly requiring full-screen video playback or embedding brief quizzes about video content, to ensure participants actually pay attention. Pre-testing for trait humility or openness could identify subgroups most likely to benefit from awe inductions. Researchers should also consider mixed method approaches, pairing self-reports with simple behavioral measures, for example, the willingness to ask for help, this may capture humility more directly. Finally, extending the design to repeated or longer awe exposures, along with follow-up assessments, would help clarify whether a single brief video is sufficient to shift self-perceptions of weakness and strength.

In conclusion, our direct replication did not find a significant effect of awe on humility, participants who watched the awe eliciting video reported a slightly lower but non significant, humility score compared to those in the neutral condition. This contrasts with Stellar et al. (2018), who observed a medium to large increase in humility following an awe induction in a controlled lab setting. The discrepancy suggests that brief, remote video exposures may be less effective at eliciting awe's self-transcendent effects than in-person or more immersive experiences, and highlights the importance of experimental context and engagement levels when attempting to reproduce seminal findings (Stellar et al., 2018)

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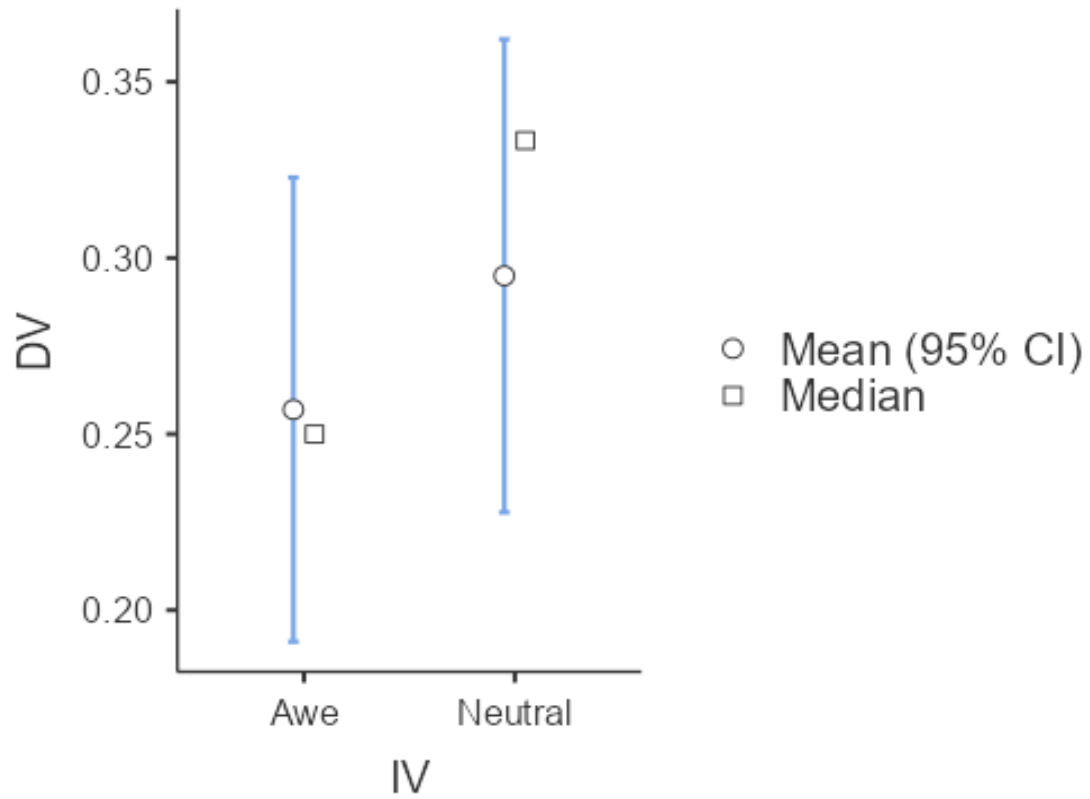
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Figure 1.

Proportions of Weaknesses to Strengths listed



Replication Study: How does the type of Norm influence Sustainable Behavior

Student 15

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PSYC 204 - Research Methods in Behavioral Science

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May 5, 2025

Abstract

Within this replication study, our goal was to confirm the validity of *Sparkman & Walton's* (2017) studies concerning normative behavior towards meat consumption. This study uses both static and dynamic norm conditions to learn more about which would people tend to confirm. A selected group of participants were given the condition to either read a dynamic descriptive norm statement or a static descriptive norm statement. These participants were recruited from an online survey distributed towards Illinois Institute of Technology's psychology undergraduate and graduate students. By replicating this study, we hope this can contribute to future research within this area.

Dynamic Norms & Sustainable Behavior: A Replication Attempt

Meat consumption is often viewed as unhealthy since it is often associated with greasy heavy foods. Yet despite its bad reputation, most have consensus on eating meat, more specifically red meat. Within just 1909 through 2007, the U.S has almost doubled its meat consumption (Daniel et al., 2011). If you also tend to eat more meat because everyone else is doing so, then you might be falling victim to pre conformity. Behavior is often based on what most people think is the norm. If people see that others are and have been striving for a change, then they might be more inclined to change into positive behavior themselves. Given this opportunity towards a greater cause, we would test if showing participants a dynamic or static norm statement would cause a greater change in behavior. Under this study, we would use meat consumption as an operationalized counter normative sustainable behavior. We are interested in manipulating norms because of the potential social impacts it could have on behavior. Norms tend to be unwritten expectations that the general public tend to follow. Individuals follow norms because they serve as guides and expectations of how to act and behave within the population. More specifically, dynamic norms targets change over time while static norms focuses on the current state of the population. Using Sparkman & Waltons 2017 study, they used two condition statements, one for each norm. Overall, they were able to find that those exposed to dynamic norms were more likely and motivated to change their behavior.

Prior research has demonstrated the impact of dynamic norms on changing behavior. In one study looking at hand hygiene during the COVID-19 pandemic, Capps et al. (2022) examined whether individuals were more likely to use hygienic products such as hand sanitizer or soap and water to wash their hands. This experiment utilized 3 different methods, gain-frame, static descriptive norms, and dynamic descriptive norms. Within the span of 6 weeks, these messages were placed in their university resident halls next to hand sanitizer dispensers. This

location choice was due to adherence from young adults within the United States during the COVID-19 pandemic. Thier results showed that people were more likely to follow dynamic norms in comparison to static norms, yet they believe the difference was not statistically significant enough to say these messages were effective to change health behavior during the pandemic.

Similarly, another study conducted during the COVID-19 pandemic examined the effect of dynamic and static norm messages on vaccines. Despite the controversial topic, especially during the early stages of COVID, this online survey was given to 2,289 unvaccinated individuals from Switzerland from the age of 16 to 60. They found no effects for either for their static or dynamic norm condition. However, they did find that those who did chose to get vaccinated, were now more strongly for vaccine intentions. One argument for why this might hold truth is due to social norms messages about COVID-19 (Geber et al., 2022). Especially during that time when COVID was at its peak, people had to think not only for themselves, but for their surroundings as well. Hense why people felt stronger about vaccinations if they were already pro-vaccine.

There have been several different studies conducted where the effect of dynamic norms static norms differs depending on different factors such as conformity, being open-minded, and even willingness to change. Despite previous studies showing otherwise, we hypothesize that those individuals who were exposed to a dynamic norm condition will be more likely to change their attitudes and more willing to reduce their meat consumption. To test our hypothesis, we conducted a replication study of Sparkman and Walton's 2017 study on students in Illinois Institute of Technology to learn more about norms and their impact on behavior.

Methods

Design

We conducted a between-subject experimental design where participants were randomly assigned either a static or dynamic norm condition as our independent variables. Our dependent variables was self-reported interest in eating less meat.

Participants

Participants were recruited within Illinois Institute of Technology through Sona. Participants received extra credit for a psychology course after completing the study. Within our study sample, there were a total of 103 participants, but we decided to omit some due to faulty or incomplete data. The ages ranged from 18-29 with a mean of 20 with standard deviation of 2.33. Demographic for this study population included 12 Black, Afro-Caribbean, or African Americans, 30 Hispanic or Latino, 20 East Asian or Asian American, 9 South Asian or Indian American, 17 Non-Hispanic White or Euro-American, 1 Middle Eastern or Arab American, and 4 were multiracial. From the 103 total participants, we eliminated 9 participants leaving 92 total participants.

Materials & Measures

Within the static norm conditions participants were given an excerpt which states “Recent research has shown that, 30% of Americans make an effort to limit their meat consumption. That means that 3 in 10 people eat less meat than they otherwise would.” On the other hand, those who were assigned the dynamic norm condition read the following excerpt, “Recent research has shown that, in the last 5 years, 30% of Americans have now started to make an effort to limit

their meat consumption. That means that, in recent years, 3 in 10 people have begun to eat less meat than they otherwise would”. After having read either one which a person was assigned to, the participants had to respond to our dependent variable which was a Likert scale of 1 (not at all) to 7 (extremely) of how interested they were in consuming meat.

Procedure

When participants clicked on this experiment via Sona, they were asked to be redirected to the website Qualtrics to proceed with the study. Once they have been redirected, they were asked to read the consent and risk factors that this study could potentially bring. After having read the consent form, they could have either agreed or disagreed to participate. If they agreed, they would have proceeded with the study, if they did not, they would have been redirected to the end of the survey thanking them for their time.

We asked patients via an online questionnaire which could have been answered with any device such as computer or cell phone. We used Qualtrics as a form to auto assign each participant to either a static or dynamic norm condition. Prior to starting the study, they were asked if they wished to take part in this study. To the people who decided to proceed with the study, they were given either the static or norm, those who did not wish to participate were redirected to the end of the survey thanking them for their participation. Once each participant has been assigned to their designated condition, they were asked several questions such as if they know or have heard about this study, are they already a vegetarian, gender, race, how much meat are they already eating, and how likely are they to refrain from consuming meat after having read their condition statement. They were also asked what best describes their personal ideology with a Likert scale of 1 (Very Liberal) to 7 (Very Conservative). Once the survey has been

completed, they were told what the study was about and more information if they wished to learn more about this topic and who to contact for further questions.

Results

We conducted an independent sample *t*-test to see individuals' intentions to eat less meat with a static or dynamic norm. We believe those who were exposed to the dynamic norm condition would have a greater mean which signifies they are less likely to consume meat. We hypothesize that those exposed to a dynamic norm condition will have a statistically significant impact compared to the static norm. As shown in Figure 1, results showed that mean dynamic norm conditions ($M = 3.89$, $SD = 2.16$) are higher than static norm conditions ($M = 3.60$, $SD = 2.19$), this difference was not statistically significant, $t(90) = 0.646$, $p = 0.520$, $d = 0.135$. The observed mean difference represents a small effect; dynamic norm conditions were 0.14 standard deviations smaller than static norm conditions.

Discussion

In all, we wished to learn and see if people were more willing to change normative behavior when given a dynamic or static norm statement. More specifically, within this replication study, we wanted to see if individuals would be more interested and willing to change their eating habits. To do so we hypothesized that the dynamic norm conditions will not have a statistically significant impact in comparison to the static norm condition. After having conducted the experiment, we found that our data supports the non-significant statistical differences between static norm vs dynamic norm. With a *p*-value of greater than 0.05, we fail to reject the null hypothesis, thus unable to replicate Sparkman and Waltons 2017 findings.

We do not believe their findings to be faulty, yet we believe our data to be flawed. There are potential limitations that could have affected our data. This includes having this study be

conducted to only Illinois Institute of Technology, more specifically psychology major students. If we could have expanded this replication study to other universities, we believe our data would have less bias and reflect higher external validity. Following the sample size was too small. The sample we used to analyze the data consisted of less than 100 participants. If we have had more participants or even started differently, we could have a higher probability of detecting a small effect showing a closer correlation to Sparkman and Waltons findings.

Another potential flaw is the fact that anyone who took the survey and completed the experiment could have lied about their true feelings. This study was also an online survey to Illinois Institute of Technology students. We have no control over what environment they were in while filling the form, any possible distractions they could have encountered, what device they used to fill the form, or even what time of day they completed this experiment. We believe this could have been improved if we had expanded the study and allowed for other universities outside of Illinois and to take it within a specific timeframe only. Next, we have the issue of a student sample. Perhaps they do not have much experience or control over what their next meal might be because they have a meal plan, or they are still eating at home cooked meals. If this experiment was distributed to older students and adults, there is a high likelihood that the results would have differed.

Despite our study being unable to replicate Sparkman and Waltons 2017 findings, we did have some strengths about this replication study. Such as the inclusion of our internal validity. We were able to control over which participants get which conditions through random assignment. There was no bias when distributing the experiment because they were automatically assigned a specific condition. Following with our construct validity, participants were able to be open about their opinions and say how likely they were to change their eating habits instead of a simple yes or no response.

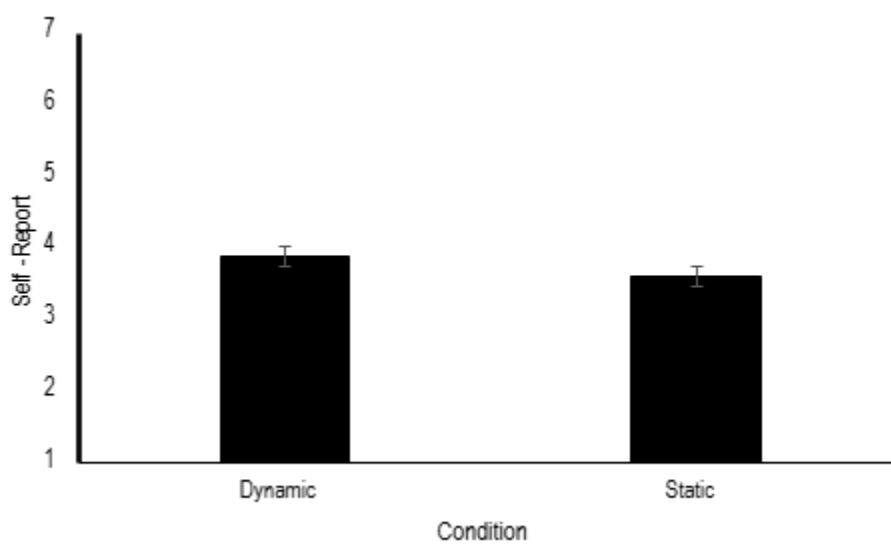
For future studies, we believe it would be best if we could control as many outside factors as possible, so we would conduct this same study again but within a lab setting. Again, we were not in control over what device they used, what time of day they might have completed this study, or even something as trivial as minimizing the possible distractions participants could have encountered. To advance this study more, a suggestion could be to have participants write out and express their feelings about their eating habits instead of rating on a scale of 1-7. Using this method of writing things out can make people express their feelings more precisely and think more about their true sentiments instead of giving it a simple number.

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Appendix

Figure 1



Replication Study: How does the type of Norm influence Sustainable Behavior

Student 16

Illinois Institute of Technology

Dr. Robert Keating

PSYC 204 - Research Methods in Behavioral Psychology

Abstract

This research paper is a replication that attempts to see the relationship between dynamic norms and sustainable behaviors. The study utilized an independent groups posttest-only design utilizing the data of 103 undergraduate and graduate students at Illinois Institute of Technology. Participants viewed a dynamic or static norm statement and were then presented Likert scales on their own meat consumption and what they thought the rest of the population does for dietary habits. We hypothesized that the dynamic norm condition would have a statistically significant positive increase in reducing meat consumption in comparison to the static condition. However, the results were that differences between dynamic and static norm conditions were not statistically significant which was reflected in recent literature regarding social norm manipulations. To which considerations of other variables effects on meat consumption along with social norms in future research.

Introduction

In understanding the different social norms people use to dictate behavior there lies a key distinction within this that can be categorized as either a static norm or a dynamic norm. Social norms being behaviors and attitudes shaped by the people around them (Chung and Rimal, 2016). This is divided into static norms, the perception of people doing the behavior, and dynamic norms which show a shift in social behavior and should be noted in how social norms are presented to people (Capps et al., 2022). With the connection between norms and behaviors being conformity when presented with this information. The current study was an attempt to replicate, Sparkman & Walton (2017), which examined the relationship between these norms and behaviors and found a statistically significant effect of dynamic norms moving people towards sustainable behaviors. And that in replicating this study it could give further insight on utilizing dynamic norms to help curb dangerous user behaviors by motivating them to help groups such as a push for quitting smoking. To which the main purpose of this research towards social norms is how people change their own behaviors upon seeing others' behaviors

Sustainable behaviors and how to best promote them have been an increasing topic within the past decade. With worries of climate change and growing economic challenges pushing for more drastic changes from corporations and people alike. Thus, the current research on the use of dynamic norms to promote sustainable practices. However, in some studies despite general improvements toward desired behavior increased with a norm statement compared to a control, the difference between the static and dynamic norm conditions were not significant such as its use in reducing the alcohol use in college students (Tylus et al., 2024). Which leaves the question of whether a dynamic norm statement is truly necessary. Similar results have also been

highlighted within studies surrounding hand sanitizer use in university students (Capps et al., 2022) to which hand sanitizer use stayed relatively similar between static and dynamic norm conditions, leading to more research on whether the findings from Sparkman and Waltons study are replicable.

All in all, these studies aim to replicate Sparkman and Waltons findings that dynamic norms have a significant impact in influencing people's behaviors. In utilizing the undergraduate students at IIT to serve as another population to study. Goals of this study are to determine whether dynamic norm states have a statistically significant difference in comparison to static norm statements. We hypothesize that the dynamic norm condition will have a higher increase in sustainable behaviors within the participants.

Methods

Design

We used a between-subjects design in our experiment with the manipulation of social norm statements and its effects on sustainable behaviors. The statements were divided into two conditions a static and a dynamic.

Participants

The study included students from the Illinois Institute of Technology who took part in exchange for course credit through SONA. A total of 103 students were recruited in total however, 10 were removed thus leaving our sample size to 93. Multiple responses were removed from the dataset. 6 Participants were removed due to incomplete answers within the survey. 3 Participants had the exact same results in comparison to the survey takers beforehand including demographic data but in different norm conditions which were excluded due to appearing as if

the participant repeated the survey. Likewise, 4 participants were categorized into multiracial for demographic purposes. 1 Participant was removed due to extreme time outliers in completing the survey. Demographic ratios were that: 13% of participants were of Black, Afro-Caribbean or African American background, 32% of participants reported being Hispanic or Latino, 22% were of East Asian or Asian American, 10% were of South Asian or Indian background, 18% reported Non-Hispanic White or Euro-American and 4% we classified as multiracial. Mean age was 20.5 with 45 women, 42 men, and 4 nonbinaries.

Materials and Measures

Participants completed the study through an online survey hosted on Qualtrics which could be accessed by any electronic device with an Internet access and IIT account. After signing on the electric consent, they would then be prompted to read one of the two norm statement conditions. In the static norm condition, participants were informed that 30% of Americans currently limit their meat consumption. In the Dynamic Norm condition, participants learned that in the past five years, 30% of Americans have started reducing their meat intake, emphasizing an ongoing shift in behavior. After which the participants would then be showed questions with multiple choice responses on scales of 1-7 which included questions such as: “willingness to reduce meat consumption” with 1 being no interest and 7 being high interest along with questions regarding gender, political ideology, age, race, and whether they were vegan or vegetarian. With their interest in reducing their meat consumption being measured for the outcome variable of whether the social norm conditions impact people’s willingness to commit to more sustainable behaviors. And those that confirmed that they were vegan and vegetarian were excluded from the dataset.

Procedure

Participants accessed the study through an online Qualtrics survey using their personal devices. Before beginning, they reviewed an informed consent form outlining the study's purpose, potential risks, and their right to withdraw at any time without consequences. The participants would then be randomly assigned to either the static or dynamic norm statement. After reading they would then answer questions regarding their: typical meat consumption in a week, and what percent that they think tries to limit their meat consumption within the population. The participants then answered where they fell on their political ideologies from very conservative to very liberal along with questions on age, gender, and race.

Results

We tested that having read the dynamic norm statement people would be more willing to lower their meat consumption. Our independent samples t-test returned the following for people within the dynamic norm condition: ($M = 3.89$, $SD = 2.16$) and the static norm condition was ($M = 3.60$, $SD = 2.19$) and showed that mean willingness to lower meat consumption was higher in the dynamic conditions versus the static condition. However, this was not statistically significant, $t(90) = 0.646$, $p = 0.52$, $d = -0.133$.

Discussion

Our experiment was trying to observe whether the presence of a dynamic social norms would influence people's decision-making to more sustainable behavior. Specifically, that the dynamic norm condition would produce a statistically significant positive difference then the static norm condition. We saw a higher mean of people willing to engage in more sustainable behaviors after reading the dynamic norm condition compared to the static condition. Moreover, the effect size was small. From this we can see that the result was not statistically significant

between the two conditions and thus fails to reject the null hypothesis. These results align with previously established literature with the current findings in the field (Tylus et al, 2024) which can mean meat consumption is not affected by social norms but by a multitude of factors and should be considered for future experiments utilizing social norm conditions.

While performing the experiment, considerations for the different validities were noted. For internal validity our efforts to remove participants who already were in non-meat consuming diets such as vegetarianism and veganism. This is to ensure that the participants had a baseline behavior to be affected by the manipulated condition. Yet, there may be confounds such as the types of meat commonly available to the students or concerns over how animals are raised that could affect their decision to lower meat consumption.

For construct validity the participants were randomly assigned into either norm statements conditions related to meat consumption practices and then were asked to rate their own meat consumption habits after. However, potential for behaviors such as desirable responding can occur. Within the United States being more inclined to other meat alternatives can be viewed as more desirable in social settings. Regarding external validity while the study was open to all the undergraduate and graduate students at IIT who may come from different backgrounds, it lacks generalizability to populations outside of this school and perhaps even the state of Illinois as it excludes the older and younger population and efforts to expand this study if redone in the future should be considered. Moreover, the statistical validity of the results was carried out through the use of a independent sample t-test. however, the result found no significant difference from the two groups as referenced in Figure 1. In which perhaps another method of sampling the data to find more minute differences may be needed. Thus, in future

research efforts to improve internal validity, external validity and attempts at getting a broader sample size should be considered.

For subsequent research on the topic of social norms other variables to be considered such as socioeconomic factors, culture, and available food options should be considered. Likewise, to improve the external validity of the study if possible, expanding the survey to not just college but perhaps high school and advertisements in public areas accessible via QR codes could be considered to obtain a much better representative sample of the population. Modification of questions to include reverse coded questions and expanding the questions to include other areas regarding sustainable behaviors can also help in finding more minute differences between the two norm statements.

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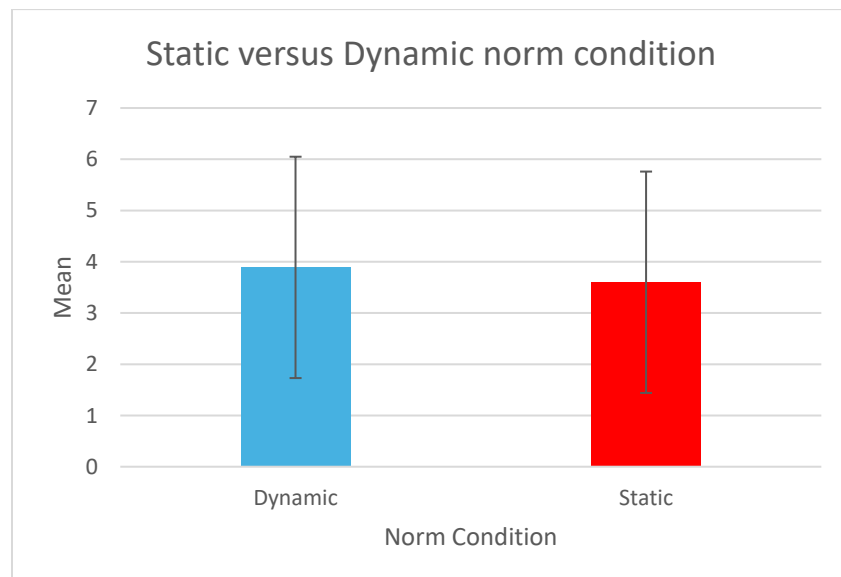
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Appendix

Figure 1

Independent Samples T-Test				
		Statistic	df	p
DV	Student's t	0.646	90.0	0.520
Note. $H_a: \mu_{\text{Dynamic}} \neq \mu_{\text{Static}}$				

Figure 2



Dynamic Norms: A Study on Sustainable Behavior

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PSYC 204 Research Methods

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May 9, 2025

Abstract

This paper is a replication study on Sparkman and Walton's (2017) study on if dynamic norms promote sustainable behavior. The report used a between-subjects experimental design to investigate whether dynamic norms are more effective than static norms at promoting interest in reducing meat consumption. Ninety undergraduate students at the Illinois Institute of Technology were randomly assigned to read either a static norm message ("Some people limit meat consumption") or a dynamic norm message ("An increasing number of people recently started reducing meat"). Participants then reported their interest in reducing meat consumption on a seven-point Likert scale. Results showed that participants exposed to dynamic norms reported slightly higher interest ($M = 3.95$) than those exposed to static norms ($M = 3.44$), but this difference was not statistically significant ($p = .270$). The effect size was small ($d = 0.24$), suggesting a trend in the hypothesized direction. These findings partially replicate earlier work on dynamic norms and indicate that dynamic messaging may modestly influence sustainable behavior.

Introduction

Social norms play a crucial role in shaping human behavior, guiding decision-making in areas such as environmental sustainability, health, and consumer habits. Traditional theories of normative influence distinguish between descriptive norms—what most people currently do—and injunctive norms—what people believe ought to be done (McLeod, 2023). Research has consistently demonstrated that descriptive norms can encourage behavioral change, but their effectiveness is often limited when the prevailing behavior is undesirable (McLeod, 2023). For example, if most people engage in a harmful behavior, simply informing individuals about the static norm may reinforce, rather than change, the behavior.

A growing body of research suggests that dynamic norms, which highlight how behaviors are shifting over time, may be a more effective means of promoting positive change. In contrast, static norms refer to the prevalence of a behavior or belief system shared within a group. Unlike static norms, which focus on the current state of behavior, dynamic norms emphasize that a growing number of people are adopting a new behavior. This perspective aligns with psychological theories of social influence and conformity, which suggest that individuals are highly sensitive to anticipated changes in societal trends (McLeod, 2023). When people perceive that a behavior is gaining popularity, they may feel compelled to mimic it for social acceptance and alignment.

Social Norms and Sustainable Behavior

This study replicates previous research on dynamic norms as a tool for encouraging sustainable behavior, particularly in the context of plant-based eating. Specifically, we examined whether highlighting an ongoing societal shift toward reduced meat consumption could enhance individuals' willingness to make similar dietary changes, even when static norms suggested meat consumption was still common. In this replication of Experiment 1, dynamic norms were used as a manipulation to test their impact on attitudes and behavioral intentions related to meat reduction. Our central research question is: Can dynamic norms effectively promote sustainable behavior by altering perceptions of social change? We hypothesize that exposure to dynamic norms enhances motivation to adopt sustainable behaviors by influencing perceptions of both personal and collective behavioral shifts.

Literature Review

Social norms messaging has become a widely used strategy to promote behavior change in areas like health, workplace equity, and environmental conservation. Studies show that **dynamic norms**, which highlight how behaviors are changing over time, can be more effective than **static norms**, which describe current behaviors. For example, dynamic norms were found to be more persuasive in anti-vaping campaigns among young adults, increasing pre conformity and reducing defensiveness (Ma et al., 2024). In contrast, static norms sometimes backfired—such as in efforts to reduce red meat consumption—by unintentionally reinforcing that the behavior is uncommon (Lee et al., 2023).

The effectiveness of norm messaging also varies by context. In a study on COVID-19 vaccination intentions, neither static nor dynamic norms had a significant direct impact, but vaccine-hesitant individuals were more open to vaccination when they perceived that others were changing their views (Geber et al., 2022). Similarly, in salary negotiations, both static and dynamic norms increased offers to women, but dynamic norms were especially effective in promoting a shared goal of equity. However, this also resulted in decreased offers for men, highlighting potential unintended consequences of norm-based interventions (Schultz et al., 2007).

Environmental studies further emphasize the importance of context. A classic study on hotel towel reuse found that messages emphasizing how many past guests reused towels—especially when localized to the same hotel room—were more effective than traditional environmental appeals (Goldstein et al., 2008). These findings underscore that the success of norm-based interventions depends on situational relevance, audience attitudes, and behavior prevalence. While dynamic norms often better support behavior change, they are not universally effective, and future interventions must be carefully tailored to avoid backfires and enhance impact.

Method

Design

This study used a between-subjects experimental design. The independent variable was the type of norm statement participants were exposed to, with two conditions: static norm and dynamic norm. The dependent variable was participants' self-reported interest in

Social Norms and Sustainable Behavior

reducing meat consumption, measured on a seven-point Likert scale (1 = "not at all" to 7 = "extremely").

Participants

The initial sample consisted of 102 students from the Illinois Institute of Technology (IIT), all over the age of 18. Participants were recruited through the SONA research participation system and were compensated with 0.5 hours of extra credit if they were enrolled in a psychology course.

After filtering out responses from individuals who did not fully complete the study, did not provide consent, or were considered outliers based on time taken to complete the study, the final sample included 90 participants. The mean age was 20.4 years. Demographic data collected included gender and political orientation.

Materials

Participants were exposed to one of two norm manipulation statements intended to influence perceptions of social behavior related to meat consumption. The first was the static norm condition, where they read, "Some people limit meat consumption." The second was the dynamic norm condition, where they read, "An increasing number of people recently started reducing meat."

These statements were designed to manipulate participants' perceptions of either a static level or a changing trend in meat reduction behavior. After viewing the norm statement, participants rated their **interest in reducing meat consumption** on a **seven-point Likert scale**.

Participants also reported the number of meatless meals they typically consume each week and answered questions about gender and political orientation using Likert-type items (political orientation: 1 = "very liberal" to 7 = "very conservative").

The study was conducted online and was available to participants over a period of two weeks. Upon accessing the study, participants first provided informed consent. They were then randomly assigned to either the static norm or dynamic norm condition and shown the corresponding norm statement.

Next, participants were asked to rate their interest in reducing meat consumption on a seven-point Likert scale, where 1 = 'not at all' and 7 = 'extremely'. Following this, they estimated the number of meatless meals they typically consume per week. Finally, they completed a brief demographic questionnaire, including items on gender and political orientation. These individual difference variables were included to assess any potential moderating effects. The entire study took approximately 2–3 minutes to complete. No experimenters or confederates were present during the study, and all participation was completed individually online.

Results

To assess whether exposure to dynamic norms increased interest in reducing meat consumption, we conducted an independent samples *t*-test comparing responses between participants in the dynamic and static norm conditions.

As shown in table 1, participants exposed to dynamic norms ($m = 3.95$, $sd = 2.13$, $n = 44$) reported slightly greater interest in reducing meat consumption compared to those in the static

norm condition ($m = 3.44$, $sd = 2.20$, $n = 45$). However, when looking at table 2, this difference was not statistically significant, $t(87) = 1.11$, $p = .270$. The mean difference of 0.51 points ($se = 0.46$) indicates a small effect size, $d = 0.24$, suggesting a trend in the hypothesized direction, though not conclusive.

Visual inspection of the bar plots and median/mean comparison (Figure 1) confirms this pattern: the Dynamic group had a higher median and mean interest level, but confidence intervals overlapped substantially between conditions. The distribution histogram (Figure 2) also shows slightly higher frequencies for higher values in the Dynamic condition.

Additional exploratory analyses revealed that women in the dynamic norm condition reported higher interest in reducing meat consumption than men in the same condition, suggesting that women may be more responsive to dynamic norm messaging. Similarly, participants who identified as politically liberal were more likely to express interest in meat reduction after exposure to dynamic norms compared to their more conservative counterparts. In contrast, race and religion were not associated with differences in responses, indicating that these variables did not significantly influence the effect of the norm condition on meat reduction interest.

Discussion

This replication study sought to determine whether dynamic norms—which emphasize changes in behavior over time—are more effective than static norms at encouraging sustainable dietary behavior. Consistent with Sparkman and Walton’s original findings, participants in the dynamic norm condition showed a numerically higher interest in reducing meat consumption

compared to those in the static norm group. However, this difference did not reach statistical significance.

There are several potential explanations for the discrepancy between this replication and the original study. First, sample size and power may have contributed to the null result. Although the sample size ($N = 90$) was comparable to Sparkman and Walton's, the observed effect size was small, and a larger sample may be necessary to detect subtle effects. Second, contextual differences (e.g., participant population, cultural attitudes toward meat consumption, or timing of data collection) may have influenced participants' receptiveness to norm-based messages.

Regarding construct validity, it is important to consider whether the manipulation of dynamic versus static norms accurately captured the theoretical construct of a perceived social shift in behavior. If participants did not interpret the norm statements as reflecting genuine change over time, the manipulation may not have effectively activated the intended construct. Future studies could improve construct validity by including manipulation checks to confirm that participants perceived the dynamic message as reflecting ongoing social change.

As for internal validity, care was taken to hold constant all elements of the experimental procedure aside from the norm condition. However, unmeasured variables—such as participants' prior exposure to plant-based messaging or pre-existing intentions to reduce meat—could have influenced responses and introduced noise. Including pretest measures of dietary attitudes or controlling for baseline meat consumption could help isolate the effect of the dynamic norm manipulation.

It is worth noting that the effect size (Cohen's $d = 0.235$) is in the expected direction and aligns with a small but potentially meaningful behavioral shift. While not statistically significant, these findings may still have practical implications for interventions targeting sustainable behavior. Future studies should explore moderators of dynamic norm effectiveness, such as personal values, baseline meat consumption, or perceived social identity relevance.

Appendix

Table 1:

Group Descriptives						
	Group	N	Mean	Median	SD	SE
DV	Dynamic	44	3.95	4.00	2.13	0.322
	Static	45	3.44	3.00	2.20	0.328

Table 2:

Independent Samples T-Test								
		Statistic	df	p	Mean difference	SE difference		Effect Size
DV	Student's t	1.11	87.0	0.270	0.510	0.460	Cohen's d	0.235

Note. $H_a: \mu_{\text{Dynamic}} \neq \mu_{\text{Static}}$

Figure 1:

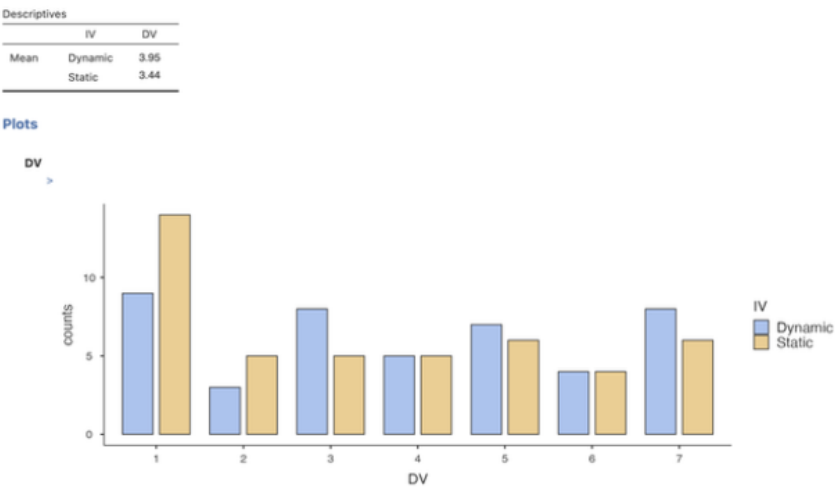
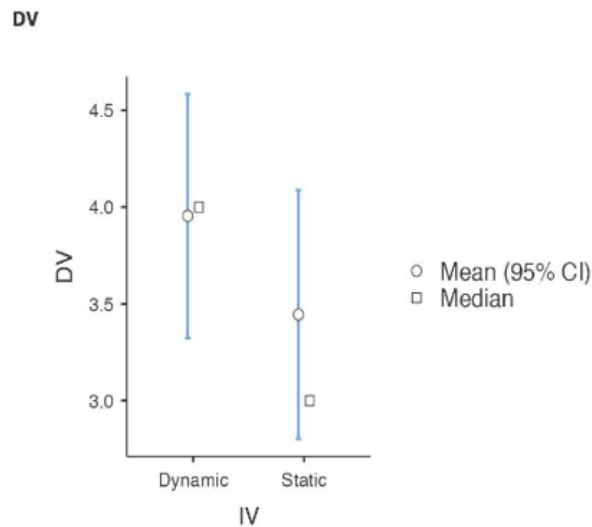


Figure 2:



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Motivation in *Stardew Valley*: An Application of Self-Determination Theory

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PSYC 312: Human Motivation and Emotion

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November 1, 2024

Motivation in *Stardew Valley*: An Application of Self-Determination Theory

Self-determination theory (SDT) is a theory developed by Richard M. Ryan and Edward L. Deci regarding how human behavior is motivated and determined. SDT contains several mini-theories that focus on innate psychological needs, intrinsic and extrinsic motivation, and goals (R. M. Ryan & Deci, 2000). Video games are interesting displays of motivational theory, since they often (whether intentionally or not) utilize research-backed ways of motivating players to complete desired tasks or have a specified experience. The game *Stardew Valley* provides ample opportunities to explore the mini-theories of SDT, including: basic needs theory, cognitive evaluation theory, and causality orientations theory.

Introduction to *Stardew Valley*

Stardew Valley is a country-life RPG originally developed by Eric Barone and released on February 26th, 2016. After inheriting their grandfather's farm plot in the small city of Pelican Town, the player can develop their farm by growing crops, raising animals, and crafting items. They can also collect resources and fend off monsters in the town's caves, fish in the local rivers and lakes, and interact with and befriend the residents of the town (*Stardew Valley*, 2024).

Stardew Valley is one of my favorite games of all time, and I chose to explore the motivational strategies used in it because I find it interesting how the game keeps me wanting to play despite carrying out what are arguably extremely boring tasks. I recently began a new farm with my partner, and the amount of freedom the game offers while still having set objectives was fascinating to me. We instinctively created our own goals to achieve that were clearly motivated by the game but never explicitly stated, making *Stardew Valley* a compelling application of need fulfillment, intrinsic and extrinsic motivation, and self-determination theory as a whole.

Basic Needs Theory: Autonomy

The basic needs theory (BNT) of SDT states that every person has inherent psychological needs that drive their behavior (Reeve, 2024). The three main psychological needs are autonomy, competence, and relatedness. Autonomy can be defined as the desire to determine one's own behavior and experiences, with a focus on individual freedom and endorsement of one's actions (Deci & Ryan, 2000). The promotion of autonomy can happen in several ways, such as by encouraging the pursuit of personal interests and goals (Reeve, 2024).

Ryan et al. (2006) investigated the link between autonomy and the enjoyment/desire to play a video game using the classic platformer *Super Mario 64*. Undergraduate students completed questionnaires before and after a 20-minute play session of the game. The researchers found that players who experienced autonomy during gameplay were more likely to continue voluntarily playing the game after the mandatory period. Analysis showed that autonomy-supportive experiences accounted for increases in enjoyment, satisfaction, and the preference for continued play (R. Ryan et al., 2006). This paper was published in the journal *Motivation and Emotion*, which has an impact factor of 1.7. While not a large value in psychology, this journal is known to be reputable and is not listed as a potentially predatory journal in Beall's List. The researchers of this study also developed their hypotheses a priori, indicating that their analyses were objective. Taken together, these factors indicate that this paper is likely highly reputable.

Autonomy is an important factor in the gameplay of *Stardew Valley*. Players are delivered "story quests" to complete around Pelican Town, which typically involve obtaining some item and delivering it to a particular villager (see Figure 1). In many games, quests like these are imperative to completing necessary objectives. However, in *Stardew Valley*, the story quests are entirely optional, and they have no time limit. Players do receive a reward upon

completing a quest, like money or friendship points with a villager. However, these rewards are not necessary to complete game objectives – moreover, there are other ways to get the same rewards, like selling crops to earn money or giving gifts to earn friendship (*Stardew Valley Wiki*, 2024). That makes story quests a way for players to pursue their interests in the game: if they find completing small errands satisfying, they can do the quests; if they find them tedious, they can ignore them.

In this way, *Stardew Valley* encourages players to play the game in the way that interests them most, supporting autonomy. While story quests are a small element of the game, this is a strong strategy to encourage player autonomy, and therefore player enjoyment, as Ryan et al. (2006) found. Personally, I love finishing these quests, while my partner tends to completely ignore story quests, focusing instead on the parts of the game she enjoys most: mining and combat. The flexibility instilled by *Stardew Valley*'s story quests therefore motivates players to play the way they want, which is just one reason I and others keep returning to play more.

Cognitive Evaluation Theory: Informational External Events

One of SDT's other mini-theories is cognitive evaluation theory (CET), which focuses on how the external environment impacts intrinsic motivation (R. M. Ryan & Deci, 2017). The theory states that external events have a controlling aspect, an informational aspect, and an amotivating aspect, which together determine how intrinsic motivation will be affected by that event. If an event emphasizes the communication of specific competence-related feedback, then it can be highly motivating; if an event emphasizes pressure or compliance, then it can be highly demotivating (Reeve, 2024).

In a foundational study, Ryan (1982) explored the effect of controlling versus informational feedback on the undermining of intrinsic motivation. Students were given a series

of puzzle problems, with half the participants receiving informational feedback after each puzzle, and half the participants receiving controlling feedback after each puzzle. Participants were then left in a room containing both puzzles and magazines, with which they could interact freely. The researchers found that the type of feedback participants received significantly affected how long they spent working on the puzzles during the free-choice time period, used as a measure of intrinsic motivation. This showed that participants who had an informational experience felt more intrinsically motivated in the following period of time (R. M. Ryan, 1982). This paper was published in the *Journal of Personality and Social Psychology*, a journal with an impact factor of 6.4, which is quite high for a psychology journal. It is not on Beall's List of potentially predatory journals, and the study itself has been widely replicated across contexts (Reeve, 2024).

Stardew Valley has several ways of communicating a player's relative success. One of the primary systems is with the in-game currency, gold. Gold can be exchanged for goods and services around town, and players can sell the items they farm, make, and find to earn gold. Players can view their current funds at any time (*Stardew Valley Wiki*, 2024). Of particular note is the "earnings screen" that is displayed every in-game night once a player goes to sleep. This screen adds up the gold earned from any items sold at the farm during the previous day, separated into five categories, with total earnings for the day shown at the bottom (see Figure 2). After being shown this screen, players wake up to start the next day on their farm.

While not explicit, spoken feedback, this screen serves to inform players how well they did that day, and in what ways. The earnings screen, therefore, can function as a form of competence-driven, informational feedback. In line with the findings of Ryan (1982), such feedback is a powerful tool in driving intrinsic motivation. I find that whether my earnings from the day were high or low, I am highly motivated by this screen in particular – if I didn't make

much money, I put together a plan to make more; if I made a lot of money, I think of what new purchases I could make to further my progress. The earnings screen therefore drives players to act in certain ways during their next day of gameplay, motivating them to continue playing.

Causality Orientations Theory: The Control Orientation

Another mini-theory in SDT is causality orientations theory, which states that individuals adopt different “orientations” to guide their behavior that are carried with them from situation to situation; these are stable personality traits. One such orientation is the control orientation, in which individuals focus on and are more motivated by social expectations and external rewards rather than internal interests (Reeve, 2024).

Hagger and Chatzisarantis (2011) tested how causality orientations impact the effect of rewards on intrinsic motivation. Undergraduate students were required to engage in an interesting puzzle, either with a reward or without a reward. After this period, participants could voluntarily interact with the puzzle, which was used as a measure of intrinsic motivation. The researchers found that control-oriented individuals spent significantly less time voluntarily engaging with the puzzle if offered a reward, showing that they were more likely to interpret an external reward in this context as pressuring, decreasing intrinsic motivation (Hagger & Chatzisarantis, 2011). This paper was published in the *Journal of Experimental Social Psychology*, a reputable journal with an impact factor of 3.5 that is not on Beall’s List of potential predatory journals.

In *Stardew Valley*, one of the primary goals of a standard playthrough is the restoration of the Pelican Town Community Center, which can be done by donating items from all parts of the game to “bundles” (*Stardew Valley Wiki*, 2024). Completing all the bundles in a room unlocks a reward, which is listed on the bundle (see Figure 3). These rewards are generally large

milestones, such as unlocking part of the map or allowing the use of a new building on a player's farm. Completing all the rooms fully restores the Community Center.

The Community Center bundle system motivates players to explore all aspects of the game, since certain items can only be obtained from certain activities (e.g., mining and fishing). As Hagger and Chatzisarantis (2011) showed, control-oriented individuals are more sensitive to the presence and quality of a reward than others. Therefore, control-oriented players may only be motivated to complete Community Center rooms because they have the best rewards, like my partner, while other players may be motivated to pursue the activities offered by the game and thus obtain Community Center items from an internal desire to explore, like me. In this way, the game has the potential to motivate both kinds of players; however, research warns that if a control-oriented player is offered a reward, it could lower their intrinsic motivation. If they have no desire to obtain an item for a bundle, they could want to stop playing.

Conclusions

Stardew Valley uses a number of evidence-based strategies to motivate its players. However, research suggests that these strategies could be improved further. For example, the findings of Ryan et al. (2006) indicate that greater autonomy corresponds with greater motivation; therefore, *Stardew Valley* could consider giving players the option to cancel a story quest entirely instead of just ignoring it, since that would allow players who have no interest in those quests to pursue the game in the way they most enjoy. Additionally, the end-of-day earnings screen could give even more specific feedback, such as “this is the most you’ve earned in a day”, since Ryan (1982) suggests that more informational feedback increases motivation. Hagger and Chatzisarantis (2011) showed that perhaps Community Center bundle items should have more ways to be obtained, in order to motivate players with all types of causality

orientations. Despite potential room for improvement, *Stardew Valley* is a fascinating case study in video games that are highly successful at motivating their players.

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Appendix

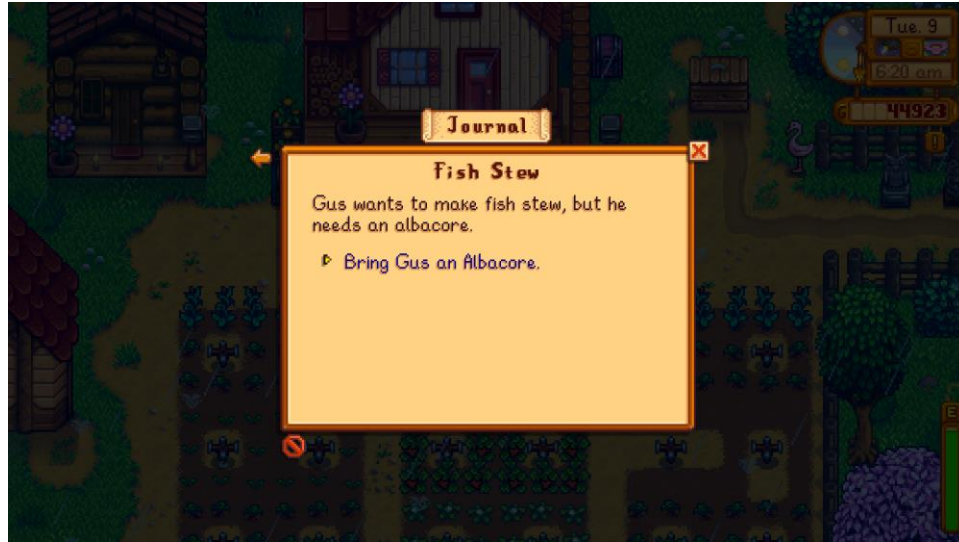


Figure 1. In-game screenshot from *Stardew Valley* showing a story quest in the journal.



Figure 2. In-game screenshot from *Stardew Valley* showing the end-of-day earnings screen.



Figure 3. In-game screenshot from *Stardew Valley* showing a Community Center bundle.

PSY312 Paper Grading Rubric Option 2

Option #2: Exploring motivation and emotion science in video games. Use motivation and emotion science to analyze strategies used in a video game of your choice, and make recommendations for improving it. Specifically, in this paper you will identify **three different strategies** used to motivate players in a video game of your choice. For each strategy / game feature, evaluate the relative strengths and weaknesses using concepts from this course and empirical evidence (findings from peer reviewed research studies on motivation and emotion, though not necessarily about video games, per se). Finally, offer some suggestions for how game developers could use evidence-based strategies to improve this video game in the future.

The Basics (20 points)

 5 / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

 5 / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

 10 / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

__5__ / 5 pts. Briefly describe the video game and why you chose it.

__20__ / 20 pts. Motivational strategy #1 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

__20__ / 20 pts. Motivational strategy #2 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

__20__ / 20 pts. Motivational strategy #3 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

__15__ / 15 pts. Conclusions. Offer some suggestions for how game developers could use evidence-based strategies from motivation science to change/improve this video game in the future (15 points)

A+ paper. Very few notes. Only minor comments added above for consideration. 100/100.

**How People Can Use Motivation Strategies to Advance Climate Change Mitigation
Strategies**

Student 13

Illinois Institute of Technology

PSY 312 - Human Motivation and Emotion

Arlen Moller

Introduction

Climate change has been an increasing problem around the world. There are people that are actively fighting climate change while there are some people that deny it is really happening. Among the people that are fighting climate change, some would appear to be more motivated than others. The reasons for these motivations are variable and can depend on different characteristics of people and their environments. Numerous examples of evidence for climate change have been shown over the last several years. The extreme flash floods in southeastern Spain is the most recent example that a natural disaster has been linked to climate change. The devastation that Hurricane Helene left is another prime example.

But does this mean that people will be instantly motivated to take action? Not exactly, as it's more complicated than it would seem to be. It can depend on what kind of mindset people grow up with. Mindsets impact motivation, performance, and even procrastination at the individual level (Duchi et. al, 2020). From the same article, they identify two types of mindsets among people. The mindsets identified are entity mindsets and incremental mindsets. Entity mindsets are otherwise known as fixed mindsets and incremental mindsets are also known as growth mindsets.

The type of mindset people have is not the only contributor to whether or not people are willing to combat climate change. Extrinsic motivation has been linked to people willing to take action on our climate, along with intrinsic motivation. Intrinsic motivation is when people decide to do something because they find it enjoyable or satisfying. Extrinsic motivation is when there are external rewards involved when doing something. Research about intrinsic and extrinsic motivation show somewhat mixed results about whether or not it can actually motivate people to

take action on climate change. Research on how mindsets are linked to people combating the climate is scarce.

This paper will dive deeper into how certain mindsets that people have are linked to climate change risk mitigation strategies and how intrinsic motivation and extrinsic motivation are associated with these strategies. We'll also go into future implications that will be impacted given certain situations.

Mindsets

A mindset can be defined as attitudes that someone has set to act about things. There are several distinct types of mindsets a person can have. However, we won't be getting into all of them here. We'll only be looking at how fixed mindsets and growth mindsets can be attributed to a person's willingness to combat climate change. A person that has a fixed mindset believes that their personal qualities are not subject to change and they believe that genes are responsible for who they are. Someone with a growth mindset believes that their qualities and overall intelligence can be changed over time through effort. A survey that was published by Duchi et. al (2020) showed that having a growth mindset about the world led to them accepting attitudes about climate change and beliefs about its risk mitigation. The same survey also found a positive correlation between having a growth mindset and having higher values, beliefs, and behavioral inclinations that people experienced after being informed about a persuasive and informative text about climate change. The same participants were later engaged in more pro-environmental activities. People that were found to have a fixed mindset were more likely to have a negative attitude towards climate change risk mitigation.

The earlier survey studied individual people but it also included certain groups of people. It found that people with fixed mindsets were more likely to be of older age and identify as a

Republican while those with growth mindsets were mostly composed of younger people and identified as a Democrat.

Lay theories can be used to interpret how people view the world around them along with other people in their environment (Soliman & Wilson, 2017). This would mean that people not only hold implicit beliefs about people or groups of people around them, but they also hold the same beliefs about the world around them. It would raise the question: is the world itself stable or changeable? There are numerous studies that have looked at different societal issues and how these lay theories can be applied.

Motivation Types

We found that having a certain mindset about things can lead to people being more motivated to take action, but it's not known what type of motivation can occur. Two types of motivation will be discussed: intrinsic and extrinsic motivation. As mentioned earlier, intrinsic motivation is when people are motivated to do something because of their internal beliefs and values, they find it satisfying. Extrinsic motivation is when they're only motivated to do something when there is a reward involved, whether it be tangible or intangible. In multiple studies by Sharpe, Perlaviciute, and Steg (2021), they found positive correlations between pro-environmental motivation, and pro-environmental behaviors along with environmental policies.

These attitudes and behaviors would associate more with intrinsic motivation because these people feel that being more environmentally friendly was something they believed was right. There were no external rewards involved in these studies, so there was no confounding variable of extrinsic motivation involved.

However, research in this area is mixed as a study published by Ortega-Egea, Garcia-de-Frutos, & Antolin-Lopez (2014) found that there was inconclusive evidence of a relationship

between motivation and pro-environmental behavior. It is also limited as there are very few studies that have examined how different types of motivation can impact climate change risk mitigation strategies.

Implications

As the planet continues to warm, it's becoming more important to take action on climate change to limit any further damage. We are heading in the right direction but more still needs to be done. The government has been very slow at getting things done for the environment so it increases the responsibility of the people to engage in pro-environmental behavior. Individuals have a central role to play to address the increasing risks that climate change poses. We can respond through adaptation and mitigation efforts (Ortega-Egea, Garcia-de-Frutos, & Antolin-Lopez, 2014).

There are already impacts of climate change that were and are unavoidable and more that could still be possible soon. It may not be possible to change the mindsets of people, especially ones with fixed mindsets as they're the most difficult to change, but we may be able to increase the pro-environmental motivation of people. If we are able to do this, we could be able to limit any further impacts of climate change.

Some people would believe that their actions would not matter, especially ones that have low intrinsic motivation and have a fixed mindset. Any action taken to combat climate change helps. Progress has been made but we need to keep doing more.

Conclusion

Growth mindsets have been positively correlated with pro-environmental motivation and pro-environmental behavior. Fixed mindsets have been negatively associated with pro-environmental behavior and motivation. The problem is that research involving intrinsic

motivation and the amount of pro-environmental behavior change is mixed and mostly scarce. More research needs to be done to determine if we can increase motivation only or if there are other variables involved in attempting to increase climate change risk mitigation.

There have been many recent examples of people denying climate change is even occurring. It has been more and more of a political issue than a scientific issue. We should focus more on science as the political side goes more into “fake” science, or pseudoscience, just to persuade people to be on their side.

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PSY312 Paper Grading Rubric Option 3

Use motivation and emotion science to advance climate change risk mitigation strategies for the sake of your and future generations' quality of life on planet Earth.

Option #3: Climate Change Risk Mitigation Strategies. Use motivation and emotion science to inform strategies for reducing climate change for the sake of your and future generations of people living on planet earth. As evidence that the planet is warming due to human behaviors increases, many scientists have put forward recommendations for governments, businesses, and individuals to help mitigate further downstream suffering. Yet, we are finding that many governments, businesses, and individuals lack sufficient motivation to follow those recommendations. A big challenge! To help address this challenge, identify **three different strategies** that you think could help motivate government officials, business leaders, or individual citizens to follow one or more of these recommendations. Then, evaluate the relative strengths and weaknesses of each strategy using concepts from this course and empirical evidence (findings from peer reviewed research studies on motivation and emotion, though not necessarily about climate change mitigation, per se). Finally, **offer some suggestions** for how the U.S. federal government, Chicago Mayors' Office, Illinois Tech, or you could use evidence-based strategies to motivate climate risk reducing action better going forward.

The Basics (20 points)

___5___ / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

___5___ / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

___7___ / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

___15___ / 20 pts. Motivational strategy #1 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

___15___ / 20 pts. Motivational strategy #2 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

_15___ / 20 pts. Motivational strategy #3 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

___15___ / 20 pts. Conclusions. Offer some specific evidence-based suggestions for improving how the U.S. federal government, Chicago Mayors' Office, Illinois Tech, or you as an individual could use evidence-based strategies to motivate climate risk reducing action better going forward.

100 minus 23

Total score 77

Motivation for Playing Video Games
Such as *Genshin Impact*

Student 14

Department of Lewis College of Science and Letters, Illinois Institute of Technology

PSY312: Human Motivation & Emotion

Dr. Arlen Moller

November 1, 2024

In today's modern era, it's safe to assume we have all played a video game at some point in our lives. Video games range from electronic consoles to educational learning. It has been estimated since the year 2017, that more than 90% of U.S. children and adolescents use some sort of video game (Calvert, 2017). If it's so prevalent within this era, what motivates people to play games? Let's take a step back and look at what is motivation. The theory of motivation science consists of three aspects, energy and the willingness to change behavior, and persistence (Reeve, 2024). We need these three subjects when dealing with motivation because, without one, it will create an imbalance. Energy is needed because it tells us how much we want something and its intensity. Direction allows us to have a bigger picture of the expected outcome. This could range from a goal someone has to what they expect to receive from the change in behavior. Lastly, we need persistence because it gives value and reason for why we want to do something. With this understanding of what motivation is, we can see how this can be applied to the topic of video games. Video games can be seen as an output for different reasons such as stress relief, entertainment, achievement-oriented hobbies, and even as a means to interact with others. We will be looking into three different strategies used to motivate players. This includes: Self-Determination Theory, Cognitive Evaluation Theory, and the Achievement Goal Theory.

Let's take, for example, the popular Chinese game produced by miHoYo, *Genshin Impact*. Genshin is an open-world, role-playing game. At the beginning of the game, we are shown two siblings who are traveling through the universe. One of these destinations includes the world of Teyvat. When the siblings tried to leave this place, they were stopped by an unknown god. Depending on which sibling a player decides to play as the other sibling gets their powers revoked and sealed away. The character you play as gets sent back to Teyvat where they are now alone and powerless. Trying to survive in this new place, the player meets Paimon. The

two of them quickly become friends and Paimon agrees to be your travel companion. Thanks to Paimon, she tells you more about how Teyvat is structured and ruled. Teyvat is composed of seven different nations, each ruled by different gods also known as Archons of different elements. After the traveler shares what happened to them and how they got to the land of Teyvat, Paimon tells the traveler to head toward the nation of Mondstadt to find clues about their missing sibling. Once there, the traveler gained new friends and helped resolve a crisis Mondstadt was facing as well as befriend the Archon. Eager to find any information about their missing sibling, the same cycle repeats for every nation they visit. The more you play the game, the more primogems are given to the player. Primogems are rewards for completing a task. You can select up to four characters at a time to play as you complete open-world quests. With primogems, a player can test their luck and see if they can win new characters or weapons.

A basic human factor is the psychological need for growth. One way to understand this is using Self Determination Theory (SDT). This theory posits that we need and strive for competence, relatedness, and autonomy (Ryan & Deci, 2004). When one has a great sense of competence, they are able to have optimal interactions with their environment and the ability to seek out more steady challenges. Following, having a greater sense of relatedness creates the need to have more interconnectedness and deep bonds with our surroundings, especially those who we share a special bond with. Lastly, with a higher standing of autonomy, we are able to have liberty and choose our own behavior. Not only can these needs be used in our daily lives, but they can also be applied to the video game Genshin Impact. SDT can be viewed in this game because these are three strategies used to motivate players. Players high in competence, are given side quests and bosses to defeat as an extension of the normal story mode to test their abilities. Players high in relatedness have a greater connection with all the details the developers use to

create an ambiance which they find themselves comfortable with. Players can also visit different people and create friendships, together they can fight the enemies or simply have conversations. Lastly, we can see Genshin Impact using SDT with the integration of autonomy because a player is able to use the characters they wish and go where they want. My recommendation as to how they may improve this game is to provide a more striving environment where they give you encouragement when you fail. An experiment where researchers were studying need satisfaction within a classroom environment showed that when given more encouragement and engagement, the classroom was overall more satisfied, therefore they participated more. Whereas, when they were put in a more autonomous and disengaged environment they reported to be more frustrated and less likely to participate (Cheon et al., 2019). Having this information about need satisfaction, we can conclude that a weakness within SDT is the amount of moderation needed to strive. All three components; autonomy, competence, and relatedness, are needed but in a very precise amount. Given, SDT is still a relatively well-founded concept because having supportive psychological needs promotes engagement and intrinsic motivation to oneself (Reeve, 2018).

Along with the Self-Determination Theory, another factor that contributes to a player's motivation is the Cognitive Evaluation Theory (CET). CET ties in relation to the three topics from SDT which are competence, relatedness, and autonomy. Instead of looking at motivation as a whole, we will look at it in terms of intrinsic and extrinsic motivation. Intrinsic motivation correlates to our strive to grow based on our interest and enjoyment (Ryan & Deci, 2004). On the contrary, extrinsic motivation refers to striving towards tangible rewards such as money or prizes. CET claims that the form in which rewards are given to an individual can affect their locus of causality impacting the three factors of SDT (Deci et al. 2017). In other words, intrinsic motivation can be undermined if the extrinsic rewards are more appealing. When rewards are

given in a highly controlling extrinsic manner, do x to receive y, it diminishes intrinsic motivation because it affects the person's autonomy. While, if rewards are given in a more informational environment, praise or compliment, it can create intrinsic motivation because it allows an individual to feel they have a higher sense of competence (Reeve, 2024). A flaw within CET is the possibility of controlling individuals. Tangible rewards could be seen as a form of bribery to get people to engage in behavior they otherwise would not conduct. Especially in professional settings where individuals are encouraged to have different ideas and autonomy, CET can overtime diminish authenticity. Back to video games, some quests may have more rewards than others, so this raises the question if this is some sort of bribery or motivation from the game. While CET may be bad in some scenarios, it does not automatically mean we should avoid it because it does come with its own strengths. If an individual reflects and has an understanding of their own locus of causality, they can enhance intrinsic motivation and have more overall motivation satisfaction. A meta-analysis with 128 participants proves if participants are rewarded with verbal awards their intrinsic motivation is enhanced (Deci et al. 2001). This could also be attributed to video games. For any game in general, there are different types of players, this does not exclude Genshin Impact. CET can be seen in the world of video games because there are different reasons why players might play a game. Video game developers need to find an intermediate where they are not controlling a player because it can make them not play a game, while also giving them just enough rewards to where a player feels they deserve the prize. I would recommend giving more praise and encouragement, especially towards new players instead of giving them a tangible reward.

Aside from the Self-Determination Theory and the Cognitive-Evaluation Theory, the last factor there is to discuss is the Achievement Goal Theory. This theory concerns why people are

motivated to do something. This can be viewed in two different forms, to have a change in our mastery goal or our performance goal. Mastery goal approach relates to making progress on the self and developing one's own competence. While those who identify with a more performance-goal approach relate to wanting to prove their abilities and demonstrate they are worthy of their own abilities (Reeve, 2024). This can be further broken down into positive or negative categories. For individuals who have a positive mastery approach orientation, they strive to learn as much as possible for their own benefit. A negative approach, mastery avoidance, tends to learn from the fear of possible failure. Moving towards performance orientations, a positive approach, performance-approach goal, tends to work to perform and show off their skills are better than others. At last, we have the negative approach, performance-avoidance goals, which consist of avoiding having the appearance that they are less capable than their peers (Wolters, 2004). Considering all these four options of how a person can pursue a goal, a weakness within this concept relates to the negative emotions that this can cause to some individuals. In a work setting or in a classroom authorities have either options, to implement a mastery or performance goal approach. A study conducted by Elliot and Church found that using performance-avoidance goals can actually generate negative emotions. Along, it can even create underperformance because they are not promoting intrinsic motivation (Elliot & Church, 1997). Not to be discouraged because this theory still supports how mastery goals can be promoted with the right environment and scenario, an individual's motivation and willingness to learn can be changed for the better. This theory can be applied to Genshin Impact because it helps describe the different types of gamers. With a more mastery-goal approach, players are more willing to make mistakes to learn from them. They are not afraid to try new character combinations and learn more about the game in general because they may be genuinely interested in what makes the game appealing

to them. Meanwhile, with a more performance goal-oriented approach, players only want to show off how many points they have received compared to other players. In relation, I would recommend that game developers should be aware that they should make a more friendly environment where players are welcome to make mistakes and not be expected to pass everything on their first try. Indulging in a more mastery rather than performance goal achievement environment can help eliminate belittling behavior from players who only point out mistakes and never give words of encouragement.

In short, many theories try to explain why and how people are motivated, this includes the motivation to play a video game. The Self Determination Theory correlates motivation to autonomy, competence, and relatedness. Cognitive Evaluation Theory expresses why our intrinsic and extrinsic motivation can impact our motivation. Finally, the achievement goal theory explains the degree of motivation one could be depending on one's goals. Using these three theories, game developers could give more words of encouragement, be mindful of how many rewards are being given, and aim to create a positive environment where players don't feel discouraged. There is no definite answer to explain motivation, but there are different theories to explain different aspects of it.

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PSY312 Paper Grading Rubric Option 2

Option #2: Exploring motivation and emotion science in video games. Use motivation and emotion science to analyze strategies used in a video game of your choice, and make recommendations for improving it. Specifically, in this paper you will identify **three different strategies** used to motivate players in a video game of your choice. For each strategy / game feature, evaluate the relative strengths and weaknesses using concepts from this course and empirical evidence (findings from peer reviewed research studies on motivation and emotion, though not necessarily about video games, per se). Finally, offer some suggestions for how game developers could use evidence-based strategies to improve this video game in the future.

The Basics (20 points)

___5___ / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

___5___ / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

___7___ / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

___4___ / 5 pts. Briefly describe the video game and why you chose it.

___15___ / 20 pts. Motivational strategy #1 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

___14___ / 20 pts. Motivational strategy #2 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).

- ___14___ / 20 pts. Motivational strategy #3 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy or game feature is used in this video game (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other gamers'). Cite at least one primary scholarly source (a peer reviewed journal article).
- ___10___ / 15 pts. Conclusions. Offer some suggestions for how game developers could use evidence-based strategies from motivation science to change/improve this video game in the future (15 points)

$$100 - 27 =$$

73

Intrinsic Motivation: I Choose You!

Analyzing Mini-Theories of Motivation in the Pokémon Franchise

Student 2

Department of Psychology, Illinois Institute of Technology

PSYC312: Human Motivation and Emotion

Dr. Arlen Moller

November 1st, 2024

Analyzing Mini-Theories of Motivation in the Pokémon Franchise

“Gotta catch ‘em all!” The question is: catch what, exactly? The iconic slogan of the Pokémon franchise refers to the classic cast of ‘pocket monsters’ that the series is named for, but perhaps it could be used in a different context. Nintendo, as game developers, need to catch every child, teen, and young adult from the 1990s to the 2020’s and get them to play Pokémon. In order to get them to do so, their games need to be interesting, engaging, and, most of all, motivate the player to keep playing. Some of the theories of motivation that can be applied to the Pokémon games are the optimal challenge of competency, implementation intentions, and relatedness satisfaction through game mechanics such as scaling battle difficulty, overworld sprites, and multiplayer interaction.

While the Pokémon franchise contains many different kinds of games, this paper will focus on the core series. Released in ‘generations’, the core games all follow a very similar structure to one another. The player character lives in a world with hundreds of fantastical creatures called Pokémon and desires to become the Champion Pokémon Trainer. Pokémon can be caught from the wild during Pokémon battles and form the player character’s team to battle various other Pokémon trainers. With almost fifteen versions released of the same structure, Nintendo needs a better way to keep players interested than just adding new Pokémon to catch, which makes for an interesting conundrum. How can the developers keep players motivated to play?

Current psychology literature suggests that there are three basic psychological needs that need to be satisfied in order to live a happy and fulfilling life: autonomy, relatedness, and competence. Competence, or feeling as though one will interact effectively and successfully, is most fully satisfied when an individual is presented with an optimal challenge. Challenges that

are too easy do not make an individual feel that they truly displayed competence and challenges that are too difficult thwart an individual's need for competence by making them feel incompetent. Individuals experience the most intrinsic motivation during activities that are optimally challenging (Reeve, 2024).

A study performed by Abuhamdeh and Csikszentmihalyi found when playing chess (an intrinsically motivated, goal-directed activity), enjoyment was strongly positively associated with perceived challenge. In this study, participants were recruited from an online chess-playing website and played a series of regular, non-tournament games together. Following the conclusion of each game, participants were asked to rate their enjoyment of the game, the perceived challenge they faced, and how well they think they performed (perceived skill). Each participant entered the study with a chess rating from the World Chess Federation (ranging from zero up to the top player in the world with 2,831 points), so a relative chess rating score was generated to represent the difference between the rating value of both players. The participants who played against individuals rated 262 points above them found the greatest enjoyment in a game (Abuhamdeh & Csikszentmihalyi, 2012). This study was published in the *Personality and Social Psychology Bulletin*, which has an Impact Factor of 3.4 and a 5-Year Impact Factor of 4.5. The journal itself is a member of COPE, or the Committee on Publication Ethics.

As the main story of a Pokémon game progresses, the player must battle through a series of gyms led by Gym Leaders in order to obtain Gym Badges. In a typical Pokémon game, there are eight gym battles, battles against the Elite Four (four high level Pokémon trainers, usually in a row), and a final battle against the current reigning Champion. The order of these battles are set, including which gym the player progresses to and when. What this means is that as the player is exploring the open world, leveling up the Pokémon in their party, obtaining better

items, and learning more about the game's mechanics, they are encountering battles that slowly increase in difficulty. This presents an optimal challenge that encourages the player to keep playing. If the player is progressing in the way as the game intended, they will be approximately evenly matched with every Gym Leader they fight. By presenting the player with continual optimal challenges, the Pokémon games intrinsically motivate the player to keep leveling up their Pokémon and keep playing in order to keep up with the increasing difficulty of the Gym Leaders they are encountering (Galar, 2024). The desire to satisfy the competency need is an internalized value, meaning that the motivation produced in this way is likely to be intrinsic, long lasting, and satisfying to the player (Reeve, 2024).

When setting a goal, there are a number of factors that can increase the likelihood of that goal being achieved. One strategy that can be used to increase motivation to accomplish a goal is called an implementation intention. Implementation intentions have two main components: a trigger and an action. When the trigger is encountered, the action is performed. This is also known as an if/then plan, such as 'if I see the stairs, then I will walk up them instead of taking the elevator' (Reeve, 2024). In a 2009 study performed in the month of March by Azjen, Czasch, and Flood, these findings were confirmed. After taking a series of personality measures, participants were given a rating form and asked to watch a newscast of their choice during the month of April and then mail the form back to the researchers the day following the broadcast. Each participant was assigned to one of three conditions regarding implementation intentions: specific, general, and none (control). In the specific condition, participants were asked to pick a specific date in April on which they would watch the newscast. In the general condition, participants were asked to pick a week in April during which they would watch the newscast. In the control condition, participants were only asked to watch the newscast during the month in

April and not plan an exact date or week. There was no significant difference in the return rate of the specific or general conditions, but there was a significant difference in the return rate between participants who had created an implementation intention and those who had not (Ajzen et al., 2009). This study was published in the Journal of Applied Social Psychology. As of 2023, this journal had an Impact Factor of 2.2. In addition, this journal accepts only 6% of the articles submitted, meaning that the articles published have gone through a rigorous vetting process.

In the main Pokémon franchise, the concept of the Pokédex is introduced very early in each game. The player character is given a virtual journal that can be accessed at any time that tracks each species of Pokémon that the player has encountered and caught. As the player progresses through new areas in the open world, they encounter new kinds of Pokémon that they can catch. Eventually, with enough perseverance, they can complete their Pokédex, which can contain hundreds of unique Pokémon (Pokédex, 2024). In the original games, overworld Pokémon encounters would occur in ‘tall grass’, where the player had a set chance to encounter a hidden Pokémon on every tile. In newer generations, Pokémon have sprites that roam the overworld. Interacting with these visible Pokémon lead to battles in which they can be caught. This establishes an if/then implementation intention in order to complete the goal of catching every Pokémon and filling the Pokédex: “if I see a new Pokémon while I am traveling, then I will catch it”. The trigger cue is seeing a new Pokémon in the wild, and the action is interacting with and catching the Pokémon. What is important to understand about the motivation to complete the Pokédex is if the player is completing it as a mastery goal (getting better at catching Pokémon) or a performance goal (getting the completed Pokédex certificate or bragging rights among friends). While motivation may be higher to complete the Pokédex, it is still important to try and make the goal itself more intrinsic.

Like competence, relatedness is known as one of the basic psychological needs. The need for relatedness can be satisfied through close, emotional bonds with other people. Humans want to feel appreciated, accepted, and as though they belong with other people (Reeve, 2024). While the act of playing a video game might seem solitary, it can actually be a source of connection for people, especially after the advent of the online multiplayer system that lets individuals play a game together without needing to be in the same room as one another. A 2006 study by Ryan, Rigby, and Przybylski sought to better understand the satisfaction of the relatedness while playing video games. To do this, they surveyed hundreds of MMO (massive multiplayer online) players to get a better sense of how it felt to play these games. Participants selected from an online gaming community filled out multiple measures of basic need satisfaction, as well as other measures including game enjoyment, post-play mood, and motivation components. These surveys found that reported feelings of relatedness while playing these games was positively associated with enjoyment of the game and intention to play the game again in the future. The motivation to form bonds with others was also very positively associated with feelings of relatedness (Ryan et al., 2006). This source is from the journal *Motivation and Emotion*, which has an Impact Factor of 1.7 and a 5-Year Impact Factor of 3.5. The study itself has been cited over four thousand times and is co-authored by Richard M. Ryan, one of the original proponents of Self-Determination Theory (Deci & Ryan, 1985).

While the main Pokémon games are not considered MMOs, there is an emphasis on playing with other people. Each core game in the Pokémon series is released in pairs or even triads, such as the original Pokémon Red, Green, and Blue; Pokémon Sun and Moon; and Pokémon Sword and Shield. In each version of a game, there are version-exclusive Pokémon that can only be obtained by trading with another player who has a copy of the opposite game. In

more recent Pokémon games, players can even join each other's worlds and explore together (*Games*, 2024). While the game can be played alone, a player misses out on the full experience by not seeking out another player to connect with. As seen in the aforementioned study, feelings of relatedness can motivate players to continue playing the game in the future (Ryan et al., 2006). It is smart of the developers to keep implementing new ways to play in multiplayer so that players are more intrinsically motivated to play by the desire to connect than by social pressures or other extrinsic forces.

One trend seen amongst this data is the increased presence of motivating factors in recent games. New mechanics are introduced or older mechanics are emphasized, likely with the goal to keep players playing. With Pokémon's business model of releasing a new game every few years, especially given the similarity in structure of these games, it is crucial to keep players intrinsically motivated to play the game. The core series of Pokémon appeals to players' sense of personal competency, sets motivational implementation intentions, and helps to satisfy the human need for relatedness while sitting alone at a console. Some ways to further increase motivation is to ensure that implementation intentions are being set to work towards mastery goals, not performance goals and to introduce more multiplayer aspects as a way to connect more with other players and increase feelings of relatedness. There is a reason these games are so popular, and it isn't just the adorable mascots. Pokémon consistently motivates players to start playing, keep playing, and ultimately, keep paying.

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Climate Change Risk Mitigation Strategies Informed by Motivation Science

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PSYC 312: Human Motivation and Emotion

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11/08/24

Climate Change Risk Mitigation Strategies

Today, climate change is a significant issue with consequences that negatively impact the well-being of individuals across the globe. Climate change refers to changes in the Earth's average temperature and weather patterns. Scientific evidence shows that human activities are increasing greenhouse gas emissions which are contributing to rising global temperatures (Servigne, 2020). Consequences of climate change have been gradually unfolding and are anticipated to become worse. Therefore, it is of the utmost importance to mitigate climate change by following the recommendations of climate scientists. However, the magnitude and psychological impact of climate change have left many individuals unmotivated to follow these recommendations (Doherty & Clayton, 2011). Given the magnitude of this challenge, this paper will identify and explore three effective strategies for motivating individuals to take action in mitigating climate change. The strategies include cultivating hope, setting difficult and congruent goals, and creating a sense of relatedness while taking climate action.

Cultivating Hope

To begin, one strategy for motivating individuals to mitigate climate change is cultivating hope. Hope is an emotion that combines a desire for a specific future outcome with the belief in our ability to achieve it through multiple pathways (Reeve, 2024). When individuals have hope they have high confidence in their ability to achieve goals and optimism from seeing different pathways to do so (Reeve, 2024). This confidence and optimism acts as a powerful motivational force to encourage individuals to make environmentally-friendly choices. Using recommendations from scientists, hope can be cultivated through climate change communications such as social media campaigns that emphasize multiple achievable goals including adopting a more plant-based diet, shopping for food locally, and composting (United

Nations Environment Programme, 2022). The messages in these campaigns will highlight potential progress of how small changes in behavior can contribute to the larger, more difficult goal of mitigating climate change. Through these messages, hope will be cultivated by creating high self-efficacy (confidence in one's ability to achieve goals) and optimism to take action towards the individual's goal. By cultivating hope, individuals will be motivated to take action through sustainable behaviors that reduce climate change.

As a strategy, cultivating hope has strengths and weaknesses for motivating individuals to mitigate climate change. One strength is that hope creates a discrepancy that drives motivation. For example, hope involves a desire for a positive future outcome which creates a discrepancy between our current state, experiencing the consequences of climate change, and our ideal state, significantly reducing the impacts of climate change (Reeve, 2024). This discrepancy further motivates individuals to take actions to reduce the gap (Reeve, 2024). However, one potential weakness is that hope may be a fleeting emotion that does not sustain motivation. Research found that hope is similar to wishing and is influenced by appraising outcomes as important but less likely (Bruininks, 2005). If progress appears too slow or mitigating climate change seems impossible then hopeful feelings may fade over time and lead to less engagement in recommended actions. This demonstrates the importance of pairing hopeful messages with progress updates to maintain hope and sustain motivation over time.

Furthermore, cultivating hope through climate change communications influences the quality and quantity of motivation to engage in sustainable behaviors. Cultivating hope can promote a high quality, intrinsic motivation by showing opportunities to satisfy a need for a sustainable future (Reeve, 2024). For example, a social media campaign that shows how composting contributes to a cleaner community will create a higher quality motivation than a

campaign offering rewards for composting (extrinsic motivation). Research shows that intrinsic motivation causes greater satisfaction, engagement, and well-being in individuals (Reeve, 2024). This suggests that messages that cultivate hope by highlighting opportunities to satisfy a need will encourage more engagement in sustainable actions. By cultivating hope and intrinsic motivation, campaign messages can motivate individuals to take action in a greater quantity of sustainable behaviors.

<<<Note – This 1st section did a nice job of defining the motivation science concept and discussing some strengths and weaknesses of using it... Something that's missing is a description of a study that showed how to increase "hope" citing the primary source where that study was first described by the researchers who ran it. I'm copying and pasting the outline for these sections that was shared on Canvas, so you can see where I was expecting to find this content. >>>>

The Main Section of your Paper (3 parts) using the Video Game Analysis as an Example

1a. Describe a mini theory, key concept, replicated pattern or effect related to motivation and emotion that was discussed in class and/or an assigned reading. In the instructions the word "strategies" is used to convey that applying "a mini theory, key concept, replicated pattern or effect" intentionally is strategic.

1b. Describe a more specific empirical finding from a published study that uses this mini theory, key concept, replicated pattern or effect. Briefly describe the methods / study design. Cite the primary source. Briefly defend the quality of this study and/or source.

1c. Describe a specific video game feature, mechanic, or affordance that could be explained using the mini theory/concept/effect and specific finding as a lens to interpret why its effective. How does the psychological science outlined above predict this game feature, mechanic, or affordance will tend to influence players' motivation and emotion (be as specific as you can about the type of motivation or emotion).

2a.

2b.

2c.

Setting Difficult and Congruent Goals

To include, setting difficult and congruent goals related to adopting a more plant-based diet, shopping for food locally, and composting is a second strategy to motivate individuals. Difficult goals are challenging yet attainable, while congruent goals resonate with an individual's personal values and reflect their sense of identity (Reeve, 2024). Research shows that setting difficult and congruent goals increases effort and persistence towards attaining that goal (. For the goal of adopting a more plant-based diet, a social media campaign might challenge individuals to reduce their meat consumption by 50% in one month while explaining how a plant-based diet improves health and mitigates climate change. This goal energizes and sustains motivation by challenging individuals and aligning with their personal values of health and reducing climate change. To increase congruence the campaign can also offer support in

achieving this goal with delicious, easy recipes. By setting difficult and congruent goals, social media campaigns can effectively motivate individuals to adopt sustainable behaviors that mitigate climate change.

As a strategy, setting difficult and congruent goals has strengths and weaknesses for motivating individuals to reduce climate change. One strength of setting congruent and difficult goals is that they both increase individuals' effort and persistence in taking actions to mitigate climate change. One study demonstrated this when researchers found self-concordant motivation improved goal attainment and adjustment which sustained long-term action (Sheldon & Houser-Marko, 2001). However, one weakness is that difficult goals risk leading to feelings of failure and negative emotional consequences if performance falls short. These emotional consequences may cause goal disengagement to avoid feelings of failure. Therefore, it's important for campaigns to balance goal difficulty with attainability and provide alternatives to prevent disengagement and promote well-being. By carefully setting difficult yet congruent goals, social media campaigns can effectively increase persistence and effort which sustains taking action to mitigate climate change.

Furthermore, setting difficult and congruent goals through climate change communications affects the quality and quantity of motivation to adopt sustainable behaviors. Congruent goals align with an individual's personal values and identity which creates intrinsic motivation, the innate desire for an individual to pursue their interests (Reeve, 2024). For example, a social media campaign challenging individuals to reduce meat consumption and improve their health creates high-quality, intrinsic motivation, while campaigns offering rewards creates extrinsic motivation. Through a plant-based diet, the campaign activates the intrinsic motivation to have a healthier diet which contributes to reducing greenhouse gas emissions from

the meat industry (United Nations Environment Programme, 2022). With intrinsic motivation, congruent goals increase sustainable actions by enhancing effort and persistence, as individuals feel more satisfied and engaged (Reeve, 2024). Thus, setting difficult and congruent goals activates intrinsic motivation which increases the quality and quantity of motivation for mitigating climate change.

Promoting Relatedness while taking Climate Change Mitigating Behaviors

Additionally, a third strategy for motivating individuals to mitigate climate change is creating a sense of relatedness through climate communications. Relatedness refers to the innate need to build emotionally fulfilling connections with others (Reeve, 2024). Climate change communications can create a sense of relatedness that effectively motivates individuals by emphasizing communities where individuals care for each other's well-being and support each other in a mutual goal. For example, a social media campaign promoting composting could invite individuals to join a local and online community that is committed to reducing waste. This satisfies individuals' need for relatedness by offering a community for high-quality relationships that accept, care, and value each other while working towards a common goal. These communications can motivate individuals to adopt composting as a way to contribute to the community and the larger goal of mitigating climate change.

As a strategy, creating a sense of relatedness to motivate climate action has both strengths and weaknesses. One strength of this strategy is that creating a sense of relatedness can create accountability and support for individuals (Reeve, 2024). For example, a campaign emphasizing a community's shared commitment in waste reduction can satisfy the need for emotional connection while encouraging accountability and support in the community. Through a community, individuals will be motivated to compost and hold each other accountable in their

goals of reducing waste which will increase engagement. However, one study found that a weakness of relatedness is that social pressure or a lack of authentic connections within a community can backfire by reducing motivation and well-being (Ryan & Deci, 2000). If a campaign emphasizes a community that is highly competitive then individuals may feel pressured, leading to inaction. This demonstrates the importance of ensuring that climate communications create a sense of relatedness as well as autonomy. Campaigns can increase autonomy by recognizing each member's individual efforts in reducing waste. By promoting a sense of genuine connection and autonomy, relatedness can effectively motivate mitigating climate change.

Furthermore, creating a sense of relatedness through climate change communications influences the quality and quantity of motivation to take action in mitigating climate change. If the campaign emphasizes relatedness in a way that also supports autonomy, then it will create a high-quality intrinsic motivation that leads to greater engagement and well-being (Reeve, 2024). However, if the campaign creates relatedness with a competitive community, then it will create extrinsic motivation that leads to a decrease in the quantity of sustainable actions by undermining performance (Reeve, 2024). Therefore, climate communications should ensure supporting autonomy while creating a sense of relatedness to optimize the quality and quantity of motivation for sustainable actions.

Conclusions

Overall, cultivating hope, setting difficult and congruent goals, and creating a sense of relatedness are effective strategies for motivating individuals to take action in mitigating climate change. By applying these strategies in climate change communications, individuals can be motivated to take sustainable actions that reduce greenhouse gas emissions and mitigate climate

change. To improve the use of these strategies going forward, climate change communications should emphasize multiple achievable goals to cultivate hope and show progress towards mitigating climate change. Second, difficult and congruent goals should be set while balancing goal difficulty with attainability and providing alternatives to prevent disengagement. Last, a sense of relatedness should be cultivated by emphasizing noncompetitive communities committed to mitigating climate change, which encourages support and accountability. By improving these evidence-based strategies, climate change communications can more effectively motivate individuals to take action in mitigating climate change.

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PSY312 Paper Grading Rubric Option 3

Use motivation and emotion science to advance climate change risk mitigation strategies for the sake of your and future generations' quality of life on planet Earth.

Option #3: Climate Change Risk Mitigation Strategies. Use motivation and emotion science to inform strategies for reducing climate change for the sake of your and future generations of people living on planet earth. As evidence that the planet is warming due to human behaviors increases, many scientists have put forward recommendations for governments, businesses, and individuals to help mitigate further downstream suffering. Yet, we are finding that many governments, businesses, and individuals lack sufficient motivation to follow those recommendations. A big challenge! To help address this challenge, identify **three different strategies** that you think could help motivate government officials, business leaders, or individual citizens to follow one or more of these recommendations. Then, evaluate the relative strengths and weaknesses of each strategy using concepts from this course and empirical evidence (findings from peer reviewed research studies on motivation and emotion, though not necessarily about climate change mitigation, per se). Finally, **offer some suggestions** for how the U.S. federal government, Chicago Mayors' Office, Illinois Tech, or you could use evidence-based strategies to motivate climate risk reducing action better going forward.

The Basics (20 points)

___5___ / 5 pts. Cover page using APA style (centered title, your name, Illinois Tech, running head). A very brief intro paragraph, briefly naming the three strategies of interest. Reference section using APA style. Should include 4-10 references in addition to the course textbook, when citing research findings, the primary/original source should be cited (i.e., a peer-reviewed journal article).

___5___ / 5 pts. Citations should appear in the body of the paper using APA style. There should be zero quotations in this paper, please paraphrase sources in your own words.

___10___ / 10 pts. Spelling & Grammar (appropriately using punctuation, no run-on sentences, etc.) And within the target length 5-6 pages (not including cover page, references section, or optional figures), double-spaced, Times New Roman 12-point font, 1" margins.

Research supported discussion (80 points)

___17___ / 20 pts. Motivational strategy #1 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

___17___ / 20 pts. Motivational strategy #2 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and empirical data from

published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

14 / 20 pts. Motivational strategy #3 identified and defined using terms from motivation research literature (5 pts). One or more specific examples of how the strategy is communicated or otherwise employed to motivate climate change risk reducing behavior (5pts). An evidence-based analysis of relative strengths and weaknesses of this strategy using concepts from this course and **empirical data** from published studies; discuss consequences for quality and quantity of motivation (yours or other students'). Cite at least one primary scholarly source (a peer reviewed journal article).

___19___ / 20 pts. Conclusions. Offer some specific evidence-based suggestions for improving how the U.S. federal government, Chicago Mayors' Office, Illinois Tech, or you as an individual could use evidence-based strategies to motivate climate risk reducing action better going forward.

100-13 = 87

Overall, a nice paper. To level up on papers like this in the future, please work on engaging more deeply with the specific details of empirical studies you're citing and be careful to distinguish between journal articles that are secondary sources (summaries of past research) vs. primary sources (new study described for the first time).

Why We Keep Coming Back to the Battlefield: Unlocking Player Motivation in Call of

Duty Mobile

Student 9

Department of Psychology, Illinois Institute of Technology

PSYC 312: Human Motivation and Emotion

Dr Arlen Moller

October 29, 2024

Introduction

Millions of gamers have been enthralled by Call of Duty Mobile's intense action, realistic visuals, and wide range of customisation possibilities. Bringing competitive play and dramatic battle experiences directly to players' hands, this mobile adaption of a well-known console franchise offers everything from solo missions to cooperative gameplay and PvP modes. This paper's goal is to investigate how different forms of motivation, especially extrinsic motivation—which emphasises the material benefits and acknowledgements players receive—play a crucial part in player engagement and retention. This research project will examine these motivating elements to determine why players keep coming back to the game and offer enhancements to increase player loyalty and happiness.

Call of Duty: Mobile (CODM), launched in October 2019, quickly rose to prominence as one of the most-played mobile games worldwide. Combining classic maps, weapons, and gameplay from various Call of Duty titles, the game has attracted over 750 million players globally since its release. This massive user base has contributed to its status as the most successful Call of Duty title ever, making it a staple in both casual mobile gaming and the competitive esports scene. The success of CODM is mostly due to its frequent upgrades, new events, and support for both multiplayer and battle royale modes, all of which make the gaming interesting and fun. With yearly championships and competitive leagues offering substantial prize pools, the game's esports component has also grown dramatically, attracting players from all over the world to compete and win. This increase is indicative of a larger trend in the growing popularity of mobile gaming, which is outperforming traditional gaming platforms in terms of player numbers and accessibility. As it continues to evolve and adjust to player preferences, CODM's popularity has established it as a

flagship game in Activision's Call of Duty brand and elevated it to a leadership position in the worldwide mobile gaming market.

Understanding Intrinsic and Extrinsic Motivation in Video Games

Key components of comprehending player engagement in video games, such as Call of Duty: Mobile, are intrinsic and extrinsic motivation. Behaviours that are motivated by internal fulfilment and pleasure—like a sense of accomplishment or immersion—without the need for an outside reward are referred to as intrinsic motivation. Intrinsic motivation in gaming frequently consists of a player's enjoyment of solving puzzles or taking in the narrative, which gives them a sense of independence and skill. On the other hand, extrinsic motivation relies on material incentives or rewards from outside sources, such as leaderboards, prizes, or social acknowledgement for achieving high scores. Levelling up, winning awards, and unlocking new features and skins are examples of extrinsic variables in CODM that promote continuous involvement by providing observable accomplishments. While motivated behaviour is often examined with respect to extrinsic reinforcers, not all actions are driven by reactions to external stimuli; some are driven by 'intrinsic' motivation. Intrinsically motivated behaviours are computationally similar to extrinsically motivated behaviours, in that they strive to maximise reward value and minimise punishment (Morris et.al, 2022).

Extrinsic and intrinsic motives both contribute significantly to player engagement, but they also interact in intricate ways, according to scientific studies. Extrinsic rewards have the potential to increase intrinsic motivation, but they can also have the opposite effect when they become overbearing and cause the player to divert their attention from having fun to reaching predetermined objectives. Studies examining the effects of various rewards on players' motivation to keep playing have extensively documented this interplay, suggesting that maintaining

engagement requires a careful balance (Deci & Ryan, 1980; Deci et al., 1999). In order to maximise player experiences and promote both strategic extrinsic incentives and intrinsic delight to sustain sustained engagement, game designers must have a thorough understanding of these interactions.

Motivational Factors in Call of Duty

Competition is a major motivator in Call of Duty: Mobile (CODM), appealing to players' needs for accomplishment, skill mastery, and social recognition. Players are motivated to enhance their in-game performance by the competitive incentives and ranked leaderboards, which provide a sense of challenge and expertise. In addition to satisfying extrinsic cravings for material rewards like high ranks and exclusive skins, this competitive dynamic increases intrinsic motivation by meeting requirements for achievement and mastery (Röhlcke et al., 2018; Vorderer et al., 2003). As players compete to establish themselves in peer communities, competition also promotes social connectedness, which increases the game's attraction for involvement.

Because players are drawn to the cooperative and social dynamics found in multiplayer situations, social connection is crucial to sustaining player engagement in Call of Duty Mobile. Online gaming is primarily driven by social contact, according to research, which satisfies relatedness requirements and creates groups that strengthen a sense of belonging (Przybylski et al., 2010). Clans, buddy lists, and team-based modes are some of the CODM elements that give players the chance to collaborate, compete, and build enduring social ties—all of which are associated with higher levels of in-game enjoyment and long-term retention (Reer & Krämer, 2019).

Przybylski, Rigby, and Ryan (2010) investigated the relationship between video game motivation and self-determination theory (SDT), particularly the psychological demands for competence, autonomy, and relatedness. The study measured how the satisfaction of these requirements affected intrinsic motivation, or the desire to play for enjoyment alone, using a

survey-based design with 1,492 gamers across several game genres. They discovered that when a game gave players a sense of competence (i.e., mastery over tasks), autonomy (i.e., making important choices), and social connection with others, they were much more likely to report high levels of engagement and emotional fulfilment. The study's findings are supported by its scientific rigour and sizable sample size, which offer important insights into how games might be designed to maximise player motivation by satisfying psychological needs.

According to SDT's framework, Call of Duty: Mobile's skill-based progression system is a very successful game mechanic. For example, by enabling players to gradually improve their skills and obtain visible rewards, ranked play modes and weapon mastering challenges directly address players' desire for competence while also boosting intrinsic drive. According to SDT, this feature encourages players to engage in mastery-driven activities that also foster good emotions like pride and fulfilment by satisfying their drive for achievement. According to the study's findings, these competence-boosting elements will encourage players to stay involved and have more fun with games.

The study by Przybylski et al. (2010) is widely regarded because it is based on the well-established motivational framework of self-determination theory (SDT) and uses a large, diversified sample. The researchers' use of standardised survey methodologies makes it possible to quantify psychological requirements across game genres with accuracy, which supports the idea that player motivation and need fulfilment are related. Furthermore, the study's conclusions have been extensively repeated in many settings, confirming their applicability in comprehending how particular game elements, such as skill advancement, captivate players via internal motivation.

As players strive for prizes that reflect their in-game proficiency and ongoing progress, progression and success are key motivators in Call of Duty: Mobile (CODM). The demand for

accomplishment is satisfied by the game's levelling (rank) systems, daily challenges, and unlockable things like skins and weapons. A sense of accomplishment is also instilled by the display of trophies and badges on a player's game card for everyone to see. This increases extrinsic motivation through observable accomplishments and intrinsic motivation by rewarding mastery (Tang et al., 2020). As players are motivated to progress and reach higher levels, research indicates that these incentive systems not only meet psychological needs for competence but also promote sustained engagement (Deci & Ryan, 2000).

Call of Duty Mobile's realistic, vivid visuals, particularly those that portray battle, weaponry, and landscapes, are crucial in drawing players in with increased excitement and immersion. According to research, players become more aroused by bright, action-packed visuals, which makes the experience more exciting and emotionally impactful (Anderson & Dill, 2000). By constructing a realistic battlefield where players are completely engrossed in high-stakes missions, CODM's visual representation of violent gaming amplifies this thrill and increases players' intrinsic enjoyment and drive to return for these heart-pounding encounters.

In CODM, where players may create their own identities and play styles by customising characters, weapons, and game cards, player autonomy is a potent motivation. Because they feel empowered by their choices and achievements, players experience a "power trip" effect that increases their intrinsic drive (Deci & Ryan, 2000). Winning with customised gear adds dimension to the joy of victory and encourages players to keep making investments in their in-game character by reinforcing autonomy and personal expression in addition to symbolising expertise.

Call of Duty also offers players gratifying gameplay that allows them to experience stress relief, assert control, and enjoy themselves through exciting combat situations. In addition to producing adrenaline, shooting and outwitting opponents gives players a sense of control and

power, which boosts their sense of self-efficacy and fulfilment (Granic et al., 2013). Additionally, the immersive aspect of gaming relieves stress by providing a haven from everyday stresses and satisfying players' cravings for excitement and success in a secure virtual setting.

Conclusion: Improving Call of Duty after Motivational Factors Consideration

Resolving server problems that impede international social connections is essential to improving player satisfaction and engagement in Call of Duty Mobile. A more seamless multiplayer experience can be created by increasing server reliability and lowering latency, which enables gamers to interact with friends and communities throughout the world without annoying disruptions. In addition to strengthening the game's social fabric, this improvement would satisfy players' innate desire for cooperation and companionship (Ryan et al., 2006).

Furthermore, player motivation and contentment can be significantly impacted by reevaluating the price of mythic gun skins. Offering these sought-after objects as incentives for finishing particular in-game goals or challenges, as opposed to charging over \$100 for them, could increase players' sense of accomplishment and mastery. This change is in line with the ideas of intrinsic motivation, which motivates players to play the game more thoroughly in order to obtain special rewards, which eventually boosts retention and enjoyment (Deci & Ryan, 2000).

Lastly, I would like to add that mobile gaming, especially shooter games like Call of Duty Mobile, is an exciting and effective way to reduce tension, in my opinion. I can play strategically because of the surge of adrenaline I get in high-stakes games, like Search and Destroy. In addition to being a popular pastime of mine, this combination of psychological relief and excitement provides a useful outlet for coping with everyday difficulties in the fast-paced world of today.

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